EXPERIMENTAL STUDIES OF A CURRENT-STEPPED Z-PINCH

SAW SOR HEOH

DEPARTMENT OF PHYSICS
UNIVERSITY OF MALAYA

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Experimental Studies of a Current-stepped Z-pinch

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Saw Sor Heoh, B.Sc.(Hons.)

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To Mother

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ABSTRACT

The current-stepping technique was first introduced by *S.Lee (1984)*. This thesis is a pilot effort to investigate experimentally the effect of a current-step on the enhancement of pinch compression as predicted by the energy balance theory.

The Universiti Malaya Current-Stepped Z-pinch, UMCSZP is designed and constructed using the energy balance slug model. Electrical signals of the plasma current and voltage and streak photographs of the radial compression were taken of the Z-pinch plasma without and with current-stepping. Magnetic field mapping along the radial direction is also carried out to obtain the evolution of the current density profile. Results obtained indicate that the assumptions of a plasma of infinite conductivity in the energy balance slug model together with a constant specific heat ratio, γ equals to 5/3 at the moment of current-stepping are not valid in this experiment.

The energy balance slug model is extended to represent more closely the real plasma condition with transient effective specific heat ratio, γ_{eff}. The transient variation of γ_{eff} with the shock speed is obtained numerically from the simultaneous solutions of the closed system formed by the 1-D shock jump equations, the thermal equation of state, the Saha equation and the caloric equation. The extended model with the consideration of the varying γ_{eff} with shock speed is called the γ-varying energy balance model.

With the parameters of UMCSZP, the new model predicts no change in the final radius of the plasma column produced by current-stepping. This is in agreement with the experimental observations. The average effective specific heat ratio, γ_{eff} of 1.2 during the first compression increases to 1.6 upon current-stepping. This decreases the compressibility of the plasma and masks the enhancement in compression from current-stepping.

Based on the new model a new current-stepped Z-pinch experiment called the Marx-line Combination Current-stepped Z-pinch, MLCCSZP is designed.

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CHAPTER 1

INTRODUCTION

1.1 Introduction

The Z-pinch is one of the earliest fusion devices proposed and studied. Unfortunately, it is plagued by inherent instabilities, predicted by ideal MHD theory and observed in early experiments leading to its early and sudden dismissal as a possible fusion device *[Kruskal & Schwarzchild, 1954; Carruthers & Davenport, 1957]*. However recent work on stability regimes such as finite Larmor-radius effect *[Haines, 1981]* and works by *Hammel et. al [1989]* and *Sethian et. al, [1989]* indicate stability as long as $\dot{I}$ is increasing. This, coupled with the rapid advances in modern high-voltage pulse generators, have rekindled interest in Z-pinch research to the extent of being considered again as a possible fusion device. This renewed status also stems partly from the growing realisation that 'conventional' approaches to fusion may not lead to reactors that will be economically competitive with other sources of energy *[Pfirsch, D. and Schmitter, K.H. ,1987]*.

Various types of Z-pinches have been proposed and investigated. These include early examples summarised by *Robson*, the gas embedded Z-pinch investigated by *Walker et. al. [1986]*, *Bolton et. al. [1986]*, *Nargundkar [1986]* and *Haines, [1987]*, the fiber Z-pinch investigated by *Scudder et. al.[1986]*, *Sethian et. al, [1985]*, *Ishii et. al [1989]* and the hybrid Z-θ pinch initiated by *Rahman, [1987]*.

The current-stepped Z-pinch proposed by *Lee [1984a]* which indicates enhanced compression by means of tailoring the current-profile is of interest here. A numerical study of the current-stepped Z-pinch including the effect of radiation cooling has been conducted by *Ali [1985]* for an ideal gas of specific heat ratio, γ of 5/3. This thesis represents the first experimental attempt to investigate the enhancement in compression predicted by the current-stepped Z-pinch.

1.2 Overview of the Experiment

The experiment is a pilot effort in the investigation of the enhancement effect of the current-stepped hydrogen Z-pinch. A Z-pinch system is chosen for its relatively large radius facilitating the observation of the reduction in radius predicted by the current-stepping theory. This experiment can be divided into 3 phases.

In the first phase, a capacitor-capacitor combination system, UMCSZP, is designed and constructed to operate as a current-stepped Z-pinch. The parameters of the pinch are : length 15 cm and initial radius 7.5 cm. The operating pressure ranges from 0.4 to 3.0 mbar. Preliminary testing is followed by electrical and magnetic field measurements and streak photography of the pinch. Analysis of the results suggests that the deviation of the experimental results from those predicted by the model may be due to the breakdown of the assumptions adopted in the model. This leads to the second phase of this experiment.

The energy balance model is extended to include the effective specific heat ratio, γ_{eff} of the plasma. The development of the plasma flow field -magnetic field coupling is also checked using the magnetic Reynolds number, $\mathcal{R}_m$. The new model predicts no significant enhancement in the current-stepped compression by UMCSZP. This is because the enhancement is being masked by the thermodynamic and magnetohydrodynamic effects of the plasma. The average compressing shock speed of the first compression is about 4 cm/μs giving an effective specific heat ratio, γ_{eff} of $\simeq 1.2$. This low speed also leads to a diffusive current sheath. Upon current-stepping, the increase in shock speed rapidly raises the effective specific heat ratio, γ_{eff} of the plasma to $\simeq 1.5$ and hence lowers the compressibility *[Lee, S., 1983a]* of the plasma. Further, the diffusive current sheath inhibits proper merging of the current sheath upon current-stepping. These phenomena produce an opposing effect which masks the enhanced compression of current-stepping.

The new model predicts that an average shock speed beyond 10 cm/μs corresponding to a peak shock speed ≥ 30 cm/μs for the first compression at the instant of current-stepping is necessary to observe the enhancement effect by current-stepping. A new system, the Marx-line combination current-stepped Z-pinch, MLCCSZP is designed and constructed on the new model. The high impedance current sources are provided by a 3-stage 150 kV Marx and a 150 kV, 1.7 Ω water-line pulse charged by another 3-stage 150 kV Marx.

1.3 Layout of the thesis

This thesis is written in 7 chapters with the first chapter as introduction. The current-stepped theory by *Lee, [1984]* will be presented in chapter 2. In chapter 3, the design of the capacitor-capacitor bank combination system called the Universiti Malaya Current-stepped Z-Pinch, UMCSZP is described. The diagnostics employed to measure the plasma parameters are also discussed. Presentation of the results obtained in the first attempt at current-stepping is found in Chapter 4. Comparison of the experimental results with that predicted from the current-stepped theory is conducted and conclusions drawn. It is found that the energy balance slug model needs to be extended to include the effective specific heat ratio, γ_{eff} of the plasma. This leads to Chapter 5 where the solution of γ_{eff} with respect to shock speed is obtained and incorporated into the current-stepped model. In Chapter 6, a new system is designed on the extended γ-varying model. The thesis is concluded in Chapter 7 with discussions on the results and suggestions for further work.

CHAPTER 2

THE CURRENT-STEPPED Z-PINCH

2.1 Introduction

The current-stepping technique proposed by *Lee [1984a]* is presented. It is a novel method to obtain pinch enhancement by tailoring the time-profile of the current source driver. The theory states that the quasi-equilibrium minimum radius ratio is dependent on the time-profile of the current and not on the absolute value of the current.

2.2 Theory

The energy balance theory *[Lee, S., 1983b]* applied to a fast radial gas compressional pinch of constant length equates the work done by the magnetic piston to the plasma internal energy.

Consider the energy in the system at the point of time represented by m. The work done per unit mass W_m by the magnetic piston in compressing the plasma from an initial radius r_o to r_m is

$$W_m = \int_{r_m}^{r_o} \frac{\mu_o I^2}{4\pi^2 r_p^{\,3} \rho_m} \, dr_p \qquad [2.1]$$

where ρ_m is the mass density, r_p the radial position of the piston and I the instantaneous current flowing in the piston.

The plasma internal energy is

$$U_m = \frac{R_o}{M} \frac{T}{\gamma-1} \chi \qquad [2.2]$$

where R_o is the universal gas constant, M the molecular weight, χ the departure coefficient, T the plasma temperature and γ the specific heat ratio of the gas.

Assuming a lossless system and equating W_m to U_m, the instantaneous plasma temperature is given by

$$T_m = \frac{M(\gamma-1)}{R_o\chi} \int_{r_m}^{r_o} \frac{\mu_o I^2}{4\pi r_p^3 \rho_m} dr_p \qquad [2.3]$$

Next, we consider the pressure relationship at this time. The plasma kinetic pressure, $P_k = (\rho_m/M)R_oT_m\chi$ exceeds the magnetic pressure, P_m by the reflected shock pressure jump factor, f_{rs} *[Lee, S., 1985a, 1988a]*. This gives

$$T_m = \frac{f_{rs}}{R_o\chi} \frac{M}{\rho_m} \frac{\mu_o I_p^2}{8\pi r_p^2} \qquad [2.4]$$

From [2.3] and [2.4], the combined energy and pressure balance condition is given as

$$I_p^2 = \frac{2(\gamma-1)}{f_{rs}} \int_{r_m}^{r_o} \frac{I^2}{r_p} dr_p \qquad [2.5]$$

Upon normalisation, equation [2.5] becomes

$$\iota_p^2 = \frac{2(\gamma-1)}{f_{rs}} \int_{k_{pm}}^{1} \frac{\iota^2}{k_p} \, dk_p \qquad [2.6]$$

where $\iota_p = I_p/I_o$

$k_{pm} = r_m/r_o$

$k_p = r_p/r_o$

$I_o = V_1/\sqrt{(L_1/C_1)}$

= characteristic peak current of the LC discharge

γ = specific heat ratio of the gas and

I_p = pinch current at quasi-equilibrium

Generally, ι^2 can be written as a product of ι_p^2 and $f^2(t)$ where $f(t)$ is a time form function ;

$$\iota^2 = \iota_p^2 \, f^2(t) \qquad [2.7]$$

Therefore equation [2.6] becomes

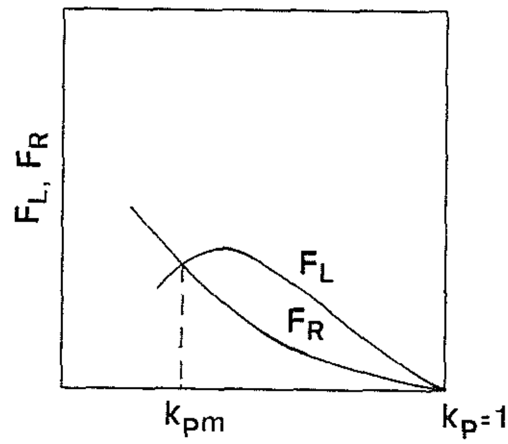
$$1 = \frac{2(\gamma-1)}{f_{rs}} \int_{k_{pm}}^{1} f^2(t) \, \frac{dk_p}{k_p} \qquad [2.8]$$

Equation [2.8] shows that the quasi-equilibrium minimum radius ratio corresponding to maximum compression at energy balance does not depend on the absolute magnitude of the current. It depends only on the time variation function of the current profile. *Lee (1983b)* has shown that for a constant length deuterium pinch, the quasi-equilibrium minimum radius ratio at maximum compression varies from 0.17 for a linearly rising current to 0.40 for an energy optimised plasma coupled LC discharge.

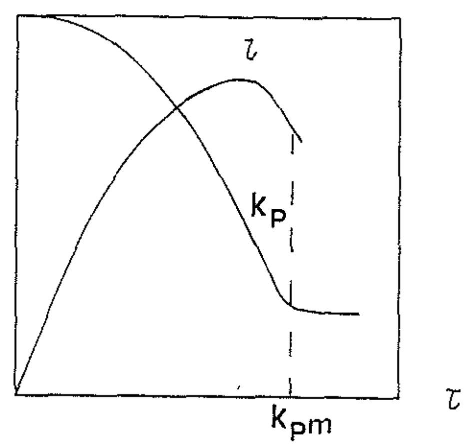
Following equation [2.6], the two functions may be referred to as $F_L = \iota^2$ and function $F_R = \frac{2(\gamma-1)}{f_{rs}} \int_{k_{pm}}^{1} \iota^2 (dk_p/k_p)$. It is observed that F_L is a function of the absolute magnitude of ι squared whereas F_R is an integral of the square of ι. Initially when k_p is less than 1, F_L is very much greater than F_R (see Fig. 2.1) but as the compression proceeds and $k_p \to 0$, F_R begins to increase faster than F_L and rises to meet F_L at the point k_{pm}. This is the quasi-equilibrium minimum pinch radius ratio at energy balance and is shown in Fig. 2.1. The corresponding normalised piston trajectory and current are also shown.

Now, if the current ι is raised rapidly just before k_{pm} is reached, then F_L would increase faster than F_R since F_L is the absolute value of the square of the current whereas F_R is the integral of the square of the current. F_L and F_R would then be diverged to meet at a smaller value of k_p, thereby enhancing the pinch compression. This is shown in Fig. 2.2.

Consider a single current-step added to the primary pinching current. The coaxial system is as shown in Fig. 2.3. When the switch S_1 in the primary circuit is closed, the capacitor C_1 initially charged to a voltage of V_1, starts to discharge the current I_1 through the inductance L_1 into the pinch with a varying inductance, $L_p(t)$. The pinching current is flowing at radius r_p with its coaxial return path at initial radial position, r_o. When the pinch approaches its quasi-equilibrium radius-ratio, the switch S_2 in the current-stepped circuit is closed so that the capacitor C_2 starts to discharge the current I_2 through the



(a)



(b)

Fig. 2.1 (a) Functions of F_L and F_R versus radius ratio without a current-step.

(b) Radial trajectory and current waveform without a current-step.

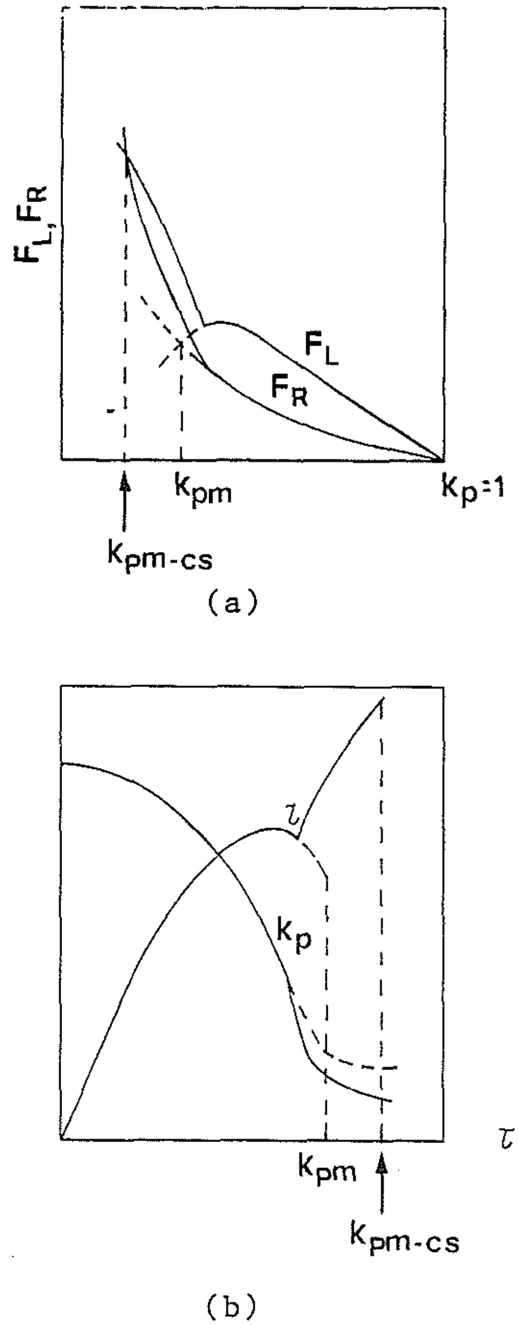


Fig. 2.2 (a) Functions of F_L and F_R versus radius ratio with a current-step.

(b) Radial trajectory and current waveform with a current-step.

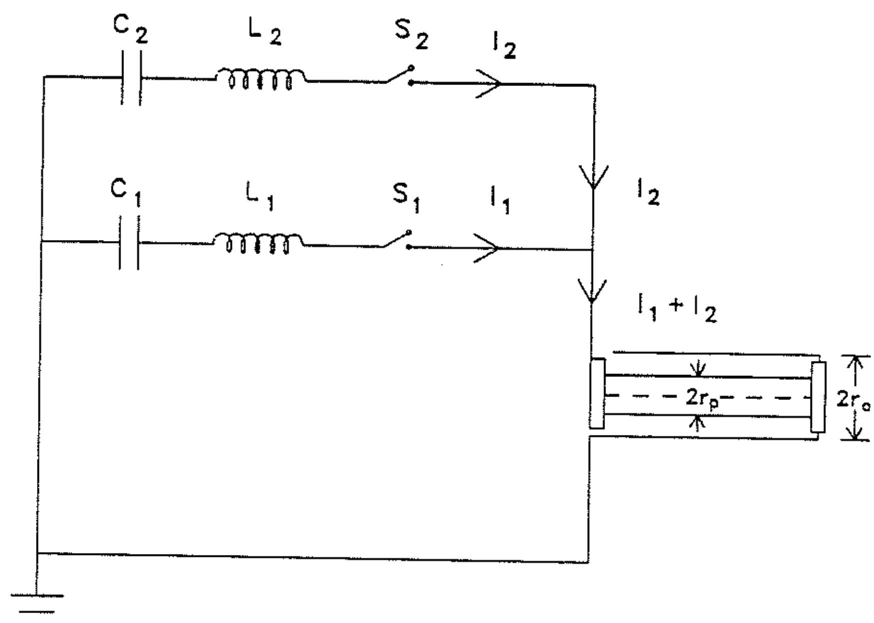


Fig. 2.3 Schematic diagram of the arrangement of a current-stepped Z-pinch.

inductance L_2 into the compressed plasma column in the pinch machine. The instant of switching of S_2 to discharge C_2 is determined by the extent of approach of the energy balance condition. This is expressed by the convergence ratio, ψ defined as

$$\psi = \frac{F_R}{F_L} \qquad [2.9]$$

Numerical studies conducted show that the value of ψ between 0.3 and 0.9 is suitable.

2.3 The energy balance slug model

The slug model originally developed by *Potter [1978]* and used by *Lee [1984b]* in a circuit-coupled form is used here to describe the dynamics of the linear gas compressional pinch. The plasma is treated as a slug of finite thickness consisting of fully ionised shock-heated particles. The slug is assumed to be of infinite conductivity such that the current only flows in a thin outer layer of the slug. The inner boundary of the slug is the shock front while the outer boundary is provided by the magnetic piston. The compression of the slug is also assumed to be axisymmetric and collision dominated. This model is chosen because of its simplicity in approach and its potential to provide a physical representation of the dynamics involved in the compressional process. The model yields description of the shock trajectory, the piston trajectory, the formation of a column of shock heated plasma and the corresponding temporal profile of the discharge current.

In the model, the compression is divided into 2 consecutive phases; namely the strong shock compression phase and the slow dynamics phase. The strong shock compression phase describes the start of the compression from the glass wall until the instant when the shock front hits the axis. This is immediately followed by the slow dynamics phase which describes the subsequent compression until when the point of maximum compression corresponding to minimum radius ratio is reached. (The point of maximum compression is determined either by the energy balance condition, ψ =1, or the intersection of the reflected shock coming out from the axis at 1/3 of the speed of the incoming shock at the axis, *[G.Guderley, 1942]* with the magnetic piston, whichever is satisfied earlier.)

2.4 The strong compression phase

In the strong compression phase, the slug of the plasma bounded by the radial inward propagating shock front and the compressing piston is assumed to be of uniform pressure across the two boundaries. This is based on the assumption that sound waves travel faster than the particle kinetic motion in the slug such that the small pressure disturbances induced by the piston travel to and fro between the shock front and the piston in a time shorter than the characteristic time of the particle kinetic motion.

Therefore by equating the magnetic pressure, P_m of the piston resulting from the driving current to the shock pressure, P_s resulting from the shock compression an expression for the shock trajectory can be obtained.

By putting

$$P_m = \frac{\mu_o I^2}{8\pi^2 r_p^2} \qquad [2.10]$$

$$P_s = \frac{2}{\gamma+1} \rho_o v_s^2 \qquad [2.11]$$

and equating [2.10] to [2.11],

$$v_s = - \frac{dr_s}{dt} = - \left[\frac{\mu_o (\gamma+1)}{\rho_o} \right]^{1/2} \frac{I}{4\pi r_p} \qquad [2.12]$$

The negative sign is introduced because the radial outward direction is taken as positive.

The piston trajectory is obtained by considering the compression as adiabatic in a volume of uniform pressure. In considering the adiabatic compression, the actual change in volume resulting from the compression, after taking into account the mass swept in by the shock front, as suggested by *Potter, [1978]*, must be used. This gives the piston trajectory as

$$\frac{dr_p}{dt} = \frac{\left[\frac{2}{\gamma+1}\right]\left[\frac{r_s}{r_p}\right]\left[\frac{dr_s}{dt}\right] - \left[\frac{r_p}{\gamma I}\right]\left[1 - \left[\frac{r_s}{r_p}\right]^2\right]\left[\frac{dI}{dt}\right]}{\left[\frac{\gamma-1}{\gamma}\right] + \left[\frac{1}{\gamma}\right]\left[\frac{r_s}{r_p}\right]^2} \qquad [2.13]$$

2.5 The slow dynamics phase

This phase starts when the shock front hits the axis.

In this phase the plasma is still assumed to be a slug bounded by the axis and the piston with a uniform pressure profile across the boundaries. A reflected shock is assumed to travel outwards from the axis while the piston compresses further inwards. This assumption is adopted to provide a reasonable representation of this further compression. The minimum radius ratio of the piston corresponding to maximum compression is obtained when the reflected shock hits the piston. It has been shown by Ali *[1984]* that the minimum radius obtained in this manner agrees fairly well with that obtained using the energy balance theory.

Therefore by equating the rate of change of enthalpy to the first law of thermodynamics for a lossless system, the slow piston compression is represented as

$$\frac{dr_p}{dt} = - \frac{r_p}{(\gamma-1)I} \frac{dI}{dt} \qquad [2.14]$$

2.6 The circuit equations

The current flowing in the pinch is provided by two parallel LCR discharge systems. A schematic of the circuit involved is shown in Fig. 2.4.

L_1, C_1 and R_1 are the inductance, capacitance and resistance respectively of the first capacitor bank system. V_1 is the initial voltage on C_1. L_2, C_2 and R_2 are the inductance, capacitance and resistance respectively of the second capacitor bank system and V_2 is the initial voltage on C_2. The circuit resistances of R_1 and R_2 are determined from the degree of damping of the short-circuit current signals of each of the capacitor

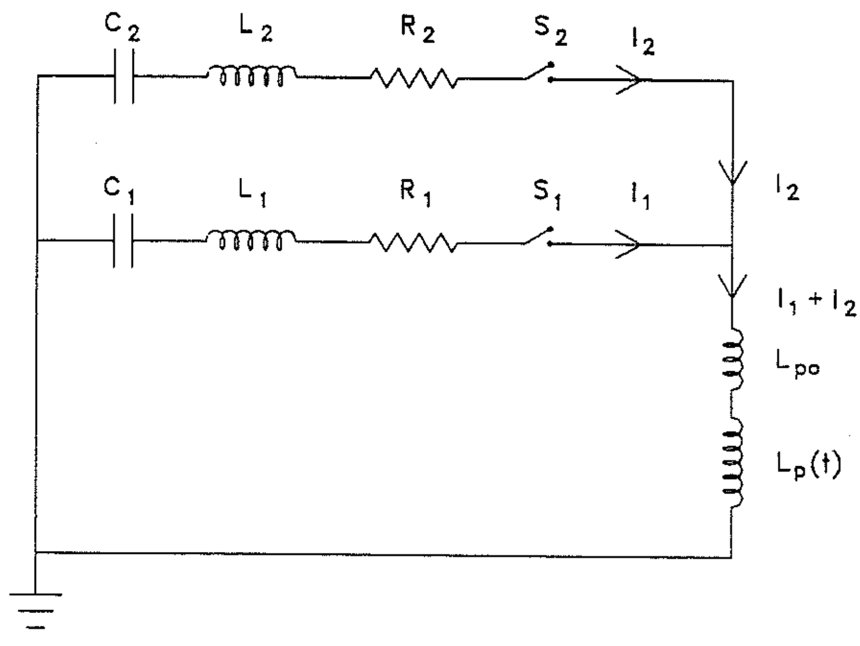


Fig. 2.4 The circuit diagram of the current-stepped Z-pinch.

banks alone. The pinch load is assumed to be a purely inductive load and since its voltage is usually measured across the anode and the cathode coaxial return, thus constituting a fixed inductance, the pinch load is represented as two inductances in series, a constant L_{po} and a time varying $L_p(t)$. L_{po} is the inductance of the pinch chamber input flange and $L_p(t)$ is the inductance of the pinching plasma. When switch S_1 is closed, C_1 discharges through L_1 and R_1, providing the first current source to drive the pinch load. The closing of switch S_2 discharges capacitor C_2 through L_2 and R_2, providing the second current source for the investigation of the enhancement in the compression by current-stepping. If either S_1 or S_2 is switched on alone, a slower or faster current source is obtained to drive the pinch load just like any other conventional gas compressional pinch. If S_2 is switched on after a suitable delay upon the switching of S_1 (as determined by the energy balance theory) to obtain a current-stepped profile for the current source to drive the pinch, an enhanced compression is expected as predicted by *Lee [1984]*. The system then operates as a current-stepped Z-pinch.

The circuit equations describing the above current sources are as follows :

(i) S_1 on and S_2 off

$$\frac{d}{dt}\left[[L_1 I_1] + [L_{po} + L_p(t)]\ I \right] + I_1 R_1 = V_1 - \frac{\int I_1 dt}{C_1} \qquad [2.15]$$

and $I = I_1$

or (ii) S_1 and S_2 on

$$\frac{d}{dt}\left[[L_1 I_1] + [L_{po} + L_p(t)]\ I \right] + I_1 R_1 = V_1 - \frac{\int I_1 dt}{C_1} \qquad [2.16a]$$

$$\frac{d}{dt}\left[[L_2 I_2] + [L_{po} + L_p(t)]\ I \right] + I_2 R_2 = V_2 - \frac{\int I_2 dt}{C_2} \qquad [2.16b]$$

and $\quad I = I_1 + I_2$

where equation [2.15] represents the case when only the first bank is switched on while [2.16a] and [2.16b] represent the condition when switch S_2 is switched on as well after a suitable delay as determined by the energy balance theory to produce the enhanced compression.

CHAPTER 3

The Capacitor-Capacitor Combination

- A First Attempt to Current-Step a Z pinch

3.1 Design considerations

The design *[Saw,S.H. et.al, 1987]* of the Z-pinch is based upon the energy balance theory *[Lee,S. 1983b]* using the slug model for the trajectory computation. From equation [2.12] upon normalisation, a characteristic time of pinch, t_p is obtained as given by

$$t_p = \left[\frac{\rho_o}{\mu_o(\gamma+1)} \right]^{1/2} \frac{4\pi r_o^2}{I_o} \qquad [3.1]$$

where ρ_o is the ambient gas density, r_o the radius of the pinch tube, $I_o = V_o/\sqrt{(L_o/C_o)}$, γ the specific heat ratio and μ_o the permeability of free space. V_o, L_o and C_o are respectively the initial discharge voltage, the total external inductance of the system and the capacitance of the bank.

The first design consideration is to match the expected pinch time to the quarter-period current rise time, t_r. For a *LC* discharge through a plasma load, the current rises sinusoidally to about 0.7 I_o. This corresponds to an average current over half a cycle of $0.45\ I_o$. From equation [3.1] the expected pinch time, t_p' becomes

$$t_p' = 2.2\ t_p \qquad [3.2]$$

Also, the current rise-time is given by

$$t_r = 0.5\pi \sqrt{(L_o C_o)} \qquad [3.3]$$

Thus for proper matching

$$t_r = t_p' = 2.2\ t_p \qquad [3.4]$$

The second design consideration in designing a pinch machine is to provide good energy coupling between the capacitor bank and the pinch machine. This is governed by

$$\beta = \frac{\mu_o \ell}{2\pi L_o} \qquad [3.5]$$

where $\mu_o \ell/2\pi$ is the characteristic inductance of the pinch tube of length ℓ and L_o the total external inductance of the system. For good energy coupling, a value of $\beta > 0.1$ is necessary. *[Lee, S., 1983c]*.

A major constraint in the design of the current-stepped Z-pinch at the present stage is to utilise technology and facilities currently available in the laboratory to avoid high cost. A capacitor-capacitor combination unit, UMCSZP is designed with the following characteristics for the first stage; hereafter refered to as the first bank,

$$V_1 = 9\ kV \qquad C_1 = 60\ \mu F \qquad L_1 = 70\ nH$$

$$t_{r1} = 3.4\ \mu s \qquad I_{01} = 251\ kA \qquad dI_{01}/dt = 1.2 \times 10^{11}\ A/s$$

and

$$V_2 = 18\ kV \qquad C_2 = 22\ \mu F \qquad L_2 = 26\ nH$$

$$t_{r2} = 1.3\ \mu s \qquad I_{02} = 463\ kA \qquad dI_{02}/dt = 5.5 \times 10^{11}\ A/s$$

for the current-stepped stage or the second bank. These parameters are chosen for a Z-pinch of length *15 cm* and radius *7.5 cm*. The inductance of the input flange is *6.8 nH* and the operating pressure is between 0.4 and 3.0 mbar of hydrogen. The construction of the Z-pinch machine is shown in Fig.3.1. The chamber is a glass cylinder of length 21 cm and diameter 15 cm with a wall thickness of 6 mm. The actual discharge length is 15 cm. The anode and cathode are made from brass with a small tapering at the edge around the inner circumference of the groove where the glass chamber sits (refer Fig. 3.1). This facilitates the formation of the current sheath at the wall of the glass chamber during the initial period of the current discharge. The current return is via a solid coaxial brass cylinder. A small slit of width 12.5 mm and length 21 cm is cut in the radial plane midway along the length of the brass cylinder. This allows side-on streak photography of the radial compression. The cathode return is made as close as possible to the glass chamber to minimise tube inductance. The anode and cathode return of the Z-pinch machine are fitted on to the anode and cathode flange of the second bank, C_2 respectively (refer Fig. 3.1). This second

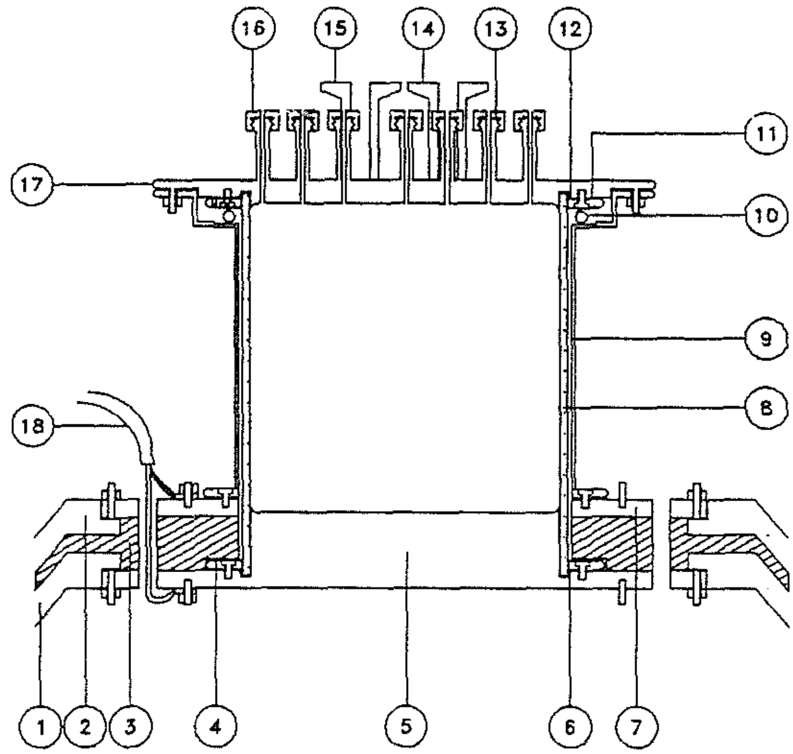


Fig. 3.1 Design of the current-stepped Z-pinch.

1. anode flange of C_2.
2. cathode flange of C_2.
3. silicon rubber and insulation sheets
4. brass flange
5. anode collector plate
6. O-ring seal
7. cathode return plate
8. glass chamber
9. cathode return cage
10. Rogowski coil
11. brass flange
12. O-ring seal
13. O-ring seal
14. gas inlet
15. to vacuum pump
16 magnetic probe inlets
17. cathode collector plate
18. coaxial cable from C_1

bank is formerly the capacitor bank use for DPF-I in KFA, Jûlich *[P.cloth & H.Conrads, 1977]* which has been installed in our laboratory as a gift from the Humbolt Foundation and KFA, Jûlich. Connection of the Z-pinch machine to the first capacitor bank system, C_1 is via short parallel coaxial cables to reduce the inductance of the system. The first capacitor bank system consists of 3 units of 20 μF, 10 kV maximum capacitors each with an equivalent series inductance of 40 nH, connected in parallel using parallel plates. A swinging cascade spark gap *[Lee, S. et al, 1988b]* is connected in series to the high voltage plate. The other end of the swinging cascade spark gap is connected to the anode flange of the Z-pinch machine via parallel coaxial cables. The schematic of the first capacitor bank system and the swinging cascade spark gap is as shown in Fig. 3.2. Fig. 3.3 shows the parallel plate and the series connection of the capacitor to the swinging cascade spark gap. The 9 MΩ resistor connected across the spark gap provides some preionisation to the ambient gas inside the pinch chamber and thus facilitates the initiation of the current sheath.

The second capacitor bank system, C_2 consists of 12 units of 1.86 μF, 60 kV maximum capacity capacitors each with an equivalent series inductance of 15 nH, connected in parallel using parallel plates in a circular arrangement (refer Fig. 3.4). Each of these capacitors is switched by two pressurised spark gaps. These spark gaps are simultaneously triggered by a master trigger pulse (refer Fig. 3.5). The master trigger pulse is provided by a -30 kV capacitor discharge system which triggers upon receiving of a 16 kV pulse by its trigger-pin. The equivalent circuit of the

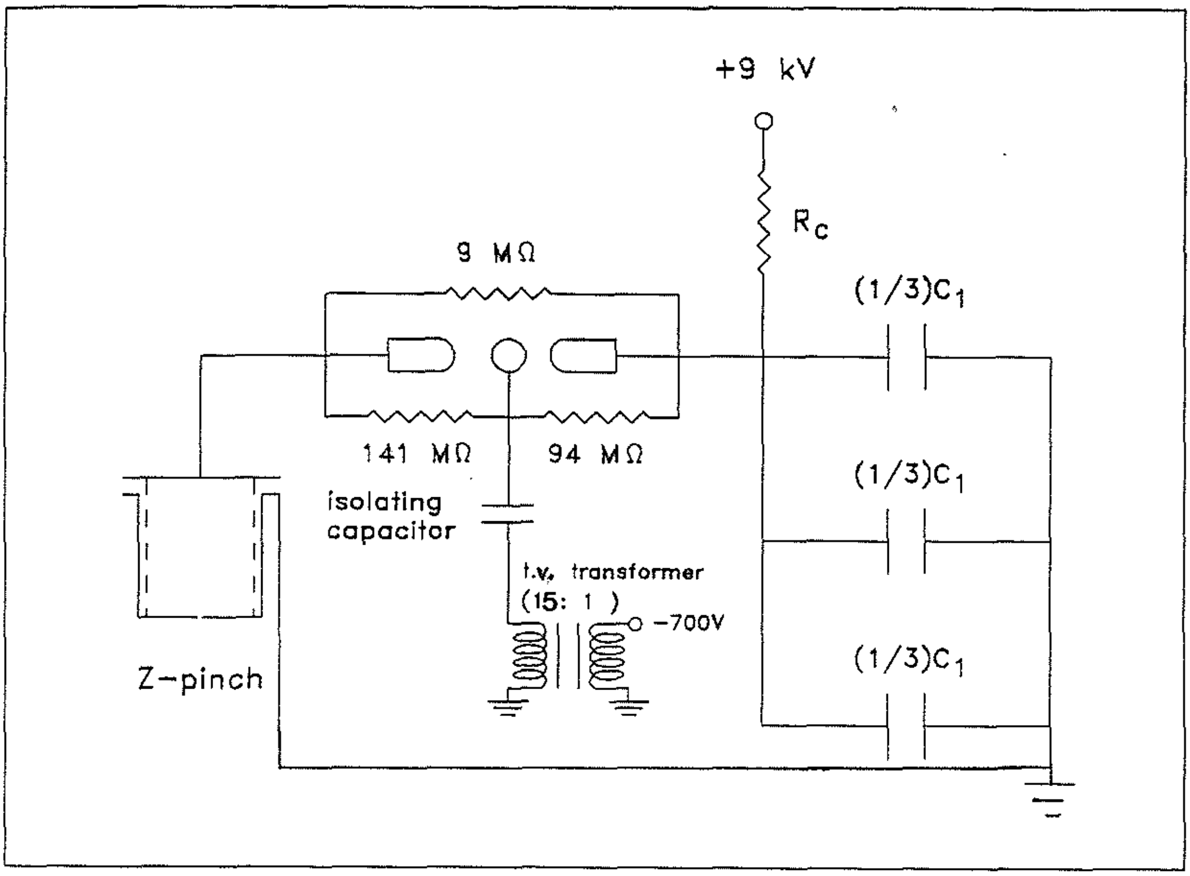


Fig. 3.2 Discharge circuitry of the first bank of the current-stepped Z-pinch. (C_1 = 60 μF).

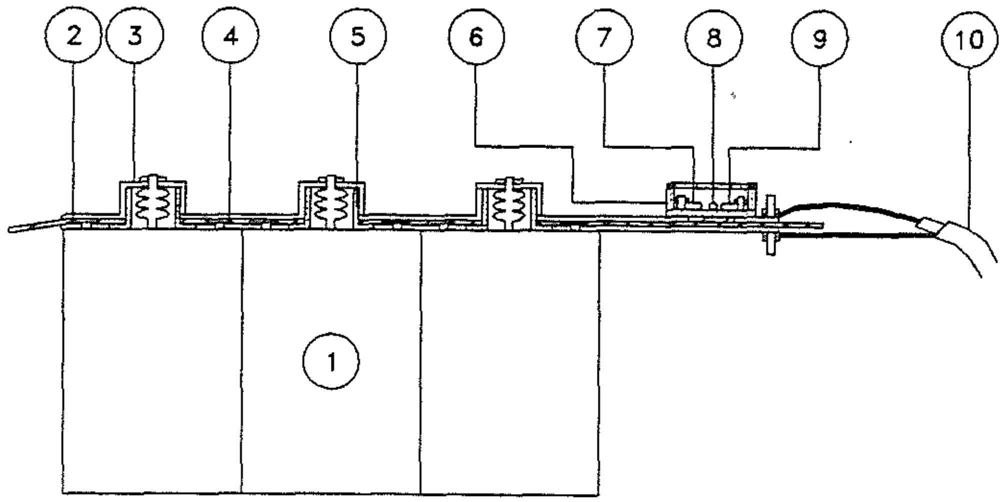


Fig. 3.3 The capacitor, C_1 and the spark-gap connections.

1. 60 μF, 10 kV
2. earth return plate
3. high voltage collector plate
4. mylar-polyethylene-mylar sheets
5. insulator cap
6. perspex box
7. spark-gap electrode
8. triggering rod
9. spark gap electrode
10. high voltage transmission cable

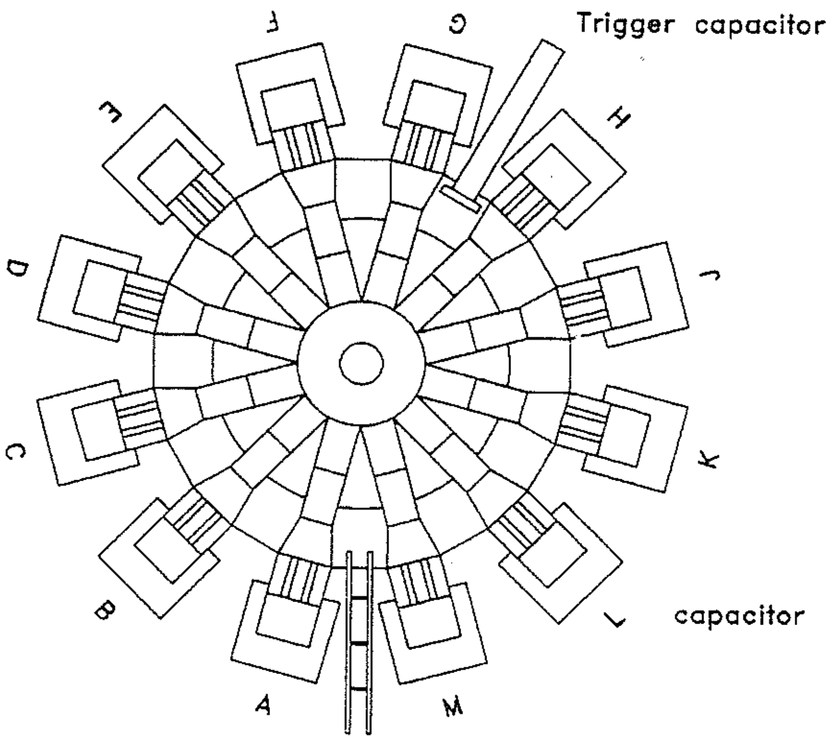


Fig. 3.4 Top view of the capacitors arrangement of C_2. (reproduced from original diagram of *KFA, FIPP, 45 00 00-1).00 00-1)*.

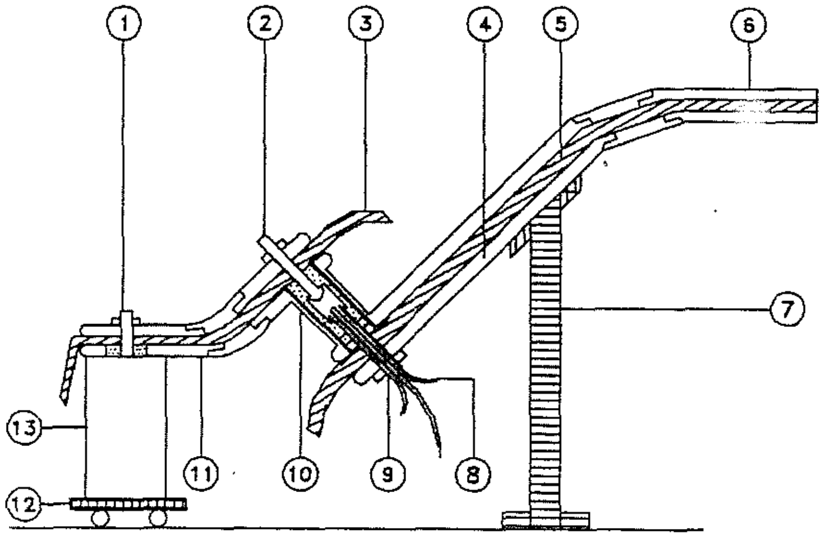


Fig. 3.5 Cross sectional view of one of the arms in Fig. 3.4 showing the connections between the capacitor, the pressurised spark gap and the anode flange. (reproduced from original diagram of *KFA, IPP, 45 02 00-1*)

1. high voltage connectors
2. high voltage connectors
3. mylar-polyethylene-mylar sheets
4. anode connector plate
5. mylar-polyethylene-mylar sheets
6. cathode connector plate
7. insulator support
8. gas inlets/outlets
9. spark gap electrodes
10. cathode return of the spark gap
11. earth plate
12 trolley
13 capacitor (1.86 μF, 60 kV)

second capacitor bank system is shown in Fig. 3.6. Fig. 3.7 shows the schematic of the overall set-up of the UMCSZP.

3.2 Diagnostics and instrumentation

3.2.1 Current measurement

The discharge current is measured by means of a Rogowski coil *[Leonard, S.L., 1965]* placed in the upper part of the coaxial cathode return between the glass chamber and the cathode return. The Rogowski coil has approximately 500 turns and it is operated as a current transformer by shorting the terminals with a small resistance, r as shown in Fig. 3.8. The calibration of the current transformer is obtained by operating the pinch machine in the non-pinching mode where the current waveform resembles that of a typical underdamped discharge. This is achieved by operating the pinch machine at a pressure of 5 mbar air to produce a sinusoidal waveform as shown in Fig. 3.9.

The first peak value of the current waveform I_1 is given by

$$I_1 = \frac{\pi C_o V_o (1+f)}{T} \qquad [3.6]$$

where C_o = capacitance of the capacitor, 60 μF

V_o = fixed charged voltage of the capacitor, 9kV

T = period of the current pulse

f = reversal ratio

$= (V_2/V_1) = 1/2\ ((V_4/V_3) + (V_2/V_1))$

The calibration factor, K_I of the coil is determined by

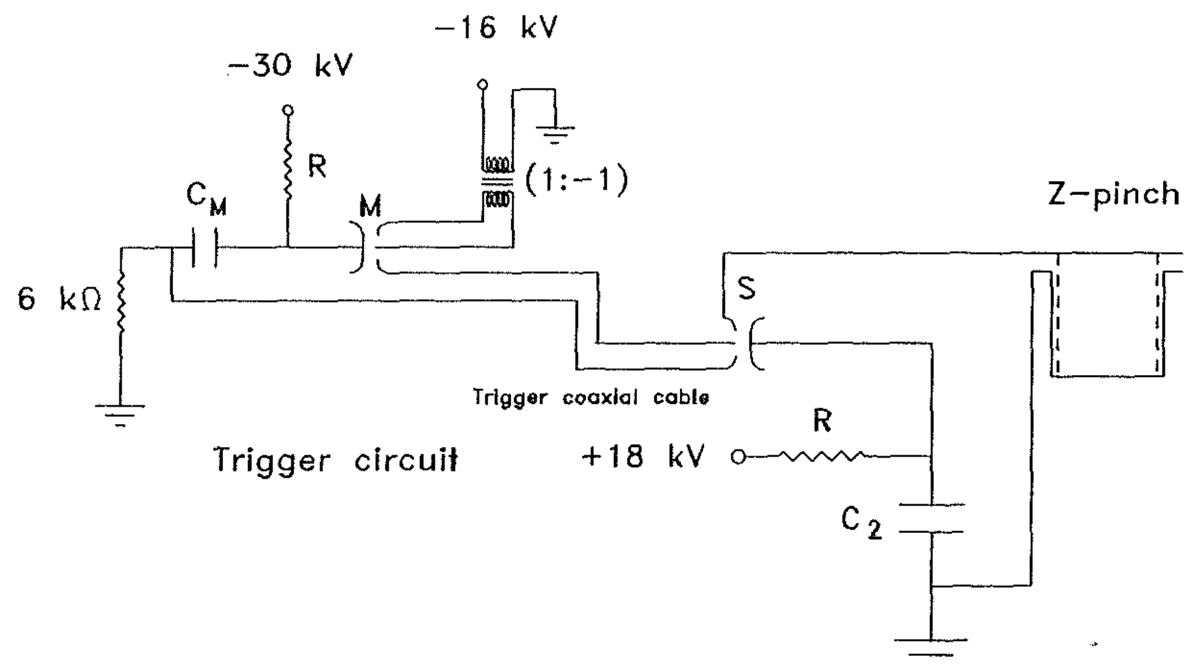


Fig. 3.6 Discharge circuitry of the second bank of the current-stepped Z-pinch. (C_2 = 22 μF, C_m = 0.6 μF and S_1 is one of the 24 pressurised spark gaps in similar parallel configuration).

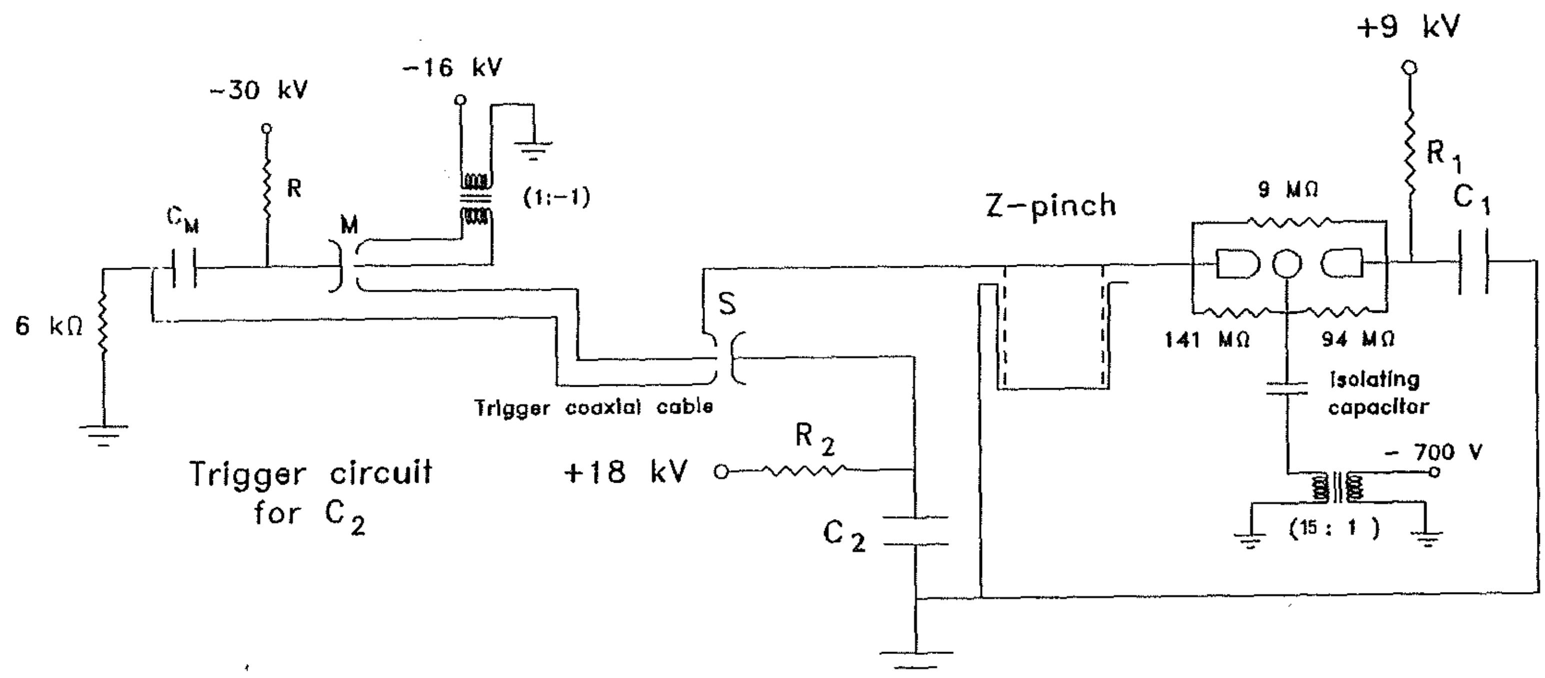


Fig. 3.7 Discharge circuitry of the UMCSZP current-stepped Z-pinch.

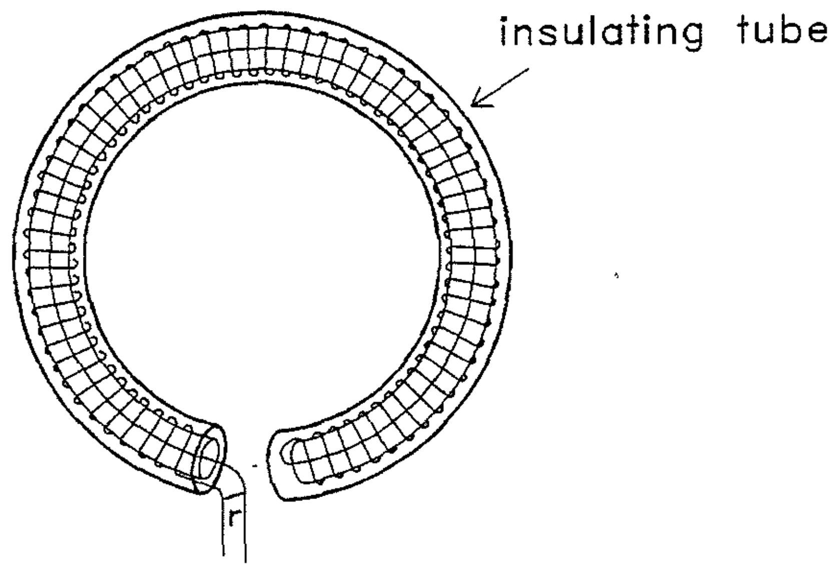


Fig. 3.8 The Rogowski coil as a current transformer.

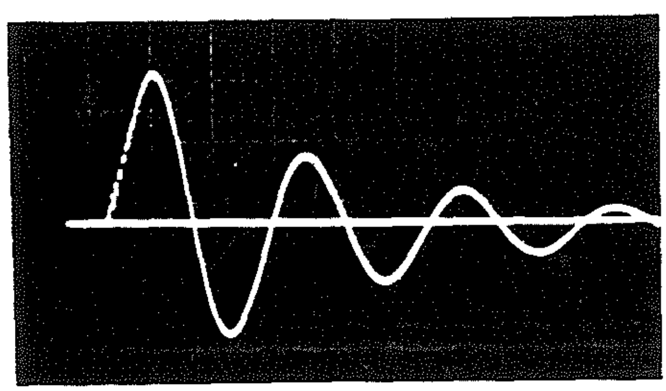


Fig. 3.9 Sinusoidal current signal for the calibration of Rogowski coil.

(Scale : 5 V/cm, 2 μs/cm at 5 mbar H_2)

comparing the value of I_1 calculated from equation [3.6] with the amplitude of the first peak of the current waveform, V_1.

3.2.2 Voltage measurement

The voltage induced across the pinch tube during the discharge is measured by means of a capacitive voltage probe *[R.Keller, 1964]*. It consists of two cascading sections as shown in the schematic diagram of Fig. 3.10. The first section is a differentiator with a time constant, R_1C_1. It also serves as a low-pass filter whereby $f_{cl} = 1/(2\pi R_1C_1)$ determines the upper response limit of the probe. An RC integrator with a time constant of R_2C_2 forms the second section of the probe. This section which is also a high-pass filter only reproduces signals of frequency equal to or greater than $f_{ch} = 1/(2\pi R_2C_2)$. The output signals of section one and section two are as follows:

$$V_1 = R_1C_1 \frac{dV}{dt} \qquad [3.7]$$

$$V_2 = \frac{\int V_1 dt}{R_2C_2} \qquad [3.8]$$

Combining [3.7] and [3.8] gives

$$V_2 = \frac{R_1C_1}{R_2C_2} V \qquad [3.9]$$

The bandwidth is given by

$$f_{cl} > f > f_{ch} \qquad [3.10]$$

3.2.3 Construction of the voltage probe

The probe constructed is mounted directly to the cathode flange of the Z-pinch unit *[Saw, S.H., Wong, C.S. & Lee,S., 1991]*. Capacitor, C_1 forming the differentiator is a coaxial capacitor made from the inner core of a 50 Ω coaxial cable and aligned facing the anode flange (or the high voltage point of interest) with an additional perspex dielectric as shown in Fig. 3.11. It is secured to the cathode plate by a coaxial copper adaptor. Since the signal is carried by a 50 Ω coaxial line to the measuring station, R_1 is fixed at 50 Ω for impedance matching. The integrator consisting of components R_2 and C_2 is housed in a module together with the $R_1 = 50\ \Omega$ for line termination. This integrating module is then connected to the input of the oscilloscope.

With this design, it is not necessary for the oscilloscope to be placed near the discharge system. The choice of values for the capacitors and resistors are determined by the bandwidth response required. In our present design, C_1 is approximately 12 *pF*, R_1 is 50 Ω , C_2 is 4700 *pF* and R_2 is 20 kΩ. The calculated frequency response is from 1.7 kHz to 0.3 GHz. The equivalent circuit of the probe system is as shown in Fig. 3.12. Calibration of the probe is done *in situ* by operating the Z-pinch machine at a known voltage, e.g. 9 kV, at a sufficiently low pressure (for H_2 gas, $P < 0.8$ mbar). The voltage signal will show a pre-discharge holding phase as shown in Fig. 3.13. The flat top of the square pulse corresponds to the initial charged voltage, V_o of the capacitor switched over and held across the pinch tube. This is used to calibrate the voltage probe.

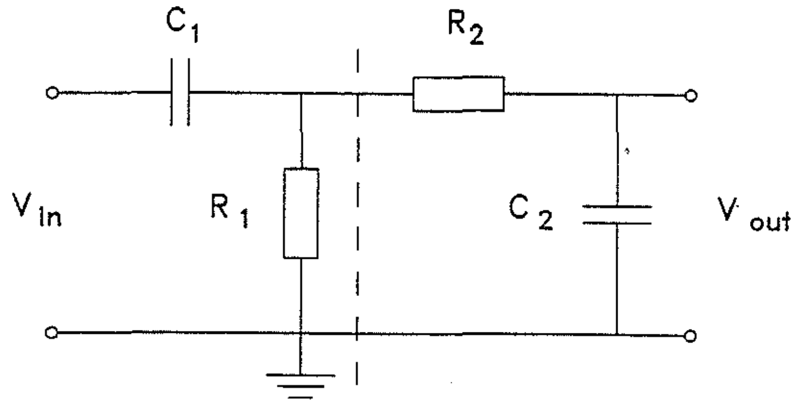


Fig. 3.10 The principle of the voltage probe.

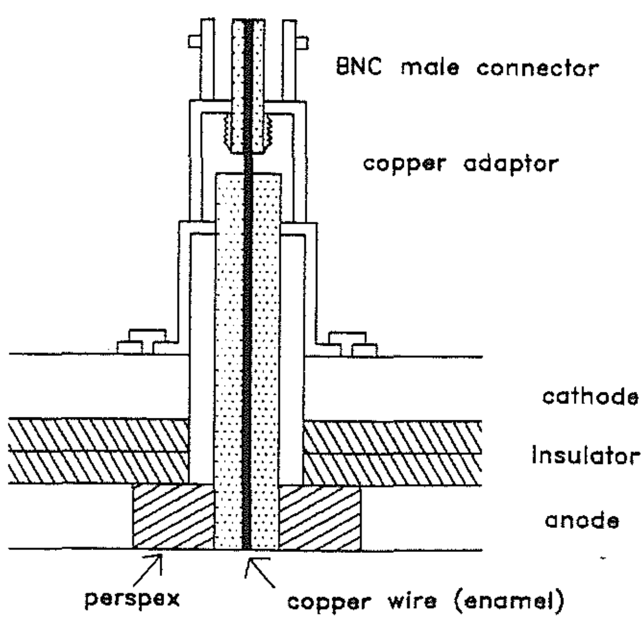


Fig. 3.11 The design of C_1.

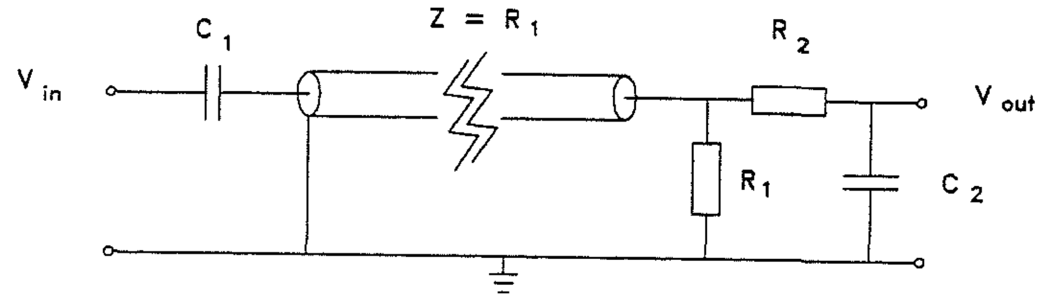


Fig. 3.12 Equivalent circuit of the voltage probe system.
(C_1 =12 pF, R_1 = 50 Ω, R_2 = 20 kΩ, C_2 = 4700 μF)

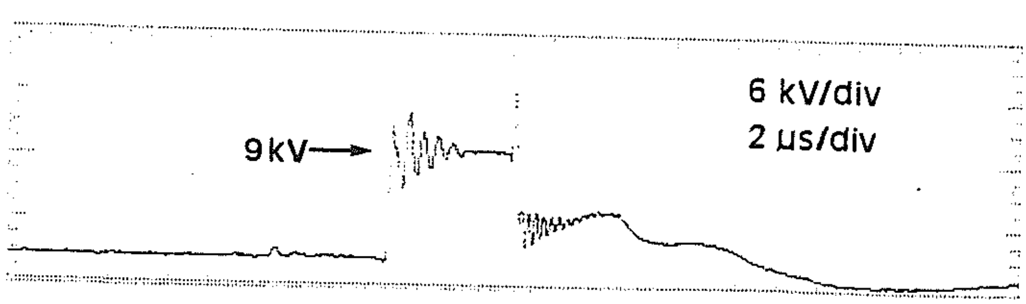


Fig. 3.13 Calibration of the voltage probe.

3.2.4 Streak photography

High speed photography of the pinching event of the plasma is obtained using the *IMACON*, a streak unit with a scanning speed from 10 ns/mm to 100 ns/mm with a streak length of 76 mm.

3.2.5 B-field measurement

The time-resolved distribution of the radial current density is obtained by mapping the $\mathcal{B}$-field in the θ direction at various radial positions. The $\mathcal{B}_\theta$-field is obtained by means of magnetic probe *[Lovberg, R.H., 1965]* which is sensitive to the rate of change of flux through the probe and hence to the turns-area of the probe and to the rate of change of the $\mathcal{B}_\theta$-field. The major factor limiting the frequency response of the coil of inductance $\mathcal{L}$ terminated with a resistance $\mathcal{R}$ is the $\mathcal{L}$-$\mathcal{R}$ time constant, $\tau = \mathcal{L}/\mathcal{R}$. Since it is necessary to carry the signal more than 10 m from the pinch system, $\mathcal{R}$ is normally taken as 50 Ω which is the characteristic impedance of the coaxial cable used to carry the signal. Therefore to keep the response fast enough $\mathcal{L}$ must be made small in the order of tens of nanohenries. For an $\mathcal{N}$-turns coil of length ℓ and cross-sectional area $\mathcal{A}$, the inductance is equal to $\mathcal{L} = \mu_0 \mathcal{N}^2 \mathcal{A}/\ell$ where μ_o is the permittivity of free space.

Furthermore, the signal which is proportional to the rate of change of the $\mathcal{B}_\theta$-field has to be integrated to obtain the $\mathcal{B}_\theta$-field. As such the magnitude of the signal will be further attenuated by the *RC* time constant of the passive intergrator, which is at least ten times longer than the fastest signal of interest. Therefore in order to keep the final integrated signal of measureable magnitude, the initial voltage carried by the line must be made as large as possible. Unfortunately this is only possible at the expense of the response of the inductive coil, since a larger magnitude requires a bigger turns-area which increases the inductance of the coil and thus decreases the response of the probe to the changing magnetic flux. A suitable arrangement can be obtained by keeping the ratio N/ℓ as small as possible in the order of 10^{-2}.

The magnetic probe constructed is as shown in Fig. 3.14. The coil is made from SWG 44 enamel coated copper wire. The length of the coil is approximately 3 mm with 20 turns wound on a PVC sleeving core of diameter 1.5 mm. It is then inserted inside a glass tubing sealed at one end. The outer diameter of the glass tubing is 4 mm and the wall thickness 1 mm. The two leads from the coil are brought out from the glass tubing as a tightly twisted pair and is connected via an adapter to a male BNC connector. A 50 Ω coaxial cable is used to carry the signal to the screen room. The signal is introduced to the oscilloscope input channel via an integrator. The voltage recorded is given by

$$V_o = \frac{N\mathcal{A}\mathcal{B}}{R_1 C_1} \qquad [3.11]$$

The equivalent circuit of the magnetic probe system is shown in Fig.3.15.

The probe is calibrated together with the integrator using a standard LC discharge through a 2-turns Helmholtz coil. For an accurate calibration including frequency response, the

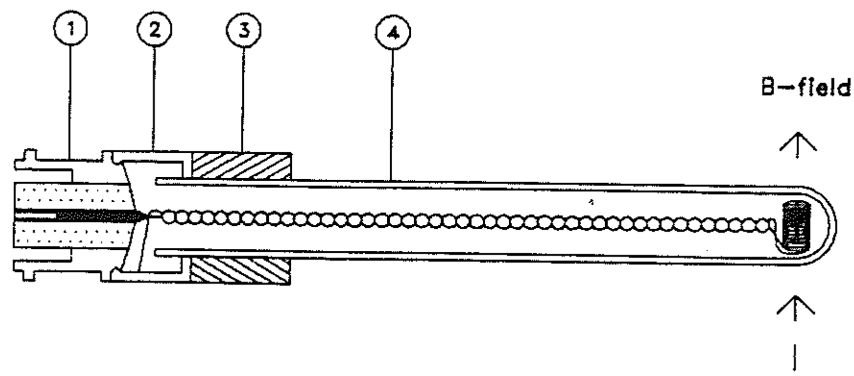


Fig. 3.14 Design of the magnetic probe.

1. BNC socket 2. copper adapter
3. perspex holder 4. glass tube

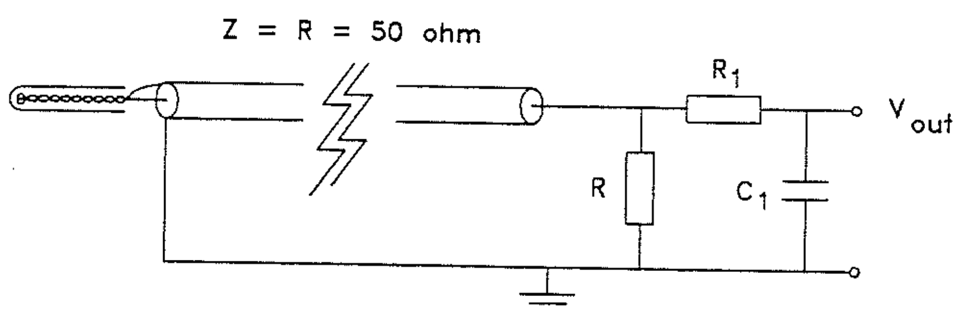


Fig. 3.15 Equivalent circuit of the magnetic probe system.

(R_1 = 20 kΩ, C_1 = 4700 μF)

discharge time of the LC system is made the same order as that of the Z-pinch system.

The radius of the Z-pinch chamber is 7.5 cm. Seven probes are introduced into the pinch chamber axially at various radial positions across the diameter of the chamber. Four probes are placed at *r=6 cm, r=4 cm, r=2 cm* and *r=0 cm.* Another three are placed on the opposite side at *r=7 cm, r=5 cm* and *r= 3 cm.* Since the number of oscilloscope channels are limited, at each discharge only signals from four probes (*r=6 cm, 4 cm, 2 cm, 0 cm or r=7 cm, 5 cm, 3cm)* are recorded simultaneously with the current and voltage signals. The time resolved radial current density is then obtained from 2 sets of probe positions obtained from two different discharges. This is possible since the discharge of the system is quite reproducible. Further only two sets of probe positions with identical current and voltage signals are used to determine the radial current density of the pinch.

3.3 Experimental set-up

A schematic diagram of the experimental set-up of *UMCSZP* to record the current, voltage and streak pulses for streak photography or magnetic probe signals for $\mathcal{B}$-field measurements simultaneously is shown in Fig. 3.16. The block *TRIG1* sends out a master pulse which controls the discharge of the whole system. It consists of a press-button operated low-voltage SCR unit which triggers a HVSCR unit via a press-button switch. Upon triggering, the HVSCR sends out a negative pulse of approximately 700V to a t.v. transformer to be stepped up to approximately 10 kV, 1 μs rise time. This pulse is used to trigger the swinging cascade

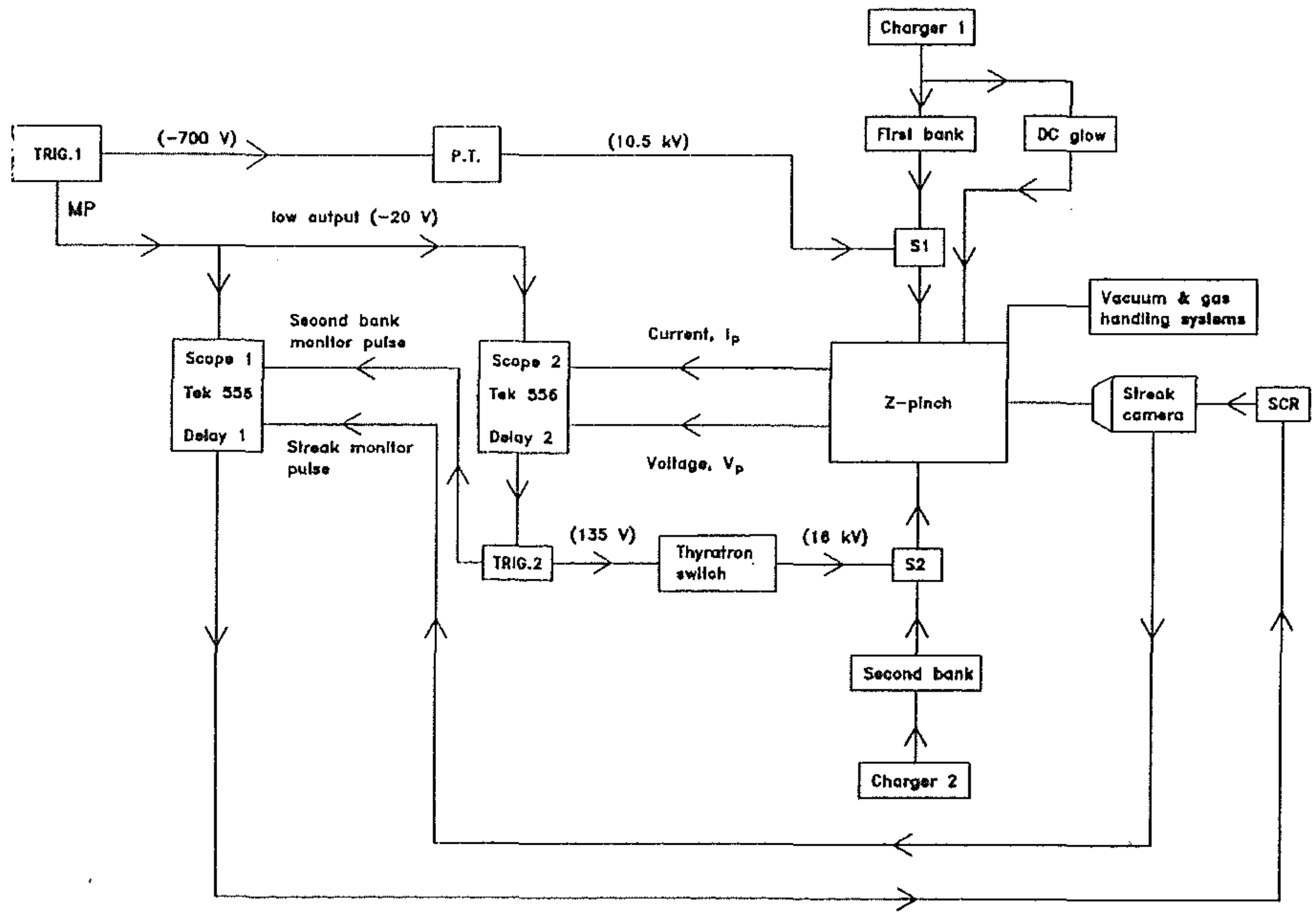


Fig. 3.16 Block diagram of the UMCSZP set-up.

spark gap and thus discharging the first capacitor bank system onto the pinch chamber.

At the same time, the HVSCR unit sends out an attenuated pulse of about -20 V (monitor pulse) to trigger the oscilloscopes *(Tektronix 556)* to record the current, voltage and the streak signals. With the triggering of the oscilloscopes, delayed pulses are send out from delay units in the oscilloscopes to switch the streak camera and the second capacitor bank system. As the magnitudes of these delayed pulses from the oscilloscopes are only about 7 V, they are insufficient to trigger the streak camera and the second capacitor bank system directly.

In the case of the streak camera, the delayed pulse is sent to another low voltage SCR unit of output 40 V which then triggers the streak camera. Upon triggering, the streak camera sends out a streak pulse to be recorded by the oscilloscope. The streak pulse is used to monitor the setting of the delay time required to switch the streak camera.

For the second capacitor bank system, the delayed pulse from the oscilloscope goes to a low voltage SCR of output 135 V. This then triggers the thyratron to switch a 16 kV pulse used to trigger the master gap. Upon triggering, the master gap switches the -30 kV and distributes through 24 lines to trigger 24 parallel spark gaps thus discharging the second capacitor bank system onto the pinch chamber. The output from the low voltage SCR is recorded by the oscilloscope. It is used to monitor the setting of the delay time required to switch the second bank with reference to the discharge of the first bank.

3.4 Experimental procedures

The pinch chamber is initially pumped to a base pressure of 10^{-4} mbar by a diffusion pump backed by a rotary pump. It is then flushed with hydrogen a number of times before filling to the required pressure. This reduces the impurities of the system. Compressions from the single discharge by the first bank alone are studied. Current, voltage and streak photography or B-field measurements are obtained simultaneously. This is followed by the implementation of the current-stepping technique to observe the compression by the consecutive discharges of two capacitor banks at different switching times as determined by the energy balance theory.

CHAPTER 4

RESULTS AND ANALYSIS

4.1 Current, voltage and streak measurements

4.1.1 Current and voltage measurements

The time resolved trajectory of the compressional hydrogen Z-pinch can be deduced from the current and voltage measurements of the pinch formed.

At any point of time t, for a fully ionised plasma where current flows in a thin outer layer, the voltage between the anode and the cathode is given by, assuming an inductive model,

$$V_p = \frac{d}{dt}\left[[L_{po} + L_p(t)]\ I_p(t) \right] \qquad [4.1]$$

where L_{po} is the inductance of the input flange which is deduced from the dimensions of the pinch chamber and the coaxial cathode return, $L_p(t)$ is the inductance of the pinched plasma and $I_p(t)$ is the current of the pinched plasma.

Therefore the plasma inductance becomes

$$L_p(t) = \frac{\int_0^t V_p(t)\ dt}{I_p(t)} - L_{po} \qquad [4.2]$$

The radial piston position, r_p of the plasma is then obtained from

$$L_p(t) = 2\ell(ln\ (r_o/r_p(t)) \quad [nH] \qquad [4.3]$$

which gives the radial piston trajectory of the pinched plasma column as

$$r_p(t) = r_o \exp\left(-L_p(t)/2\ell\right) \qquad [4.4]$$

where r_o is the initial radius of the plasma corresponding to the inner radius of the glass chamber assuming that the plasma lifts off from the glass wall and ℓ the length in cm of the pinch corresponding to the distance between the anode and the cathode flanges of the pinch chamber.

4.1.2 High speed photography

The final phase of the radial compression can also be observed in streak photography using the high speed image converter camera, the *Imacon*. The final radius achieved by the compression is measured from the streak photograph. The average shock speed is also measured from the streak photograph and the plasma temperature deduced from the shock jump equations.

4.1.3 Discharge with first bank alone

The current and voltage signals and the corresponding streak photograph of a typical hydrogen discharge at a filling pressure of 0.8 mbar are shown in Fig. 4.1. The top trace is the voltage signal while the bottom trace is the current signal of the Z-pinch plasma. Using equation [4.2] the pinch inductance is obtained from the voltage and current measurements taken. The radial trajectory of the compressing plasma is then obtained from

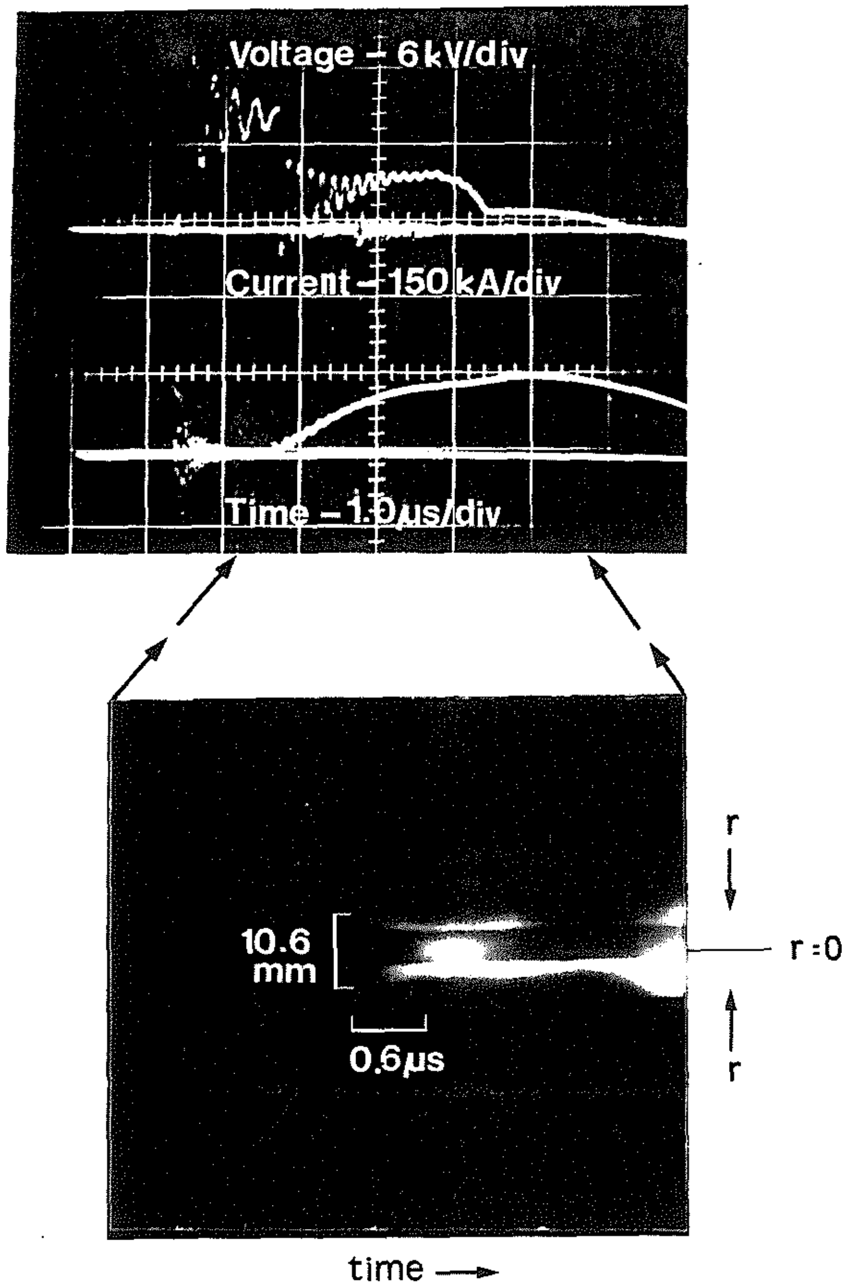


Fig. 4.1 The electrical signals and the corresponding streak photograph showing the radial compression of a hydrogen pinch at 0.8 mbar driven by the first bank alone (without current-stepping).

following the above procedures. Superimposed is the expected radial trajectory numerically computed using the energy balance slug model.

It is observed that the piston seems to compress in faster experimentally compared to that from theoretical prediction, especially so during the initial phase. This is also observed by *P.Choi [1983]* where it is suggested that the assumption made by *Bodin et. al [1960]*, which is also adopted here, that the plasma column is purely inductive with infinite conductivity and with the current flowing in an infinitesimally thin layer is only satisfied at high temperature corresponding to high compressing shock speed. At low plasma temperature the plasma voltage should be represented as

$$V_p = L \frac{dI}{dt} + I \left[\frac{dL_p}{dt} + R \right] \qquad [4.5]$$

The resistance of the plasma column, R can be approximated by assuming the plasma current to flow in an annulus column of thickness δ at radius r, with the skin depth of the plasma current estimated by assuming a fully ionised plasma and the plasma resistivity as given by the transverse Spitzer resistivity *[Spitzer, Lyman Jr., 1967]*. Fig 4.3 shows the estimated plasma resistance and the rate of change of plasma inductance for the present pinch as a function of plasma temperature. It is observed that initially when the plasma is cold, the plasma resistive component dominates. Therefore ignoring it will falsely indicate a faster compression than that obtained from electrical measurements. Furthermore, since

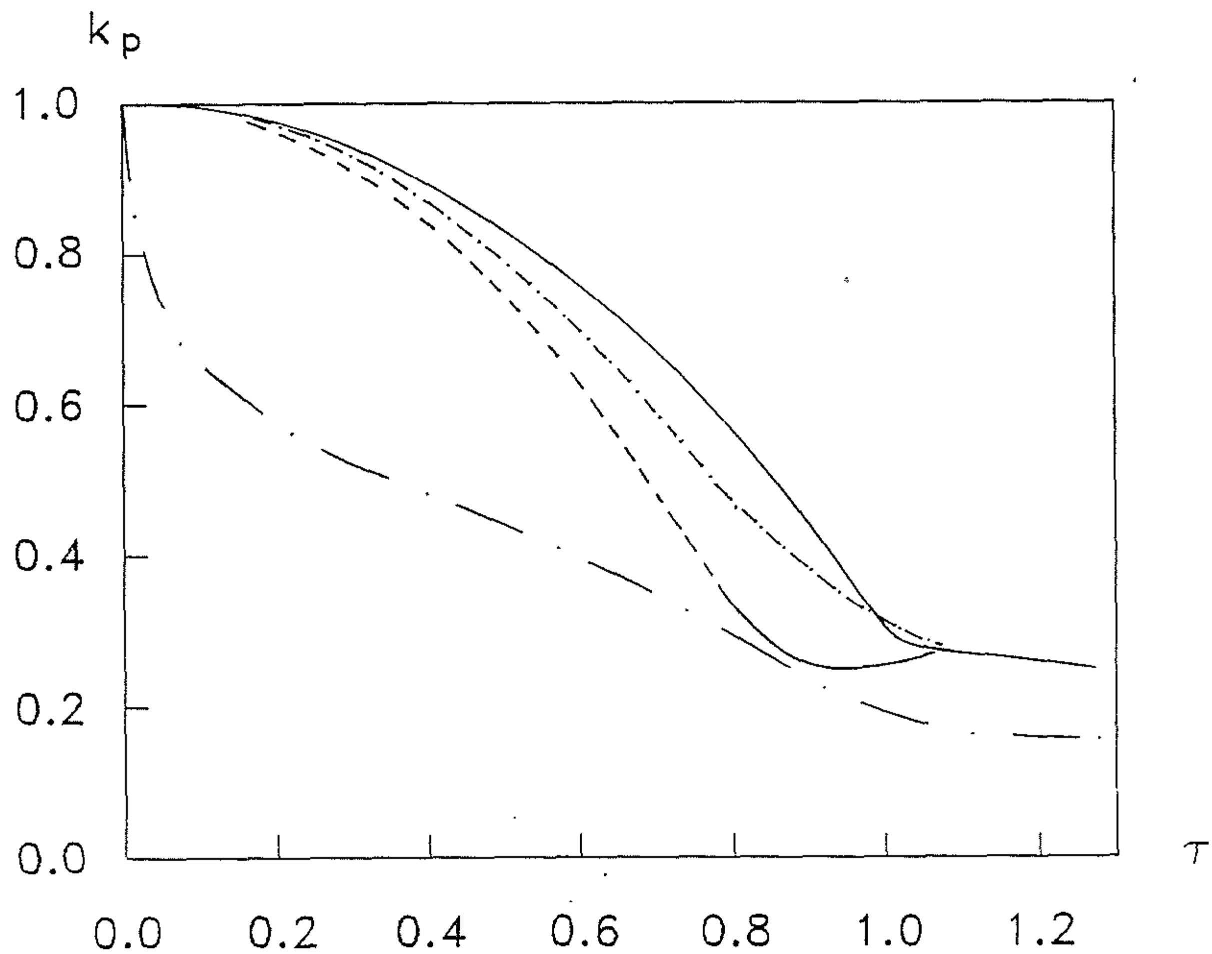


Fig. 4.2 Normalised radial trajectories of the magnetic piston, k_p vs normalised time, τ of a 0.8 mbar hydrogen pinch driven by the first bank alone.

— · — from IV measurements
—·—·—·— from IV measurements (corrected)
– – – – extrapolated from streak photography
———— from theory

the plasma current flows in an annular column of finite skin depth, a relative error will result from the contribution of the magnetic field inside the annular current. By assuming a uniform current density J_o in the annulus of thickness δ, the relation between the relative error in the radius and the ratio of δ/r as a function of T_e is shown in Fig.4.4. If these effects of skin depth of the plasma current are taken into consideration, a slower initial compression which is closer to that obtained from theoretical simulation is obtained. This is superimposed in Fig. 4.2.

An average shock speed is calculated from the measured t_p as the time taken for the compressing shock to hit the axis. The peak shock speed associated with this average shock speed can be obtained from numerical computation. The plasma temperature can then be estimated from the peak shock speed. For a fully ionised shock heated plasma with $\gamma = 5/3$ and the departure coefficient $\chi = 2$, the plasma temperature is related to the shock velocity, v_s by $T_e = 1130\ v_s^2$ where v_s is in cm/μs and T_e in K *[S. Lee, 1985b]*.

From the analysis of the electrical signals of the plasma from the discharge of the first bank alone, it is observed that the pinch is formed at $t_p = 2.1\ \mu$s producing a column of radius ratio 0.30. This indicates an average shock speed of $\simeq$ 3.4 cm/μs. From the numerical computation, the peak shock speed is about 10 cm/μs. The plasma temperature at this speed is 1.1 x 10^5 K.

From the analyses of the side-on streak photographs taken of the compression, the pinch is observed to form at t_p =

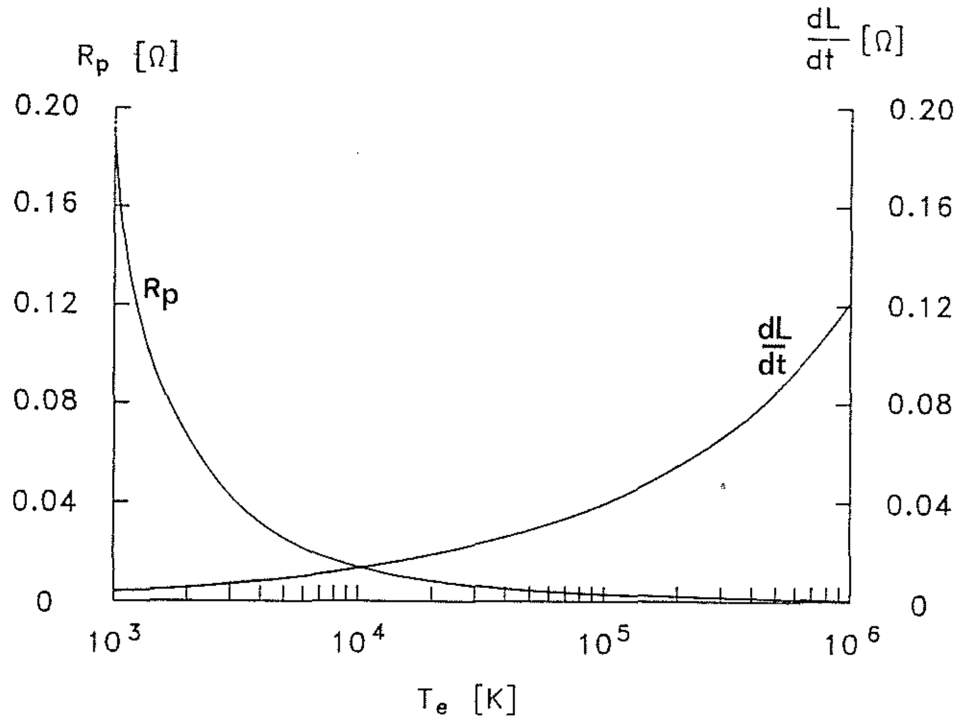


Fig. 4.3 Variation of plasma resistance, R_p due to skin depth and $\frac{dL}{dt}$ versus plasma temperature, T_e.

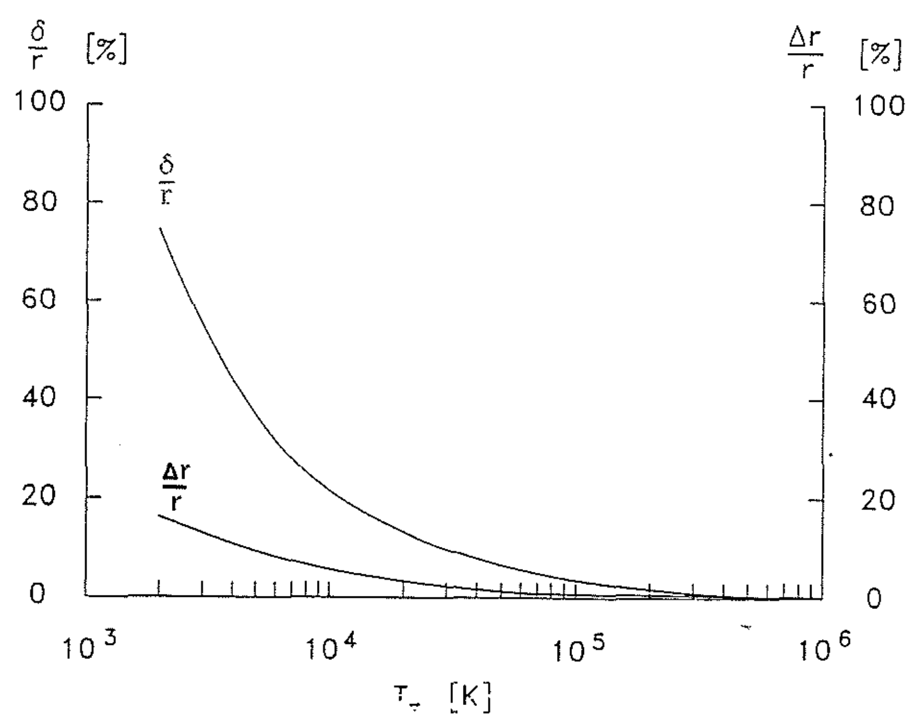


Fig. 4.4 Relations between the ratio of skin depth to radius of the plasma column, δ/r and percentage error in the plasma radius, ($\Delta r/r \times 100\%$) as a function of plasma temperature, T_e.

2.0 μs with a minimum radius ratio of 0.25. The average shock speed is 3.8 cm/μs. The peak shock speed measured from the photograph is about 6.5 cm/μs corresponding to a plasma temperature of 4.8 x 10^4 K. It is believed that the measured peak shock speed is a little low due to the difficulties involved in identifying the shock front near the axis where it begins to merge.

4.1.4 Current-stepped Z-pinch discharges

The current and voltage signals and the corresponding streak photograph for a 0.8 mbar hydrogen current-stepped compression for different switching times, t_{cs} of the second bank are shown in Fig. 4.5. The piston trajectories obtained from the analyses of the electrical measurements and those predicted by the model are shown in Fig. 4.6. A summary of the results obtained is tabulated in Table 4.1. It is observed that the minimum radius ratios obtained upon current-stepping for different switching times do not differ much from that obtained with the first bank alone, infact they are slightly larger than that by the single stage compression. The corresponding streak is reproduced in Fig. 4.7 and the results derived are summarised in Table 4.2. The temperatures deduced from the shock jump relations are also shown.

Therefore it appears that current-stepping does not enhance the pinch compression contrary to the prediction by the energy balance model. This could be due to the introduction of other factors upon current-stepping which have the effect of decreasing the compressibility of the plasma and thus masking the

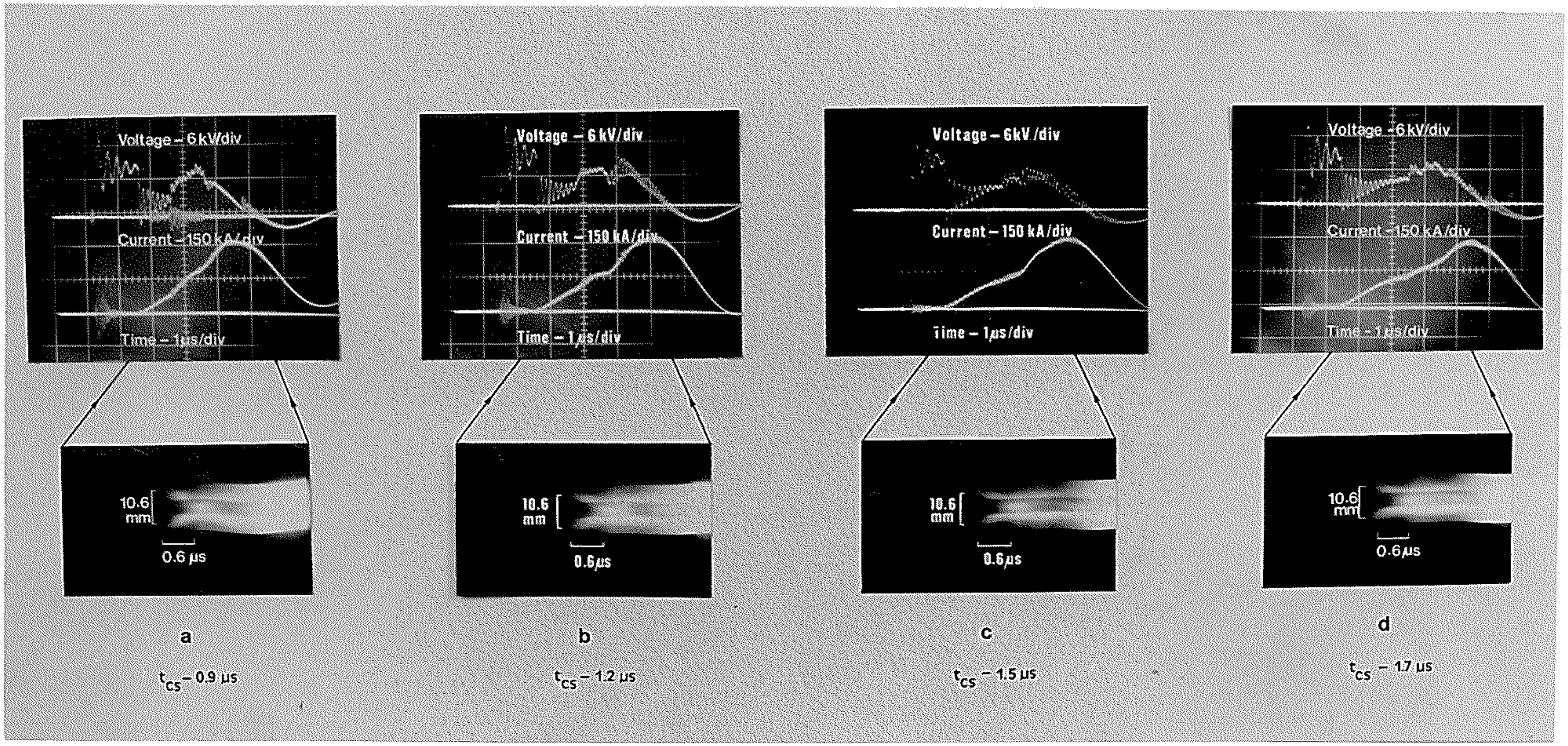
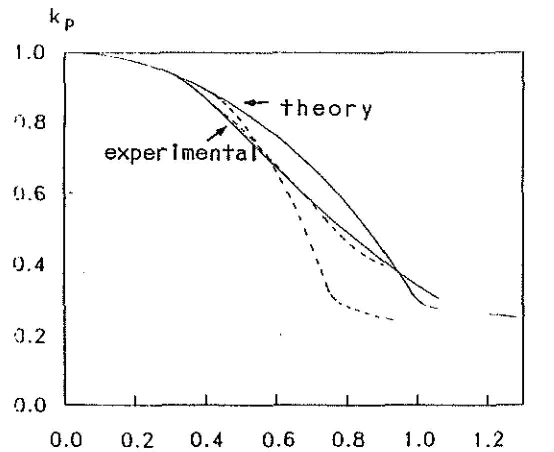
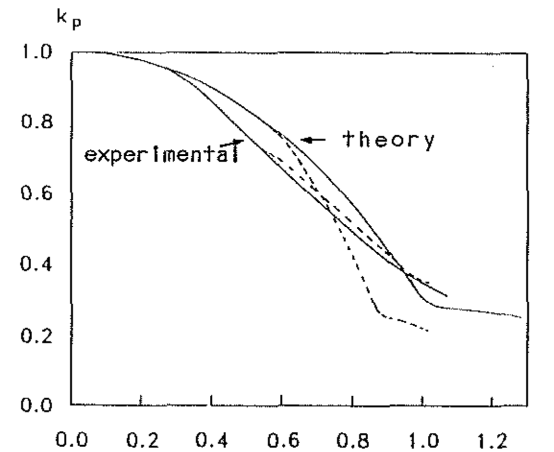


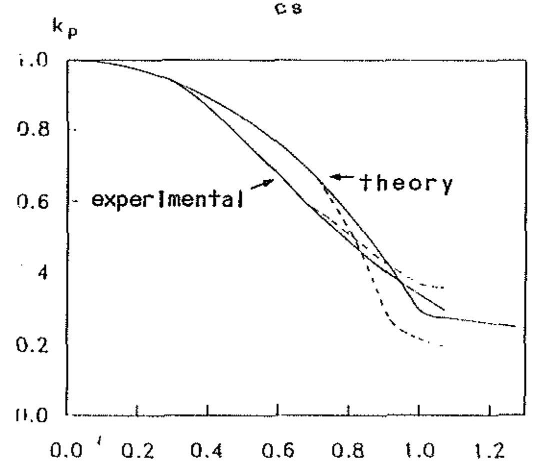
Fig. 4.5 Electrical measurements and the corresponding streak photographs of the radial current-stepped compressions for UMCSZP at 0.8 mbar hydrogen with different times of current-stepping, t_{cs}.



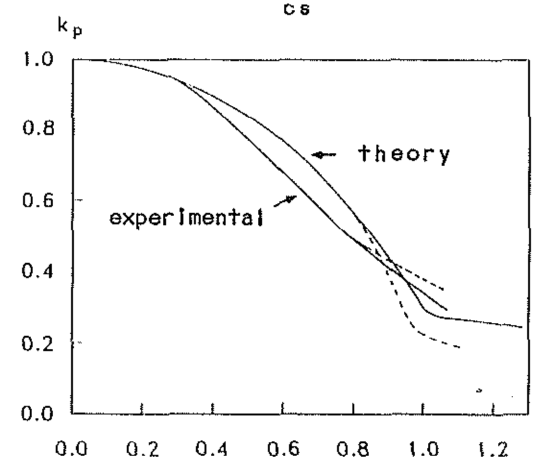
(a) $\tau_{cs} = 0.42$



(b) $\tau_{cs} = 0.56$



(d) $\tau_{cs} = 0.79$



(c) $\tau_{cs} = 0.70$

Fig. 4.6 Piston trajectories of UMCSZP with different times of current-stepping.

(—— without current-stepping, - - - with current-stepping, $\tau = t/t_o$, $t_o = 2.15\ \mu s$)

		Theoretical (energy balance slug model with $\gamma = 5/3$)	Experimental (from IV signals – corrected)
		k_{pm}	k_{pm}
First bank alone		0.24	0.30
	t_{cs} (μs)	k_{pm-cs}	k_{pm-cs}
Current-stepped Z-pinch	0.9	0.21	0.40
	1.2	0.19	0.34
	1.5	0.19	0.36
	1.7	0.18	0.35
Density compression enhancement factor,	$\left[\frac{k_{pm}}{k_{pm-cs}}\right]^2$	(1.3 – 1.8)	–

Table 4.1 Theoretical and experimental quasi-equilibrium radius ratios, k_{pm} for a simple gas compressional Z-pinch and a current-stepped Z-pinch.

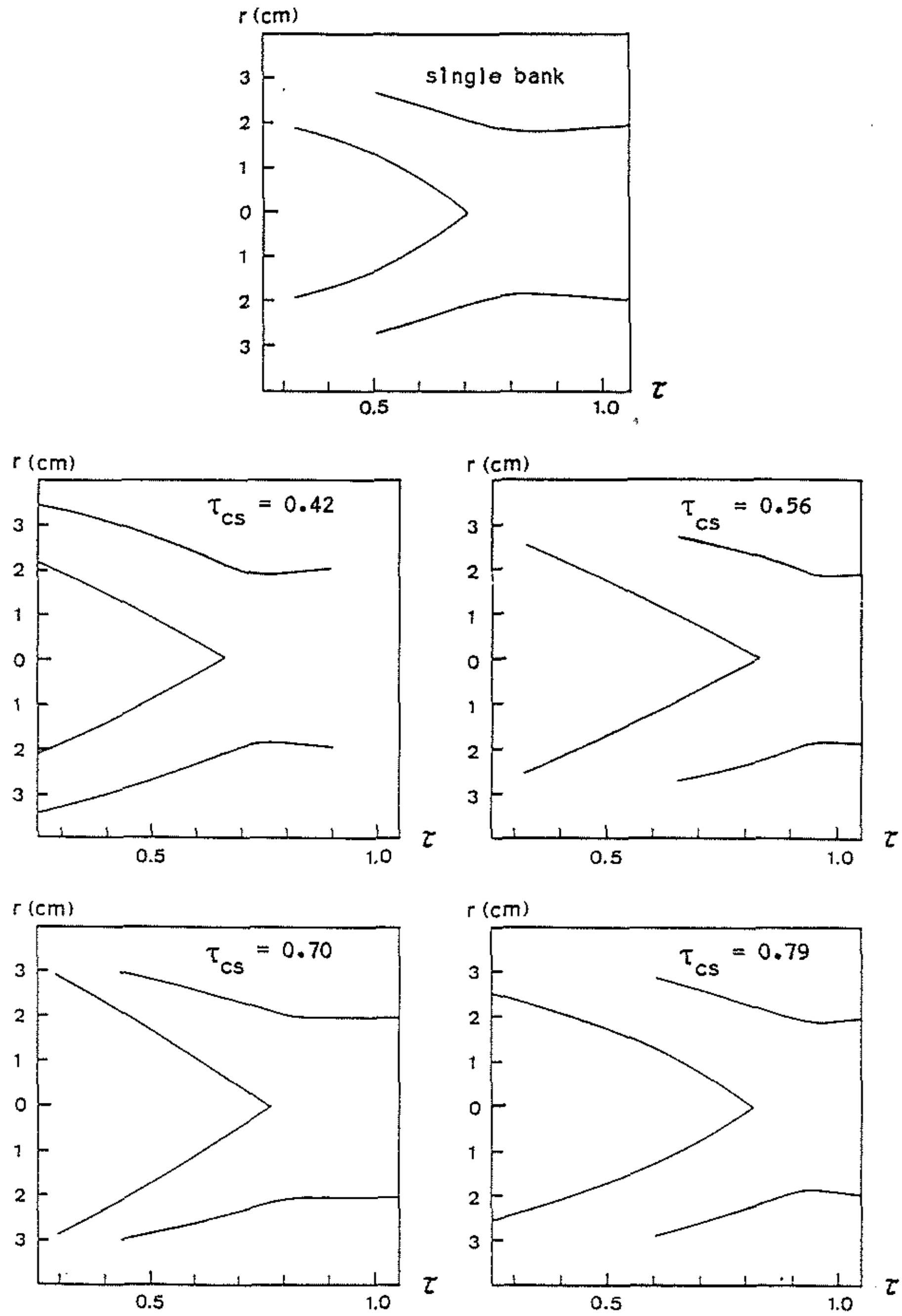


Fig. 4.7 Reproduction of the streak images of the final stage of the UMCSZP for single compression and current-stepped compression with different times of current-stepping.

	t_{cs} (μs)	t_p (μs)	k_{pm}	Shock speed Average v_s (cm/μs)	Shock speed Peak v_p (cm/μs)	Plasma Temperature T_e ($\times 10^4$K)
First bank alone	–	2.0	0.25	3.8	6.5	4.8
Current-stepped Z-pinch	0.9	1.9	0.25	3.9	7.0	5.5
	1.2	2.1	0.25	3.5	6.0	4.1
	1.5	2.1	0.27	3.5	6.9	5.4
	1.7	2.3	0.25	3.3	7.1	5.7

Table 4.2 Results obtained from streak photography

effect of enhancement in the compression expected from current-stepping. One of the factors which could result in an increase in the radius ratio is an increase in the effective specific heat ratio, γ_{eff} of the real gas. This is because γ_{eff} determines the compressibility of the gas. A larger γ_{eff} corresponds to a lower compressibility while a lower γ_{eff} indicates a higher compressibility. Hence if γ_{eff} rises upon current-stepping, the reduced compressibility may overshadow the enhancement effect of current-stepping. It is therefore necessary to compute the variation of γ_{eff} with temperature and shock speed. This will be done in the next chapter.

4.2 Magnetic field measurements

4.2.1 Introduction

In order to observe the formation of the radial current density profile and its development upon current-stepping, magnetic probe signals are taken at various radial positions together with the current and voltage signals for both cases of the plasma formed from the first bank alone and for the plasma formed from current-stepping.

Assuming that the current flows in a hollow shell parallel to the cylindrical axis, the radial current density can be related to the axial B-field as given below:

$$j_z(r) = \frac{1}{\mu_o r} \frac{d}{dr} (rB_\theta) \qquad [4.6]$$

4.2.2 Discharge with first bank alone

The magnetic probe signals for the hydrogen discharge at 0.8 mbar of the first bank alone are shown in Fig.4.8. Two different discharges at the same pressure and with similar current profiles are used to map the radial current density with the aid of equation [4.6]. The evolution of the radial current density is shown in Fig. 4.9. It is observed that the current sheath is rather diffusive and thick, indicating a poor coupling between the plasma flow field and the magnetic field.

4.2.3 Current-stepped Z-pinch discharges

The magnetic probe signals for the hydrogen discharges at 0.8 mbar for the current-stepped Z-pinch with the second bank switching at 1.9 μs after the first bank are shown in Fig.4.10. Two different discharges at the same pressure and with similar current profiles are again used to map the radial current density with the aid of equation [4.6]. The evolution of the radial current density is shown in Fig. 4.11. It is observed that upon current-stepping, the current density profile broadens before commencing to increase further. This indicates a delay in the merging of the current sheath upon current-stepping resulting in a pulling back effect on the piston. This is also observed from the radial trajectory obtained from electrical measurements presented in Fig.4.7.

4.3 Conclusion

The experimental results and those predicted by the model seem to disagree. This may be attributed to a combination

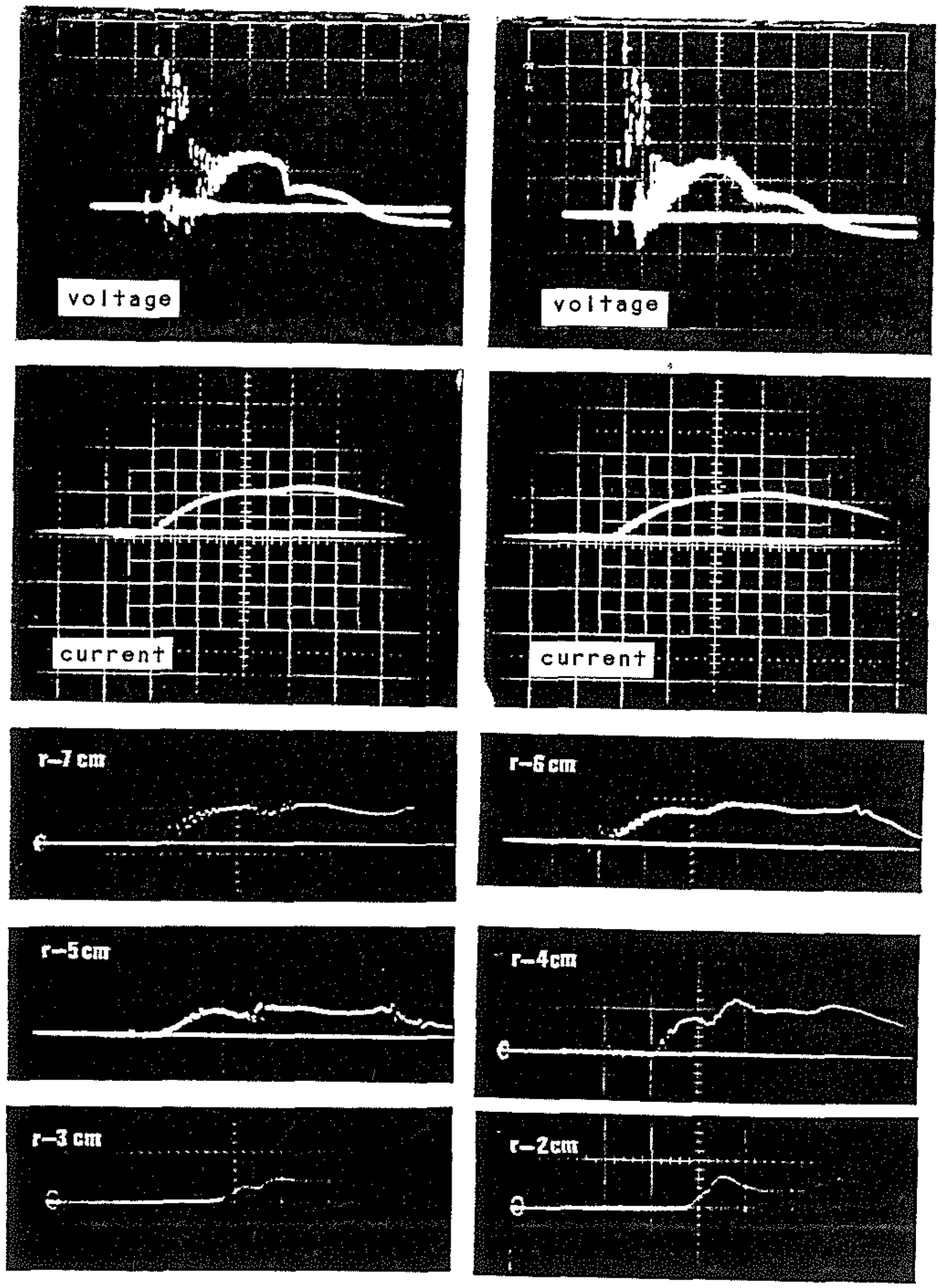


Fig.4.8 Electrical and magnetic field signals of a 0.8 mbar hydrogen pinch driven by the first bank alone.

(Scale : voltage : 3kV/div, current : 150 kA/div,
B-field - r = 7 cm : 0.767 T/div, r = 5 cm : 1.015 T/div,
r = 3 cm : 1.075 T/div, r = 6 cm : 0.729 T/div,
r = 4 cm : 0.789 T/div, r = 2 cm : 0.972 T/div,
time : 1 μs/div).

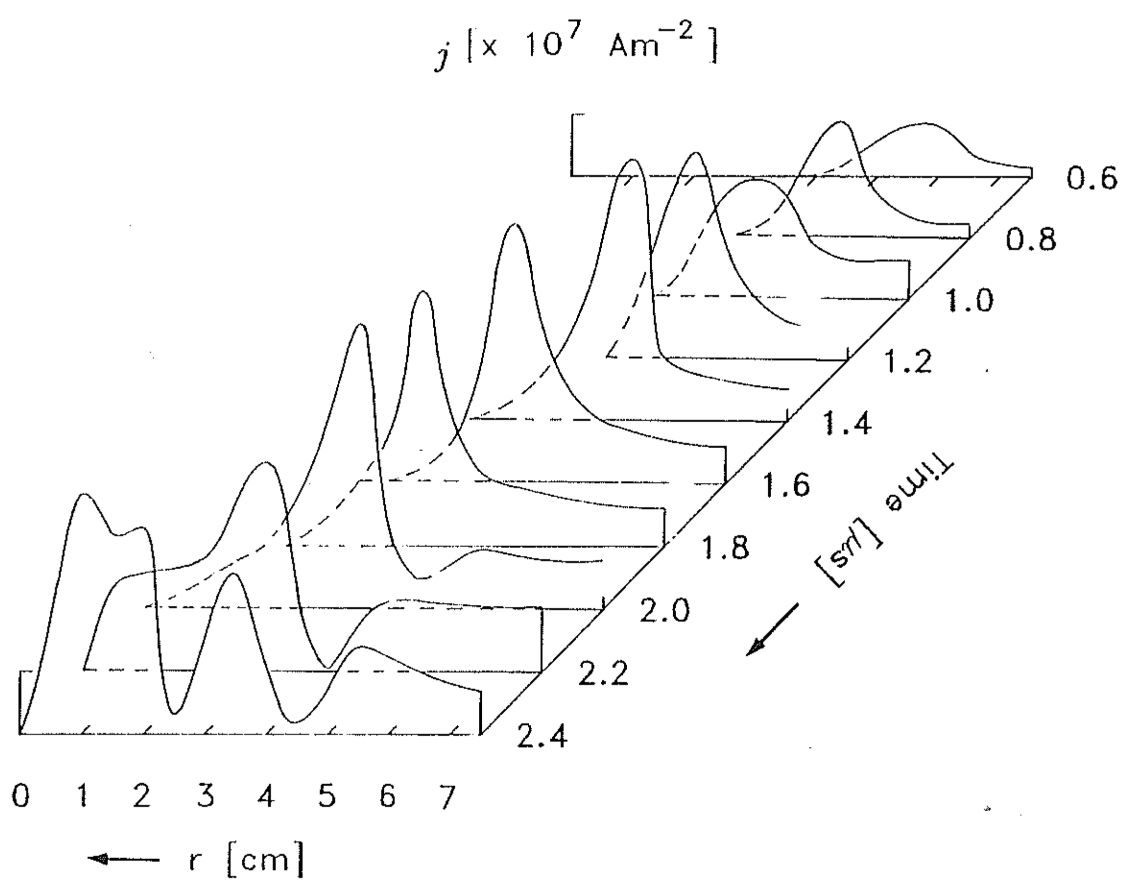


Fig. 4.9 The development of the current density of a 0.8 mbar hydrogen pinch driven by the first bank alone. ($t = 0$ is the start of the current).

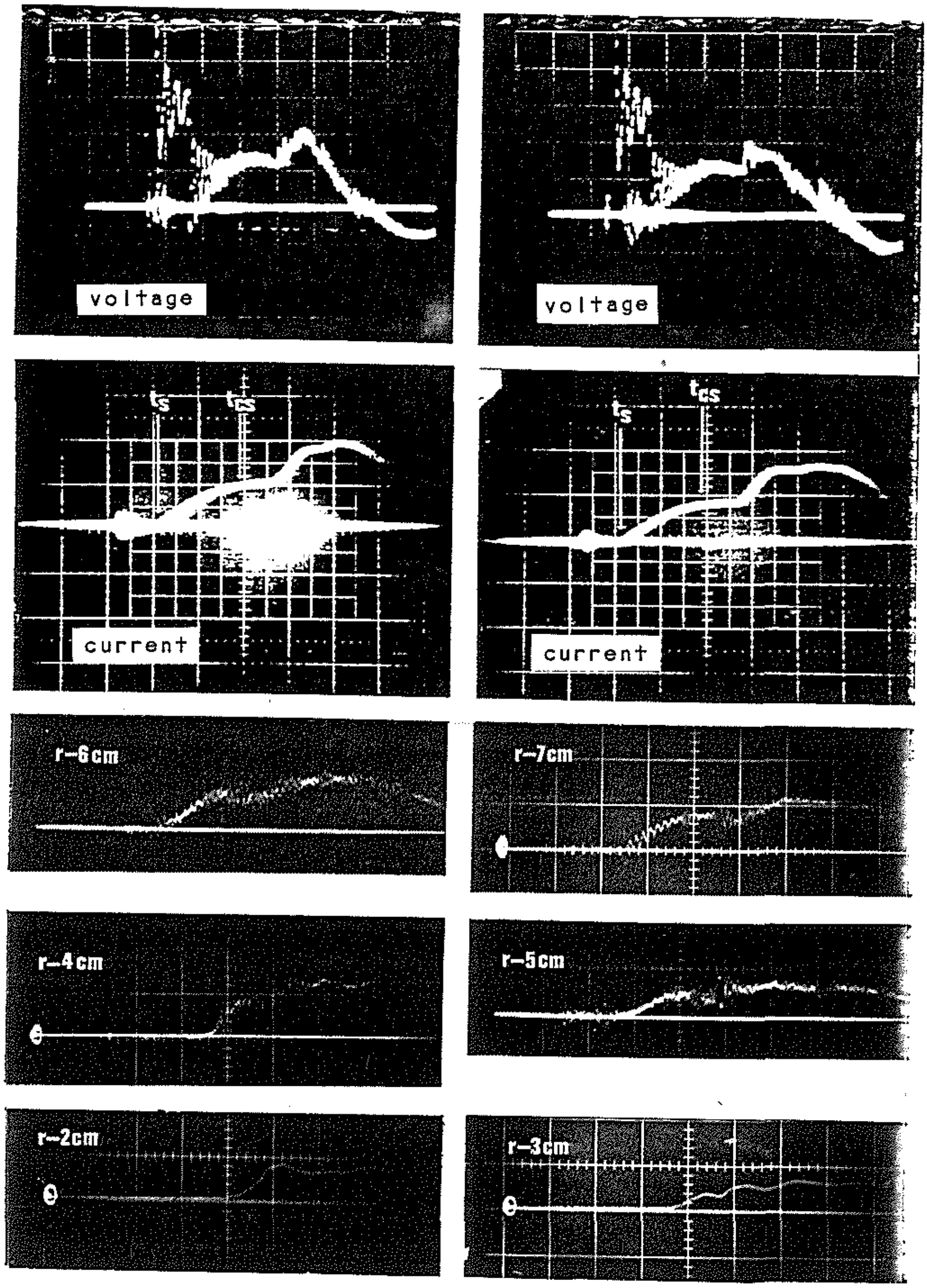


Fig. 4.10 Electrical and magnetic field signals of a 0.8 mbar hydrogen current-stepped Z-pinch (t_{cs} is 1.9 μs).
(Scale : voltage : 3kV/div, current : 150 kA/div,
B-field - r = 7 cm : 0.767 T/div, r = 5 cm : 1.015 T/div,
r = 3 cm : 1.075 T/div, r = 6 cm : 0.729 T/div,
r = 4 cm : 0.789 T/div, r = 2 cm : 0.972 T/div,
time : 1 μs/div).

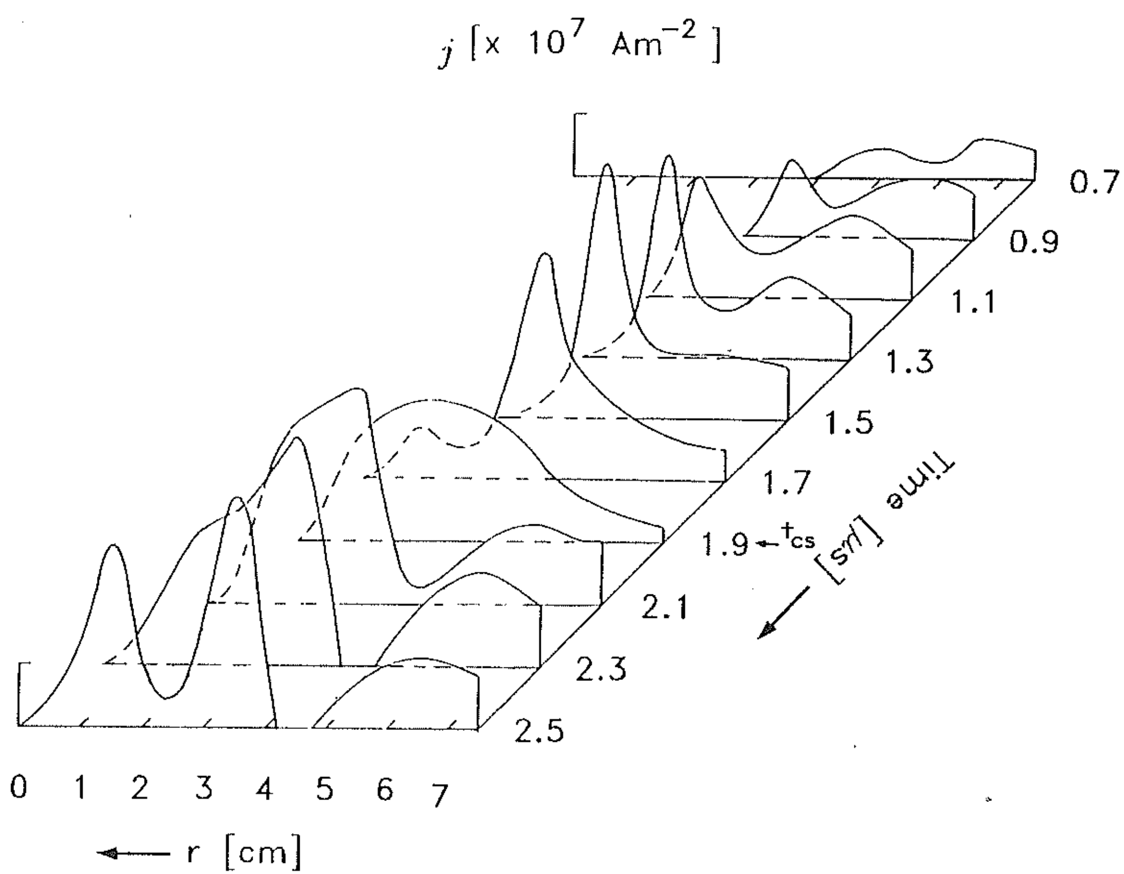


Fig. 4.11 The development of the current density of a 0.8 mbar hydrogen current-stepped Z-pinch (t_{cs} is 1.9 μs).

(t = 0 is the start of the current).

of several effects. However, 2 major effects will be discussed, namely the thermodynamic and electromagnetic effects of the plasma.

In our modelling so far, the thermodynamics of an ideal gas is assumed with the specific heat ratio of the gas taken as a constant and equal to 5/3. If the deviation of the real gas system from the ideal gas condition is small, a good approximation can be obtained without requiring the knowledge of the effective specific heat ratio, γ_{eff} of the real gas. However for shocked compression the variation of γ_{eff} with shock speed becomes important. We may make a preliminary consideration in the following way. For hydrogen atom, the estimated shock speed corresponding to the ionisation potential of 13.6 eV is about 4 cm/μs. Therefore for shock speed below 4 cm/μs, γ_{eff} will be considerably below 5/3. As the shock speed increases from 4cm/μs and as the translational modes dominate the internal energy of the gas, γ_{eff} will approach the ideal constant equal to 5/3.

In the present experimental set-up, the shock speed of the hydrogen Z-pinch during the first compression is about 4 cm/μs. This implies that the effective specific heat ratio, γ_{eff} during the first compression will be significantly below 5/3. Upon current-stepping, the shock speed will increase and γ_{eff} will be raised. Hence the compressibility of the plasma will be affected. It has been shown by *Lee [1983a]* that the radius of an argon pinch depends on the effective specific heat ratio of argon. The dependence of the final radius on the specific heat ratio of the gas has also been discussed by *Potter [1978]*.

Therefore, it appears that current-stepping raises the effective specific heat ratio and thus reduces the compressibility of the plasma. The enhanced compression expected from current-stepping has to compete with this reduced compression. The enhancement by current-stepping is being masked by the γ-effect.

A second major factor also needs to be considered. In our modelling, the plasma current is assumed to flow in an infinitesimally thin layer of plasma with infinite electrical conductivity. From the considerations of the MHD, this requires that the magnetic Reynolds number, $\mathcal{R}_m \gg 1$, representing a condition whereby the plasma flow field convection predominates over the magnetic field diffusion due to the electrical resistivity of the plasma column. This condition is fully satisfied if the electrical conductivity is large enough, since the magnetic field diffusion will be very small then. However, if the resistivity is large enough such that $\mathcal{R}_m < 10$, then the magnetic field diffusion component may be significantly large and a thick and diffusive layer of current may result. The plasma resistivity is a temperature dependent function, varying inversely to the power of 3/2. The magnetic Reynolds number, $\mathcal{R}_m$ is then dependent on V_s^4 or the shock speed to the power of 4 *[Lee, S., 1985b]*. For a plasma of characteristic length, $\ell = 1$ cm, and taking ln (Λ) equal to 10, then $\mathcal{R}_m$ is given by $(4 \times 10^{-3}\ V_s^4)$ where V_s is in cm/μs. Therefore for $V_s = 4$ cm/μs, $\mathcal{R}_m$ equals to 1. This indicates that the magnetic field diffusion is large and of equal significance to the plasma flow field convection. Hence the plasma generated is comparatively more resistive than assumed. In order to obtain a good plasma flow field-magnetic field coupling, it is necessary for $\mathcal{R}_m$ to exceed 10 or a shock speed beyond 8 cm/μs.

From the above discussion, the indication is that the enhancement effect by current-stepping is being masked by both the thermodynamic and the magnetohydrodynamic effects of the moving plasma. The disagreement between the experimental results and those predicted by the model could be due to the breakdown of the assumptions adopted in the model; namely, that the specific heat ratio is constant and equal to 5/3 and the plasma conductivity is infinite. An improvement in the model is necessary to provide a better representation of the plasma produced in the present experimental conditions. This is realised by introducing the effective specific heat ratio, γ_{eff} into the existing model. The magnetic Reynolds number, $\mathcal{R}_m$ should also be estimated at each stage to observe the extent of field diffusion of the current sheath.

CHAPTER 5
The effect of variation of γ_{eff}

5.1 Introduction

As mentioned in Chapter 4, it is necessary to extend the current-stepping model to consider the effective specific heat ratio, γ_{eff} of the plasma since the condition of the plasma may deviate from the assumed constant specific heat ratio of 5/3. In this chapter, the effective specific heat ratio, γ_{eff} of hydrogen and its variation with shock speed is obtained *[Saw, S.H., Lee, S. & Wong, C.S., 1988]* from the solution of the 1-D plane shock jump equations *[A.G.Gaydon,1963]*, the thermal equation of state, the Saha equation and the caloric equation of state *[Cambell, A.B., 1963]*. Introduction of this variation into the energy balance slug model allows a closer representation of the hydrogen plasma produced. In this extended model, the development of the plasma flow field-magnetic field coupling of the plasma and its development in time is also checked using the magnetic Reynolds number, $\mathcal{R}_m$.

5.2 Calculation of γ

For shocks faster than Mach 10 the 1-D shock jump equations relating the shocked condition to the ambient condition measured in the shock fixed coordinates may be written as

$$\rho_1 q_1 = \rho_2 q_2 \qquad [5.1]$$

$$\rho_1 q_1^2 = \rho_2 q_2^2 + P_2 \qquad [5.2]$$

$$\tfrac{1}{2} q_1^2 = \tfrac{1}{2} q_2^2 + \mathbb{H}_2 \qquad [5.3]$$

$$P_2 = \frac{R_o}{M} \rho_2 T_2 \chi \qquad [5.4]$$

where the subscript 1 refers to the ambient condition and subscript 2 refers to the shocked condition. ρ, q, P and $\mathbb{H}$ are the density, particle velocity, pressure, and enthalpy respectively. R_o is the universal gas constant and M is the molecular weight of the gas. Both χ and γ are temperature dependent. χ(T) is the departure coefficient of the gas describing the degree of dissociation and ionisation in the thermal equation of state.

The enthalpy corresponding to the shocked condition expressed in terms of shocked pressure and density is given by

$$\mathbb{H}_2 = \frac{\gamma}{\gamma - 1} \frac{P_2}{\rho_2} \qquad [5.5]$$

From equations [5.4] and [5.5], it is observed that γ can be solved as a function of shocked temperature once $\mathbb{H}_2$ and χ are known. $\mathbb{H}_2$ can be obtained by taking the summation over all energy levels while χ(T) is obtained from the thermal equation of state and the Saha equation.

The corresponding shock speed can then be obtained from

$$T_2 = q_1^2 \frac{M}{R_o} \left[\frac{\Gamma - 1}{\Gamma} \right] \frac{1}{\chi} \qquad [5.6]$$

where Γ is the compression ratio and is given by

$$\Gamma = \frac{\rho_2}{\rho_1} = \frac{\gamma + 1}{\gamma - 1} \qquad [5.7]$$

For diatomic hydrogen, the equilibrium state for dissociation and ionisation reactions can be represented as

$$H_2 + U_D \longrightarrow 2H$$

$$H + U_I \longrightarrow H^+ + e$$

where U_D is the dissociation energy (4.5 eV) of the H_2 molecule and U_I the ionisation energy (13.6 eV) of the H atom.

The thermal equation of state is given by

$$P_2 = \frac{R_o}{M_{H2}} \rho_2 T_2 \ (1 + \delta + 2\alpha) \qquad [5.8]$$

where δ, the degree of dissociation is the ratio of the number of H_2 molecules dissociated to the total number of H_2 molecules initially present in the system, N_t. The degree of ionisation, α is the ratio of the number of H^+ ions to $2N_t$. The departure coefficient, $\chi(T)$ is equal to $(1 + \delta + 2\alpha)$.

The degrees of dissociation and ionisation are described by the Saha equations.

For dissociation, this is given by

$$\frac{4\ (\delta-\alpha)^2}{(1-\delta)} = 0.44\ M_{H_2} \frac{T_2}{\rho_1} \frac{Z^2_{eH}}{(Z_r Z_v Z_e)_{H_2}} exp(-U_D/kT_2) \qquad [5.9]$$

where $(Z_r Z_v Z_e)_{H_2}$ *[Herbert, G., 1950, Moore, C.E., 1949]* are the rotational, vibrational and electronic partition functions of the H_2 molecule respectively. Z_{eH} is the partition function of the hydrogen atom. These quantities are computed from the general expression for the partition function as given below;

$$Z = \sum_i g_i\ exp(U_i/kT) \qquad [5.10]$$

where U_i is the i-excitation energy level and g_i the corresponding statistical weight. *[Pointon, A.J., 1967]*.

The diatomic molecule here is treated as a homonuclear rigid rotator and a harmonic oscillator *[Barrow, 1962]*. M_{H2} is the atomic weight of the hydrogen molecule.

And for ionization,

$$\frac{2\alpha^2}{\delta-\alpha} = 8.0x10^{-6} \frac{M_{H2}}{\rho_1} T_2^{3/2} \frac{Z_{eH^+}}{Z_{eH}} exp(-U_I/kT_2) \qquad [5.11]$$

where Z_{eH^+} and Z_{eH} are the electronic partition functions of the hydrogen ion and atom respectively. These quantities are computed from [5.10]. From equations [5.9] and [5.11], δ and α are solved for each value of T and hence $\chi(T)$ can be determined.

The value of the enthalpy may be computed as follows;

$$\mathbb{H} = \frac{5R}{2M_{H2}} T_2(1+\delta+2\alpha) + \frac{(1-\delta)U_{vib}}{M_{H2}} + \frac{(1-\delta)U_{rot}}{M_{H2}} + \frac{\delta U_D}{M_{H2}} + \frac{\alpha U_I}{M_H}$$

$$+ \frac{(\delta-\alpha)U_{eH}}{M_H} + \frac{(1-\delta)U_{eH2}}{M_{H2}} + \frac{\alpha U_{eH^+}}{M_{H^+}} \qquad [5.12]$$

where the computed quantities U_{eH} , U_{eH2} and U_{eH^+} are the electronic excitation energies of the H atom, the H_2 molecule and the H^+ ion respectively. U_{rot} and U_{vib} are the rotational and vibrational excitation energies of the H_2 molecule.

γ is then computed as a function of T_2 using equations [5.5] and [5.12].

$$\left[\frac{\gamma}{\gamma-1}\right] = \frac{5}{2} + \frac{\left[(1-\delta)U_{rot} + (1-\delta)U_{vib} + \delta U_D + (1-\delta)U_{eH2}\right]}{R_o T_2(1+\delta+2\alpha)}$$

$$+ \frac{\left[\frac{\alpha U_I}{M_H} + \frac{(\delta-\alpha)U_{eH}}{M_H} + \frac{\alpha U_{eH^+}}{H^+}\right]}{\frac{R_o T_2(1+\delta+2\alpha)}{M_{H2}}} \qquad [5.13]$$

Finally the shock speed is obtained as a function of T_2 from Eqn [5.6] & [5.7], using the values of γ and χ computed for T_2.

5.3. Numerical solutions

The solutions of the effective specific heat ratio, γ_{eff} for hydrogen with an ambient pressure, P = 0.8 mbar are presented.

The dependence of the degrees of dissociation and ionisation on shock speed is shown in Fig. 5.1. The variation of γ_{eff} with shock speed is shown in Fig. 5.2.

For low shock speeds at room temperature, only the translational and rotational degrees of freedom are excited. This results in an effective specific heat ratio, γ_{eff} equal to 1.4. At shock speeds of about 0.7 cm/us, the hydrogen molecule starts to dissociate. The degree of dissociation then rises rapidly with shock speed followed by a relatively gradual rise which finally slows as the degree of dissociation approaches 100%. During this process, the effective specific heat ratio, γ_{eff} drops from 7/5 to a minimum of 1.1 at about 2.2 cm/μs, which then rises towards 1.2 as full dissociation is reached. This increase in γ_{eff} continues until the shock speed reaches 2.8 cm/μs and ionisation begins to set in. The increase in the degree of ionisation with shock speed is less rapid compared to that for dissociation. This is due to the higher ionisation potential. Similarly, the increase in the degree of ionisation with shock speed levels off with the approach of full ionisation. Correspondingly, γ_{eff} drops from 1.2 to a minimum of 1.14 at a shock speed of 6 cm/μs. At about 7 cm/μs, hydrogen is 99% ionised. With further increase in the shock speed, γ_{eff} rises asymptotically towards 5/3.

The variation of the temperature with shock speed is shown in Fig. 5.3. Initially the shock temperature increases rapidly with shock speed until dissociation sets in. During this process of dissociation energy is absorbed and hence the temperature rises very little between 1 cm/μs and 2 cm/μs. Once the hydrogen molecules are fully dissociated, the temperature

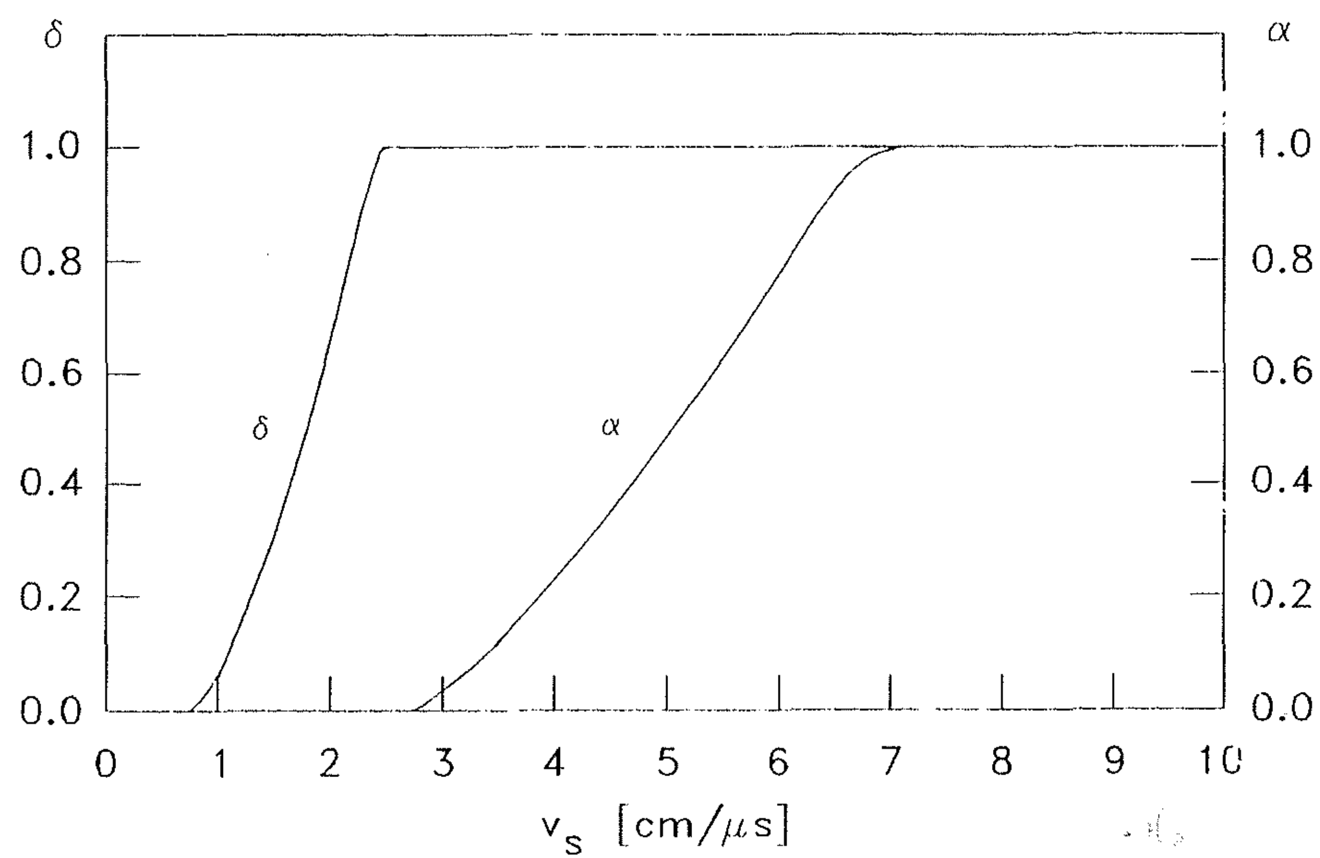


Fig. 5.1 Degree of dissociation, δ and degree of ionisation, α versus shock speed, v_s for hydrogen gas at 0.8 mbar.

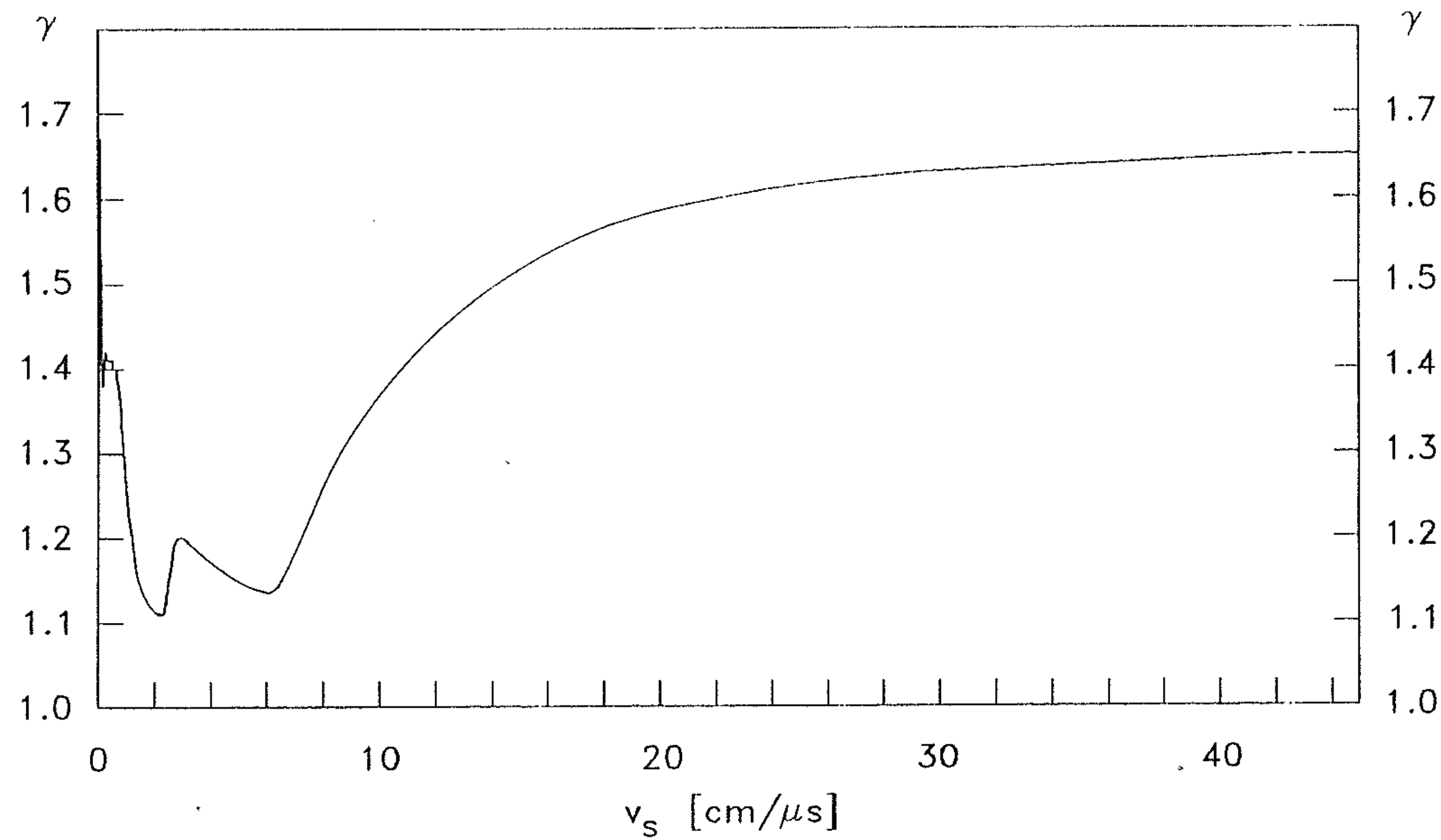


Fig. 5.2 Specific heat ratio, γ versus shock speed, v_s for hydrogen gas at 0.8 mbar.

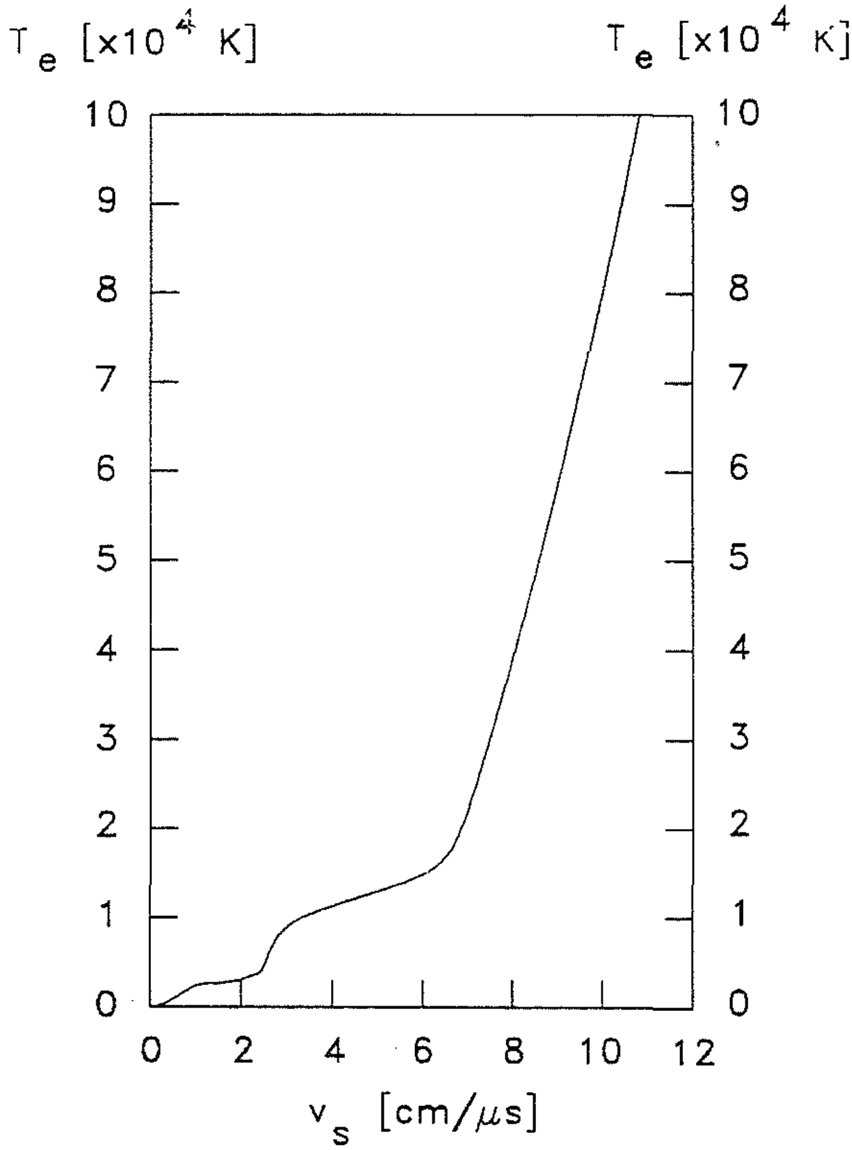


Fig. 5.3 Temperature, T_e versus shock speed, v_s for hydrogen gas at 0.8 mbar.

rises rapidly again with the shock speed until ionisation commences. Again the temperature rises very slowly with shock speeds between 4 cm/μs and 6 cm/μs until 99% ionisation is achieved at about 7 cm/μs. Beyond this, the temperature increases with the square of the shock speed.

5.4 Improved energy balance slug model with varying gamma, $\gamma(t)$.

The energy balance slug model used to describe the compressional pinch dynamics as described in Chapter 2 is extended to include the effective specific heat ratio, γ_{eff} of the plasma. For hydrogen with an ambient pressure of 0.8 mbar, this is necessary for shock speeds below 40 cm/μs. This is because at these shock speeds, γ_{eff} deviates from the constant ideal gas value of 5/3. It varies with the shock speed from a value of 1.11 at 2 cm/μs, rising to 1.5 at 14 cm/μs in contrast to the ideal gas γ which is constant. Therefore it is necessary to include the effective specific heat ratio, γ_{eff} in the energy balance slug model for the UMCSZP.

Hence the equations describing the shock and piston trajectories for the strong compression phase and the slow dynamics phase as given in [2.12] - [2.14] now contain a time varying effective specific heat ratio, $\gamma(t)$ in place of the ideal constant value of 5/3. The equations involved become as follows:

I. The strong compression phase

[i] The shock trajectory :

$$v_s = - \frac{dr_s}{dt} = - \left[\frac{\mu_0[\gamma(t)+1]}{\rho_o} \right]^{1/2} \frac{I}{4\pi r_p} \qquad [5.14]$$

[ii] The piston trajectory :

$$\frac{dr_p}{dt} = \frac{\left[\frac{2}{\gamma(t)+1}\right]\left[\frac{r_s}{r_p}\right]\left[\frac{dr_s}{dt}\right] - \left[\frac{r_p}{\gamma I}\right]\left[1 - \left[\frac{r_s}{r_p}\right]^2\right]\left[\frac{dI}{dt}\right]}{\left[\frac{\gamma(t)-1}{\gamma(t)}\right] + \left[\frac{1}{\gamma(t)}\right]\left[\frac{r_s}{r_p}\right]^2}$$

[5.15]

II The slow dynamics phase

[i] The piston trajectory is :

$$\frac{dr_p}{dt} = - \frac{r_p}{[\gamma t)-1]\ I} \frac{dI}{dt} \qquad [5.16]$$

where $\gamma(t)$ is a time-varying function solved as shown in section 5.2.

The circuit equations governing the formation of the plasma remain unchanged as in equations [2.15] or [2.16a] and [2.16b] corresponding to the compression without and with current-stepping respectively.

5.5 Computation procedures

The computation procedures adopted for the γ-varying current-stepped model are as follows. First, the computation of the effective specific heat ratio, γ_{eff} corresponding to the experimental discharge pressure or initial density is computed. The values of the shock speed and the corresponding effective specific heat ratio, γ_{eff} obtained are stored in a data file. This is followed by the computation of the energy balance slug model where the varying γ_{eff} corresponding to the instantaneous shock speed is read from the data file. Values in between the stored points are interpolated using cubic spline interpolation. The flow chart of the extended current-stepped model incorporating the effect of the varying γ_{eff} is as shown in Fig. 5.4. The program has options to treat γ as constant or as time varying.

With the extended current-stepped model a closer representation of the experimental compression is obtained. The effective specific heat ratio, γ_{eff} and the temperature are simulated as well. In addition, the development of the plasma flow field-magnetic field coupling describing the degree of plasma flow field convection over the magnetic field diffusion due to electrical resistivity of the plasma is also checked using the magnetic Reynolds number, $\mathcal{R}_m$.

5.6 Numerical results

Using the designed parameters for the UMCSZP, the following numerical results were obtained for the single stage compressional Z-pinch driven by the first bank alone and the current-stepped Z-pinch driven by the capacitor-capacitor combination current source. Section 5.6.1 presents the results of the first bank alone. The results with current-stepping are

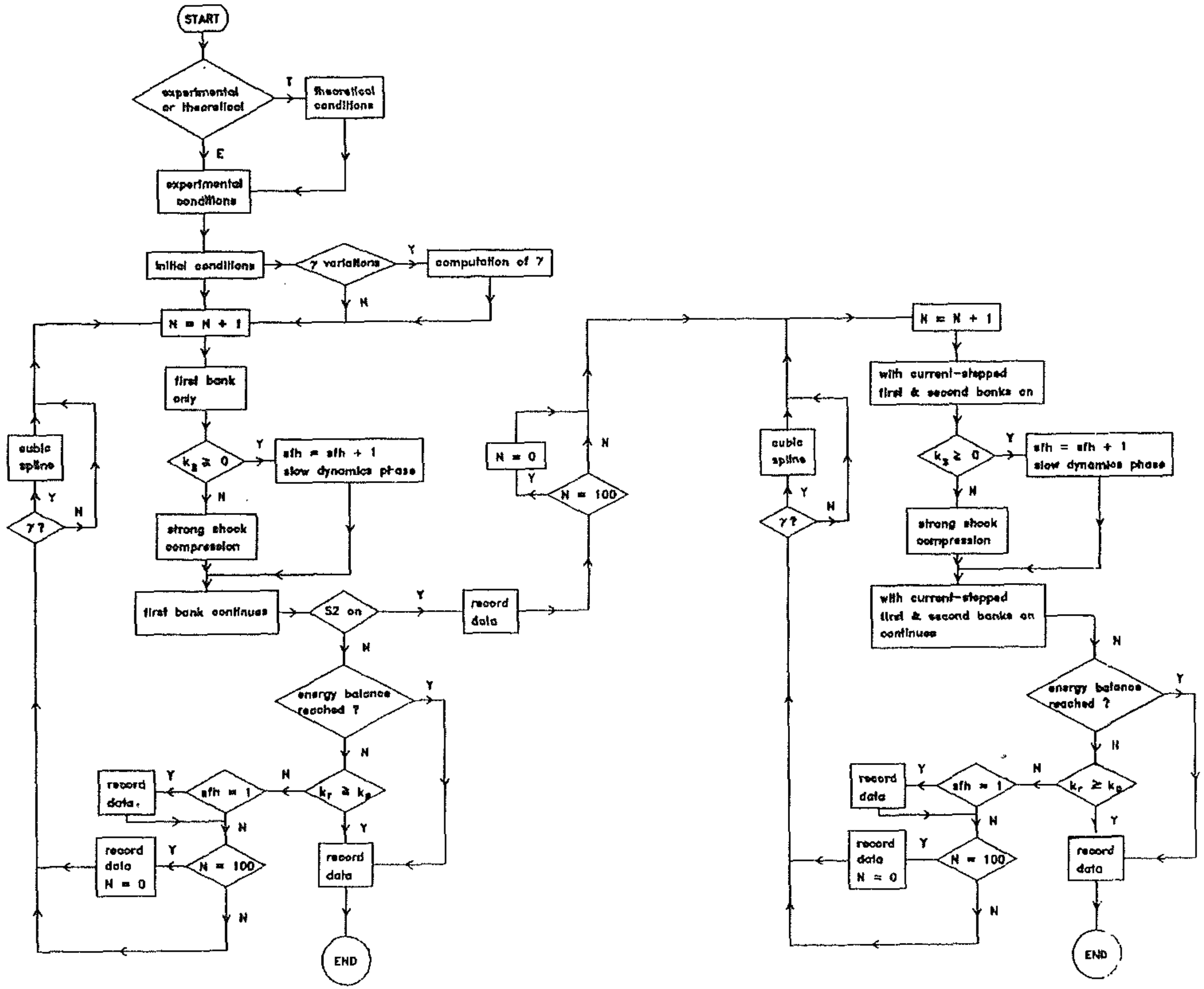


Fig. 5.4 Flow chart of the extended current-stepped model.

presented in section 5.6.2.

5.6.1 First bank alone - without current-stepping

With the parameters of the designed pinch of the first bank (as given in section 3.1) and the Z-pinch operated at 0.8 mbar H_2, the new model predicts a smaller final radius ratio than that predicted by the γ-constant model. This can be understood in terms of the increased compressibility resulting from the lower varying γ.

A set of results obtained with the extended energy balance slug model is presented. The variation of γ_{eff} is shown in Fig. 5.5. Initially at $\tau = 0$, γ_{eff} drops from 5/3 to 1.1 at $\tau = 0.46$ at which time the compressing shock speed is 2.3 cm/μs. As the plasma proceeds toward full dissociation, γ_{eff} starts to rise to 1.2 at $\tau = 0.58$. At this time, the compressing shock speed reaches 3 cm/μs and ionisation begins to set in. γ_{eff} starts to drop again with the onset of ionisation until the hydrogen gas reaches 80% ionisation at $\tau = 0.88$, γ_{eff} is then 1.14. The corresponding shock speed is 6 cm/μs. As the degree of ionisation approaches 100%, γ_{eff} starts to rise again until γ_{eff} = 1.58 at τ =1.02 at which time the shock front hits the axis. During this time the shock speed rises from 6 cm/μs to 20 cm/μs. The variation of the shock speed during the compression is presented in Fig. 5.6.

The development of the temperature of the shock heated hydrogen plasma is shown in Fig. 5.7. The plasma temperature rises slowly from room temperature at $\tau = 0$ to 1.3×10^4 K at $\tau = 0.9$. It then rises sharply to 4.1×10^5 K when the shock

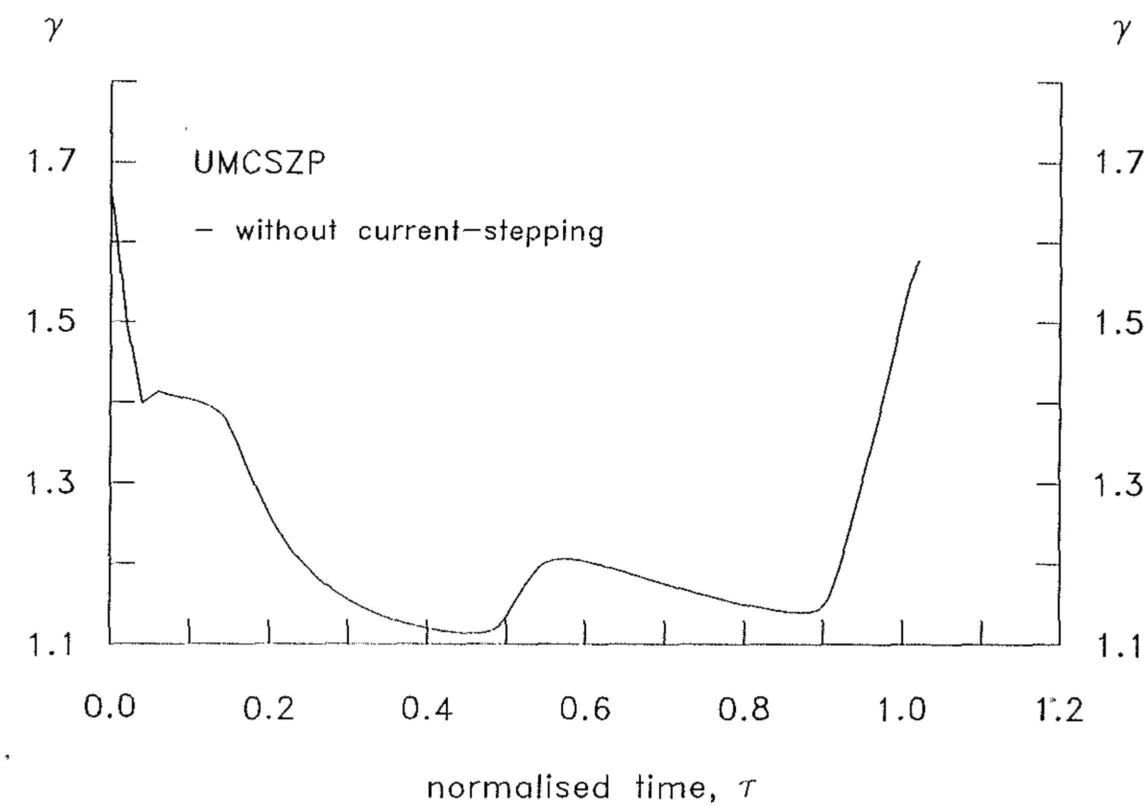


Fig. 5.5 The variation of γ_{eff} for hydrogen discharge at 0.8 mbar driven by the first bank alone (without current-stepping).

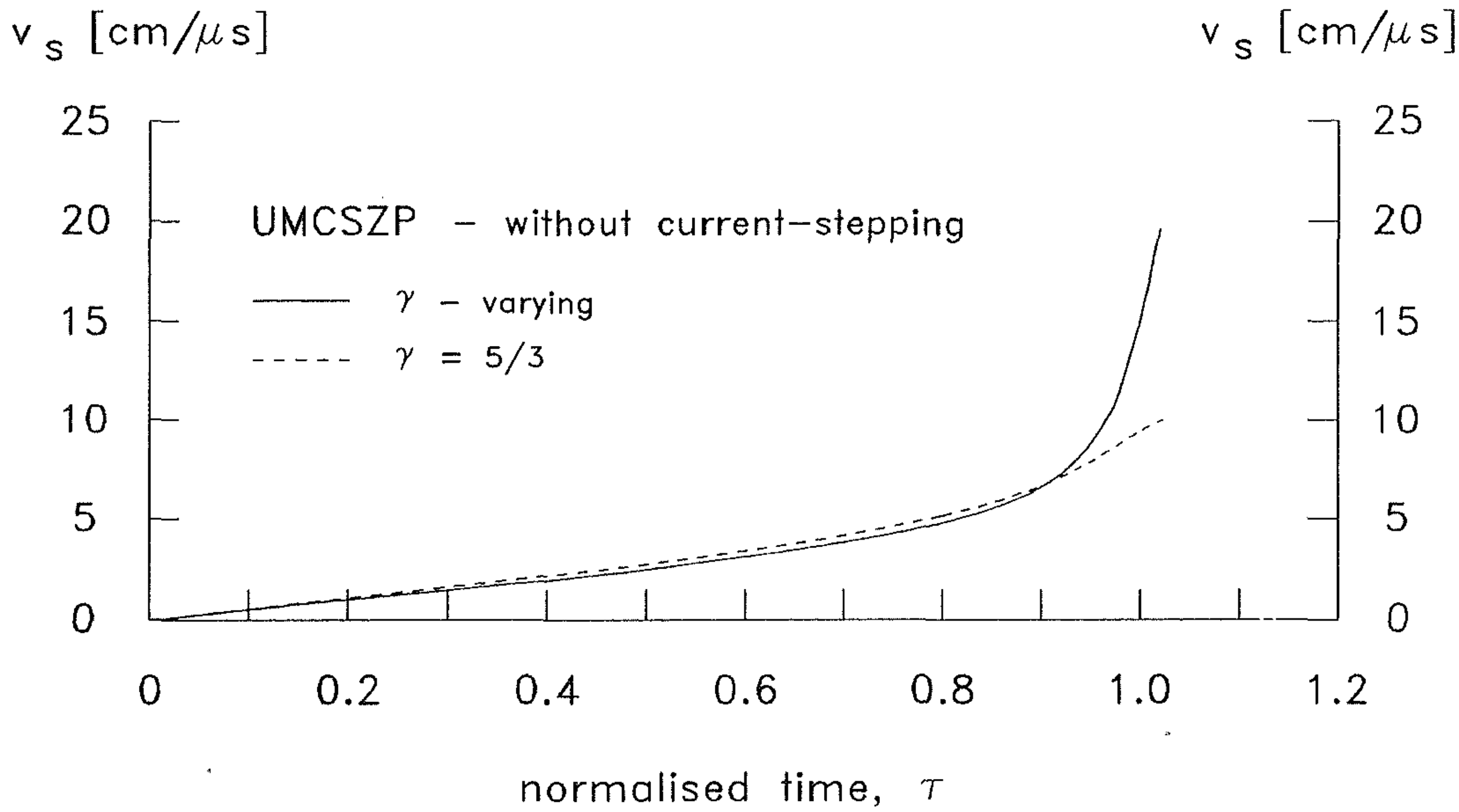


Fig. 5.6 The variation of shock speed, v_s for a hydrogen discharge at 0.8 mbar driven by the first bank alone (without current-stepping).

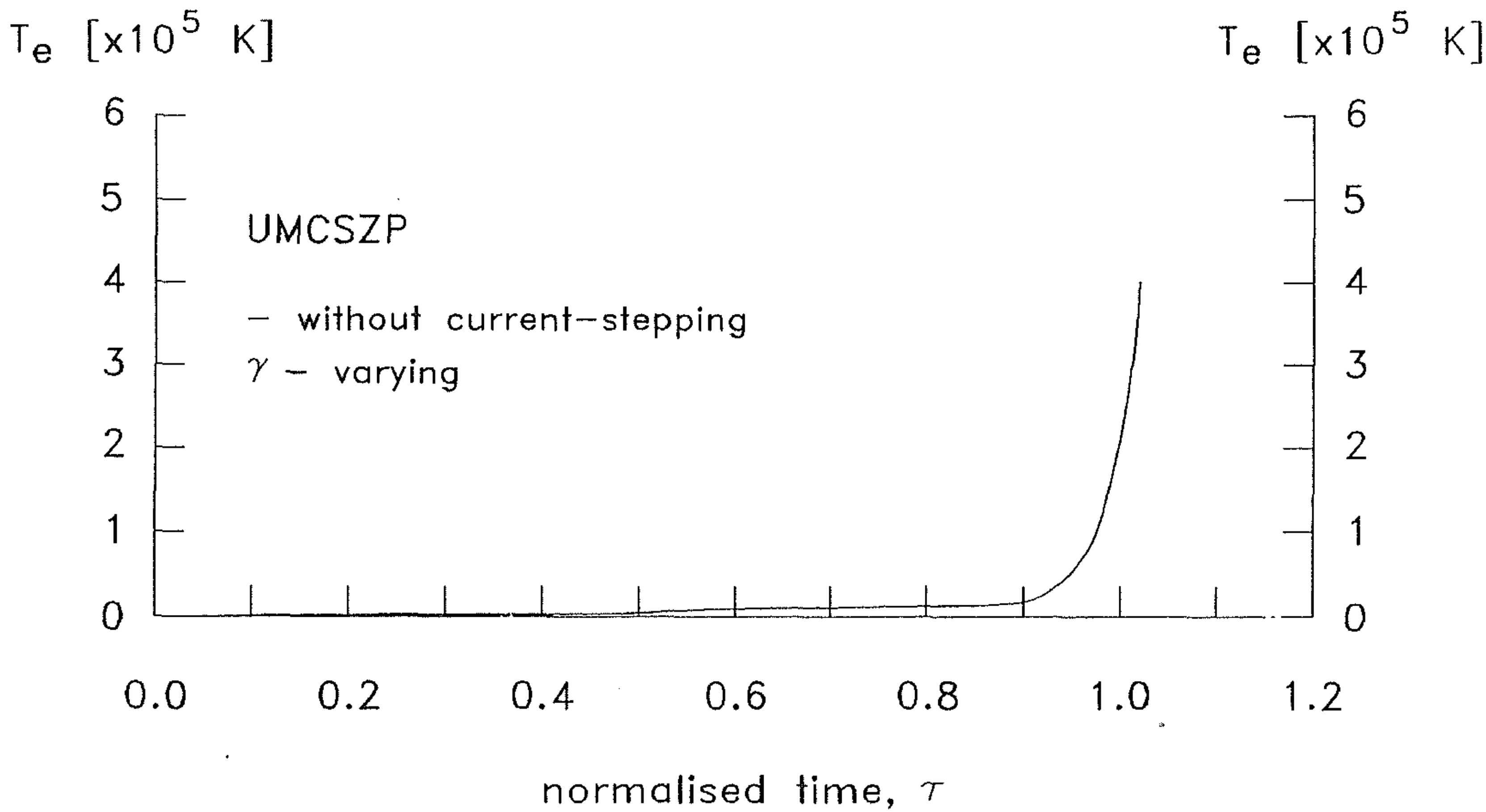


Fig. 5.7 The development of the temperature of the shock heated plasma of a hydrogen discharge at 0.8 mbar without current-stepping.

front hits the axis at $\tau = 1.02$. This increase in plasma temperature is due to the increase in the shock speed arising from the smaller effective specific heat ratio.

The deviation of γ_{eff} below 5/3 results in an increase in the piston speed. The shock speed does not vary much until at $\tau = 0.5$ when the smaller radius of the piston results in an increase in shock speed. These give rise to a thinner current sheath and a smaller final column of plasma. The radial piston and shock front trajectories are shown in Fig. 5.8. Those obtained from the γ-constant model are also shown in broken line for comparison. The final radius ratio predicted for a single stage Z-pinch by the γ-varying model is 0.11 while that of the γ-constant model is 0.24.

The corresponding plasma currents and voltages are shown in Fig. 5.9. The increase compression results in a reduced column and hence a higher plasma voltage is shown. The plasma inductance is shown in Fig. 5.10. The plasma inductance rises slowly at first until $\tau = 0.88$ at which time it begins to rise sharply towards 77 nH at $\tau = 1.02$.

The development of the plasma flow field-magnetic field coupling described by the magnetic Reynolds number, $\mathcal{R}_m$ during the compression is presented in Fig. 5.11. It is found that during the early period of the compression from $\tau = 0$ to $\tau = 0.9$, $\mathcal{R}_m$ is less than 5. It only rises to 14 at $\tau = 0.95$ and above 500 at $\tau = 1.02$ when the shock front hits the axis.

Thus the new model demonstrates that for the major duration of the compression, γ_{eff} oscillates around 1.14. It only rises towards 5/3 during the last 200 ns of the compression. In

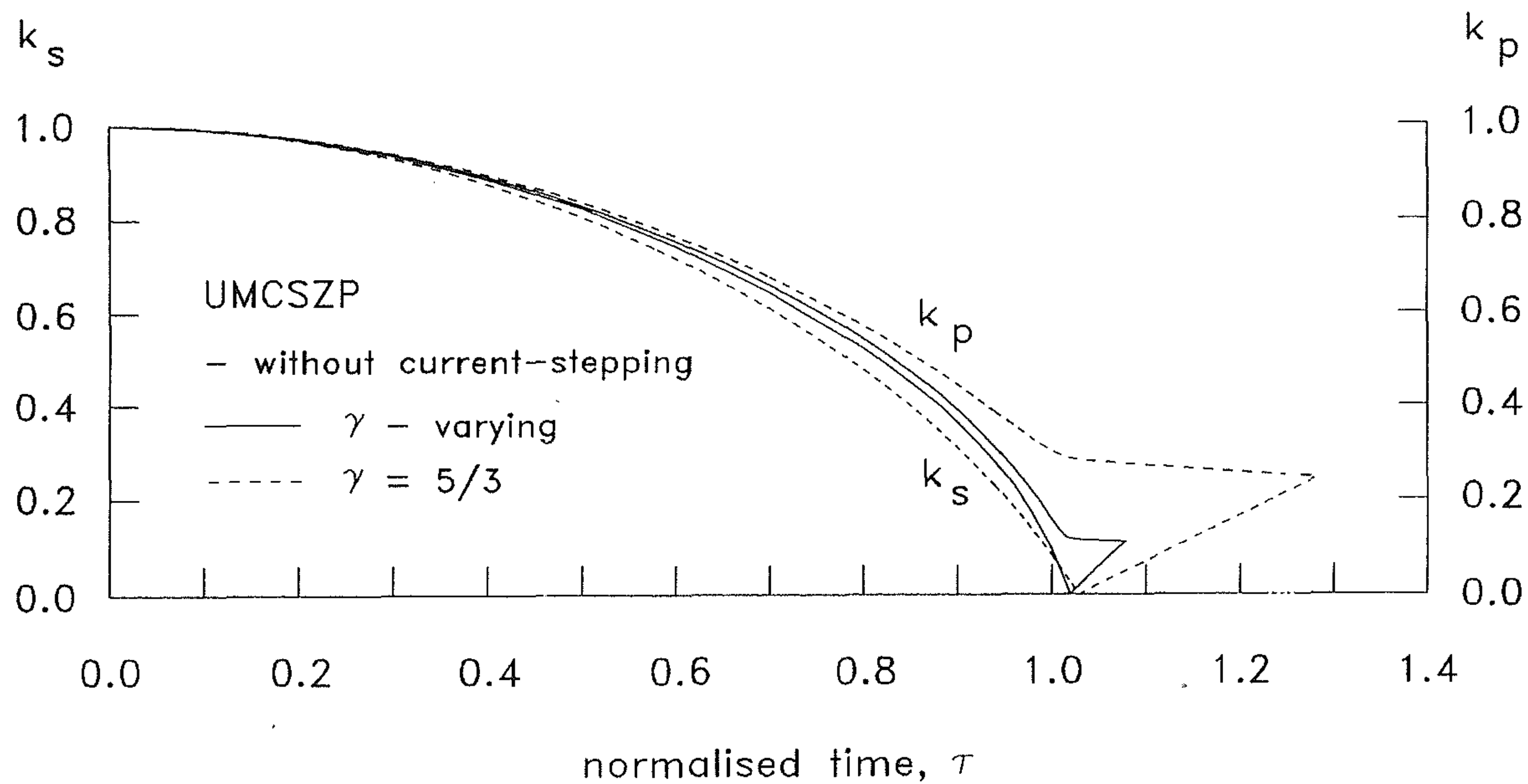


Fig. 5.8 The corresponding shock and piston trajectories of the hydrogen pinch without current-stepping predicted by the γ-varying model and the γ = 5/3 model. (P = 0.8 mbar hydrogen).

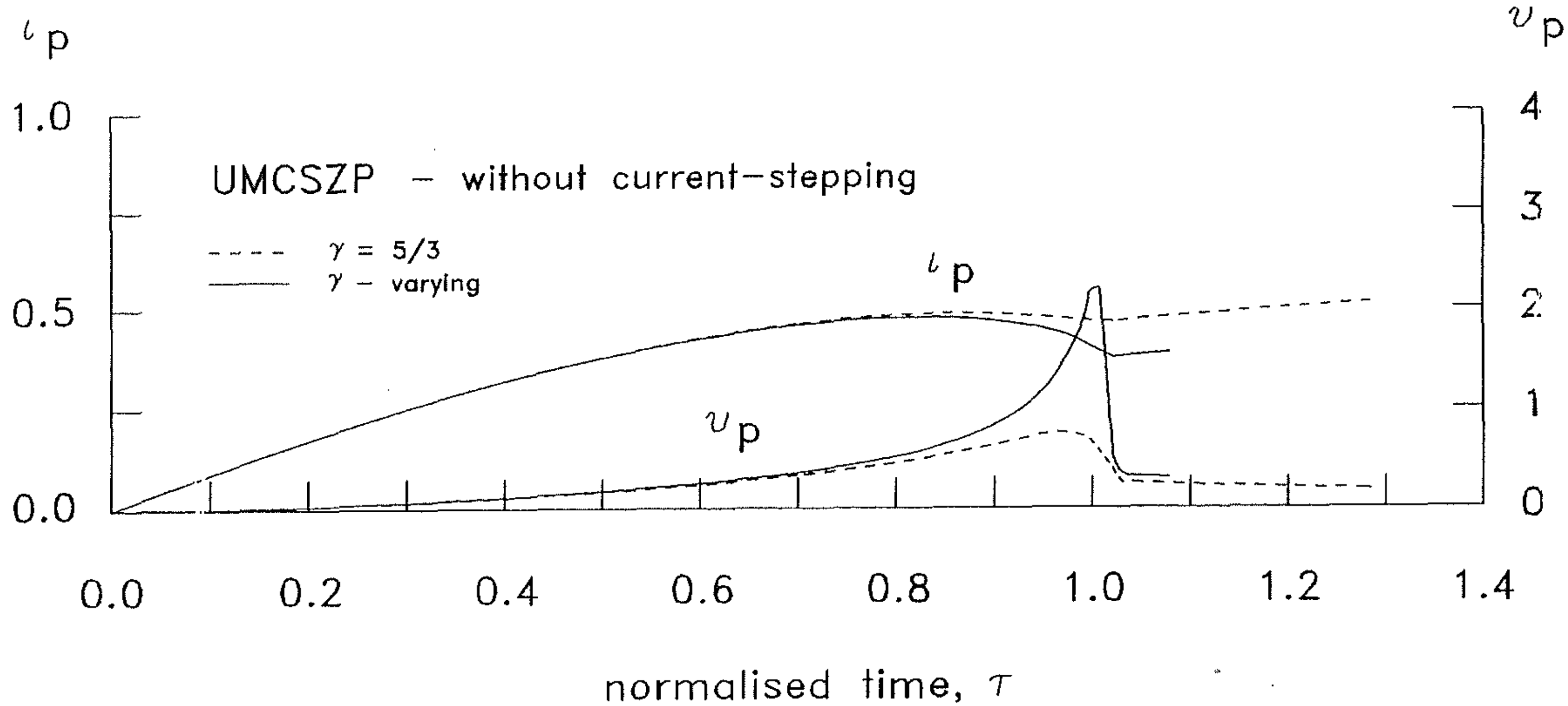


Fig. 5.9 The corresponding plasma current and voltage of the hydrogen pinch without current-stepping predicted by the γ-varying model and the $\gamma = 5/3$ model. (P = 0.8 mbar hydrogen).

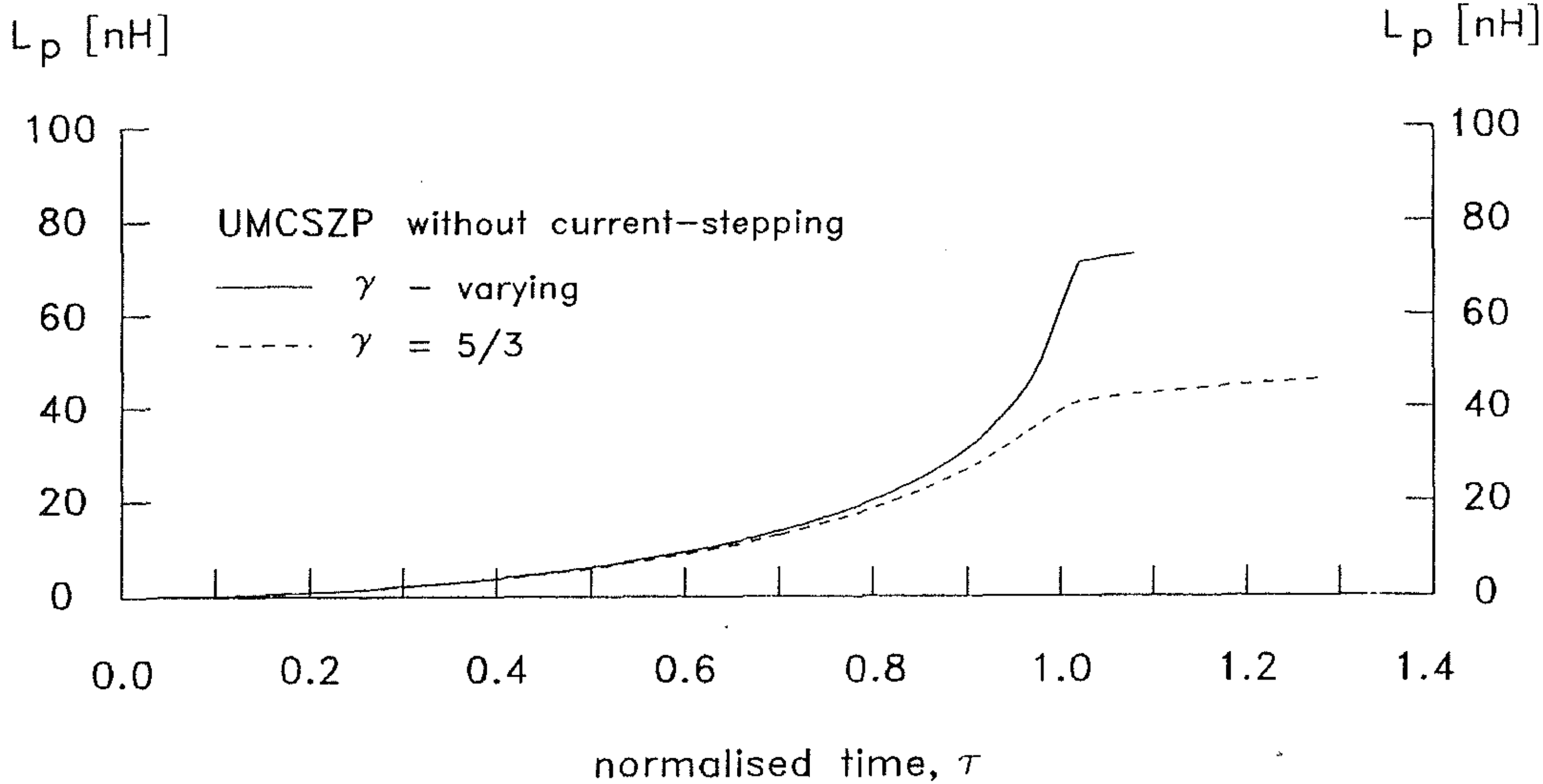


Fig. 5.10 The corresponding plasma inductance, L_p of the hydrogen pinch without current-stepping predicted by the γ - varying model and the $\gamma = 5/3$ model. (P = 0.8 mbar hydrogen).

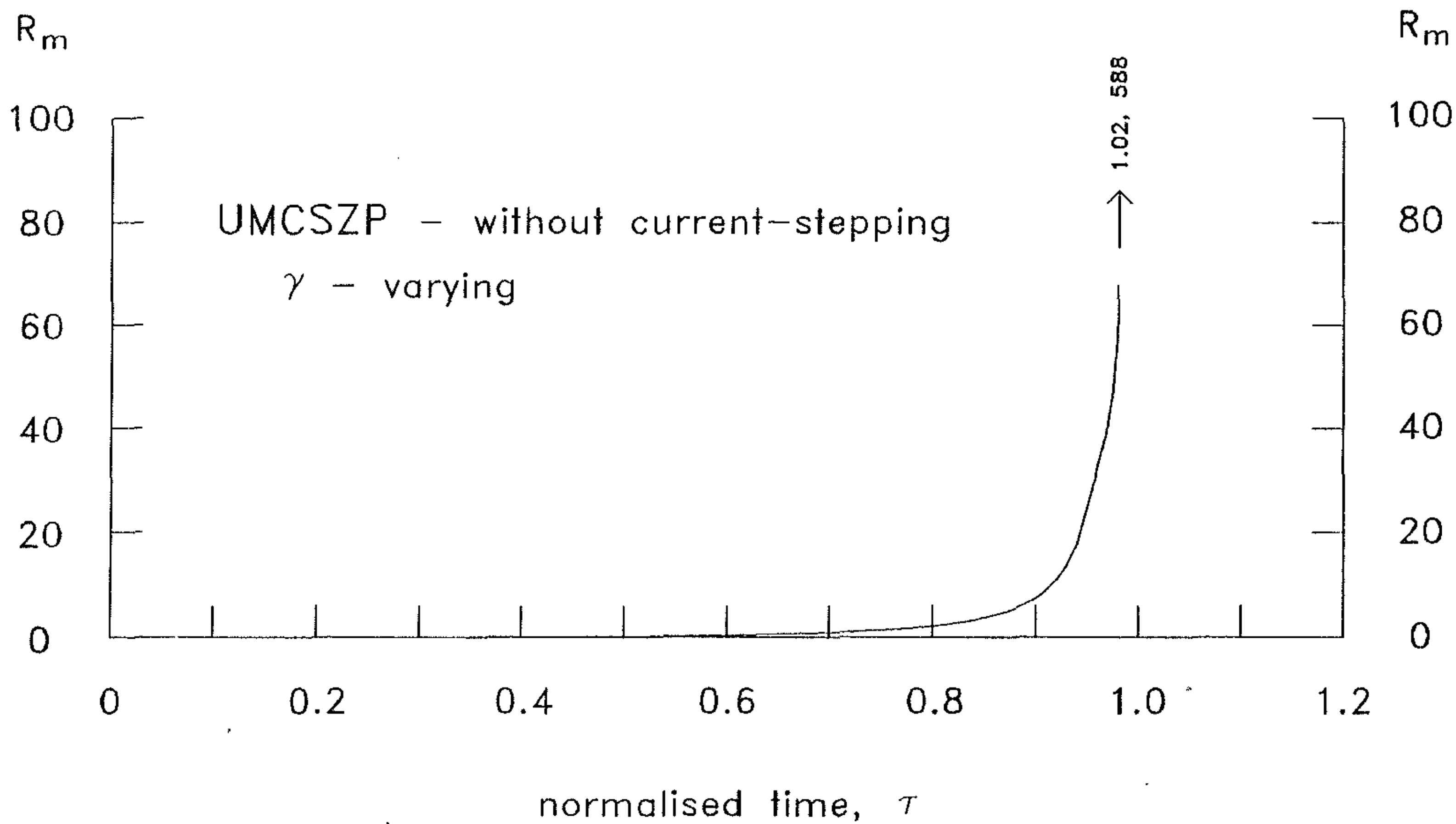


Fig. 5.11 The development of the magnetic Reynolds number, R_m of the hydrogen pinch without current-stepping predicted by the γ - varying model.

accordance with the effect of γ_{eff} on the compressibility of the plasma, a smaller final radius is predicted. The γ_{eff}-variation mechanism is found to reduce the final radius ratio of a single stage compressional Z-pinch of UMCSZP from 0.21 to 0.11 as shown in Fig.5.8. Also the plasma flow field-magnetic field coupling is poor until $\tau > 0.9$.

5.6.2. Current-stepped Z-pinch with varying specific heat ratio, $\gamma(t)$

Using the designed parameters of the UMCSZP as given in section 3.1, and with the Z-pinch operated at 0.8 mbar H_2 the effect of γ_{eff} on current stepping is investigated using the extended model. The numerical results show that the expected enhancement predicted by the γ-constant model upon current-stepping is insignificant here. The extended model indicates no significant enhancement in the current-stepped compression. The final radius ratio upon current-stepping remains approximately the same.

Results obtained with the extended current-stepped model for various times of switching of the second bank are shown in Fig. 5.12 to Fig. 5.18. Fig. 5.12 and Fig. 5.13 show the numerically simulated plasma current and voltage responsible for the shock and piston trajectories presented in Fig. 5.14 and Fig. 5.15. The variations of γ_{eff}, shock speed and plasma temperature are shown in Fig. 5.16, Fig. 5.17 and Fig. 5.18 respectively.

The results from the extended current-stepped model indicate no enhancement in current-stepping for UMCSZP system. This is because of the low value of γ_{eff} for the major duration of

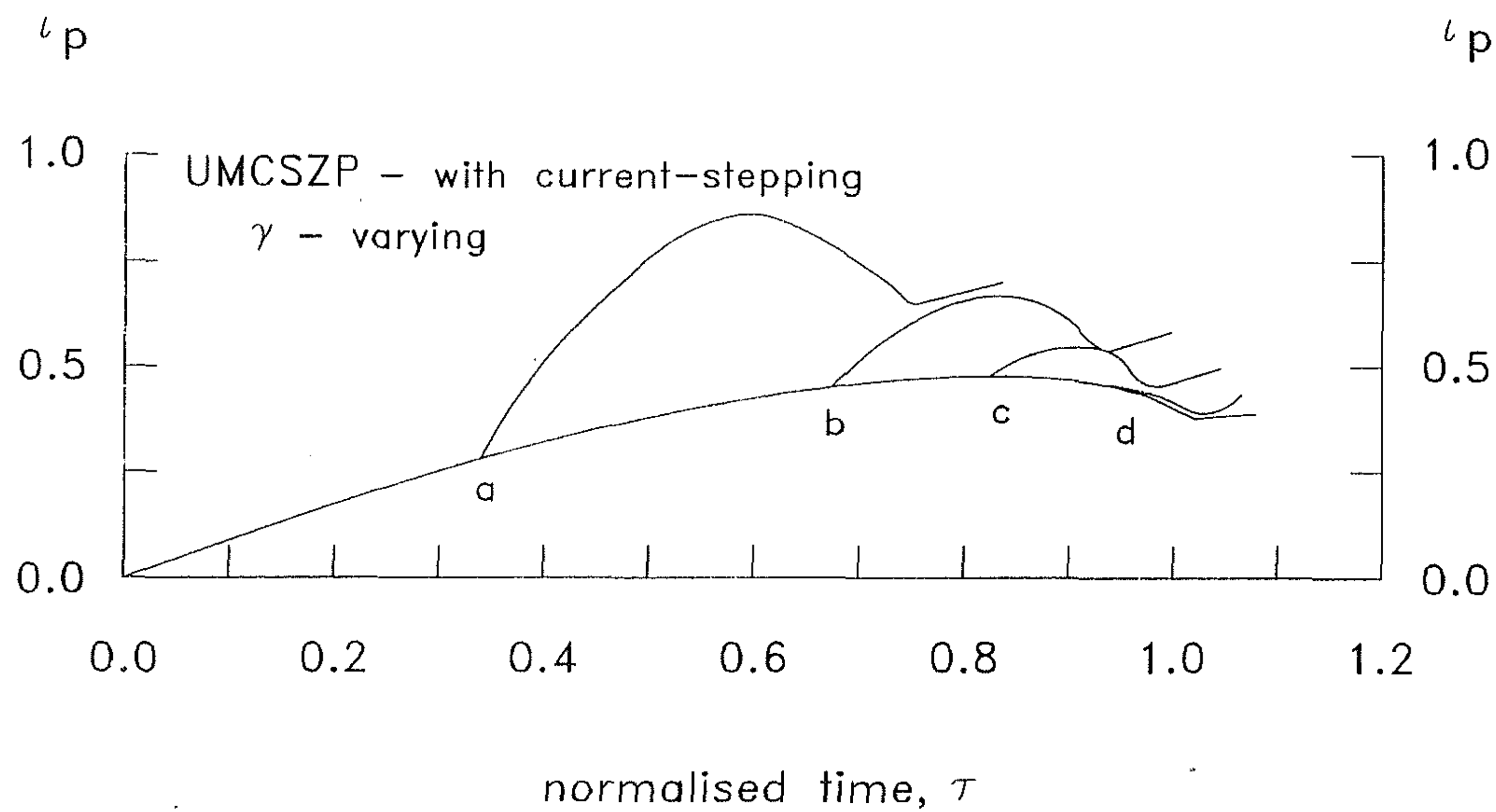


Fig. 5.12 Normalised plasma current versus time predicted by the γ - varying model for UMCSZP with various times of current-stepping (0.8 mbar hydrogen).

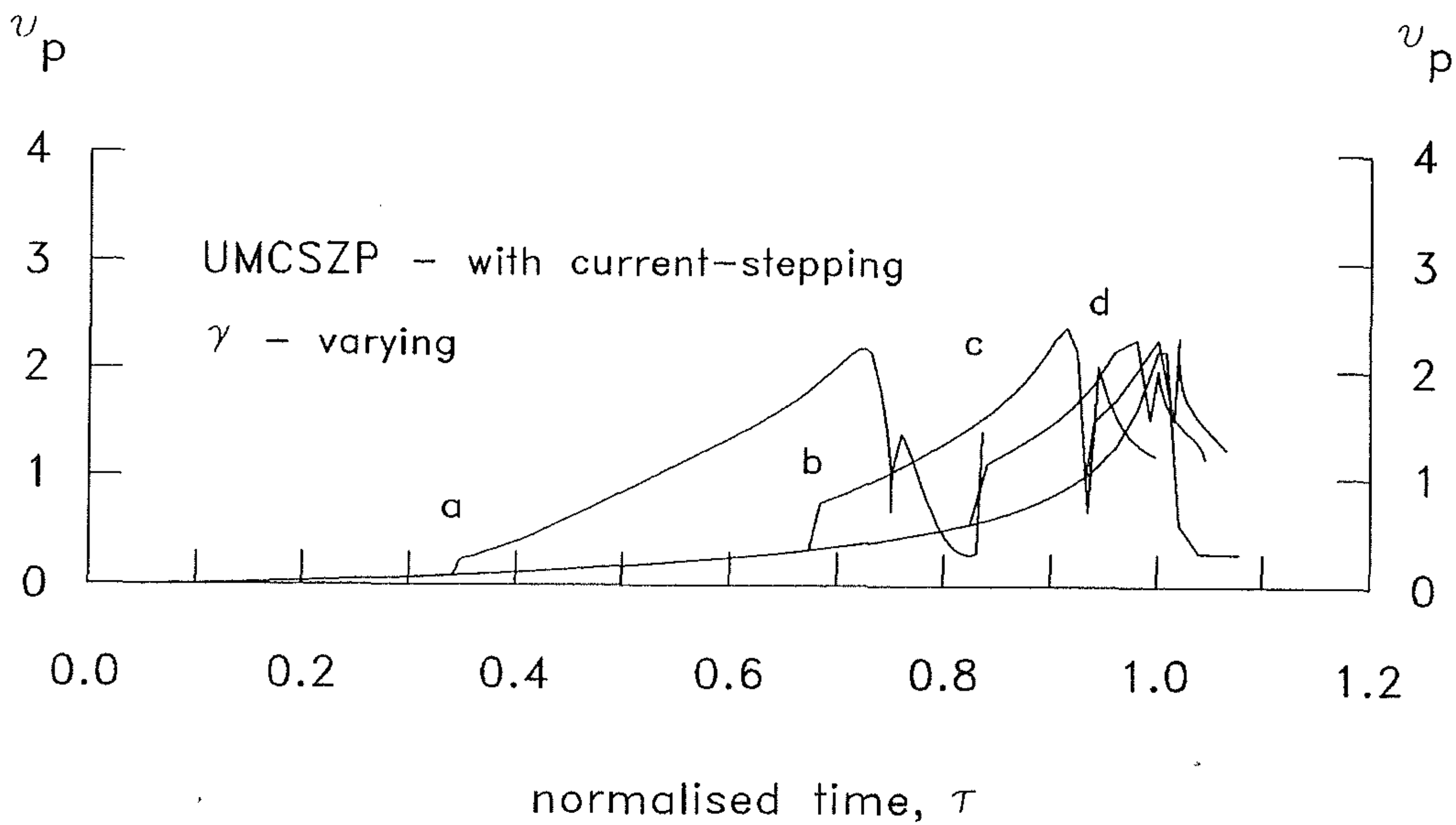


Fig. 5.13 Normalised plasma voltage versus time predicted by the γ- varying model for UMCSZP with various times of current-stepping (0.8 mbar hydrogen).

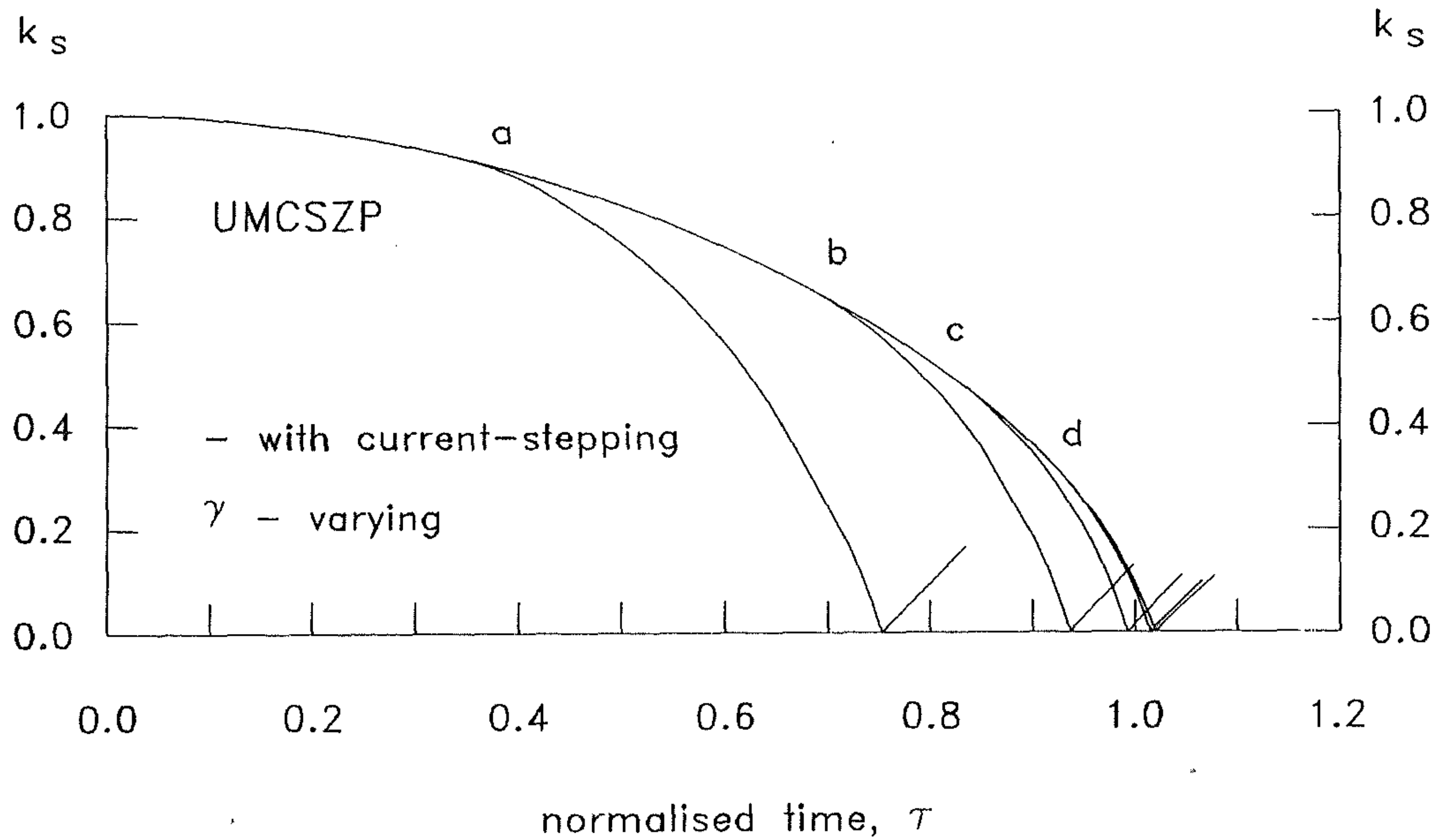


Fig. 5.14 Normalised shock trajectories predicted by the γ - varying model for UMCSZP with various times of current-stepping (0.8 mbar hydrogen).

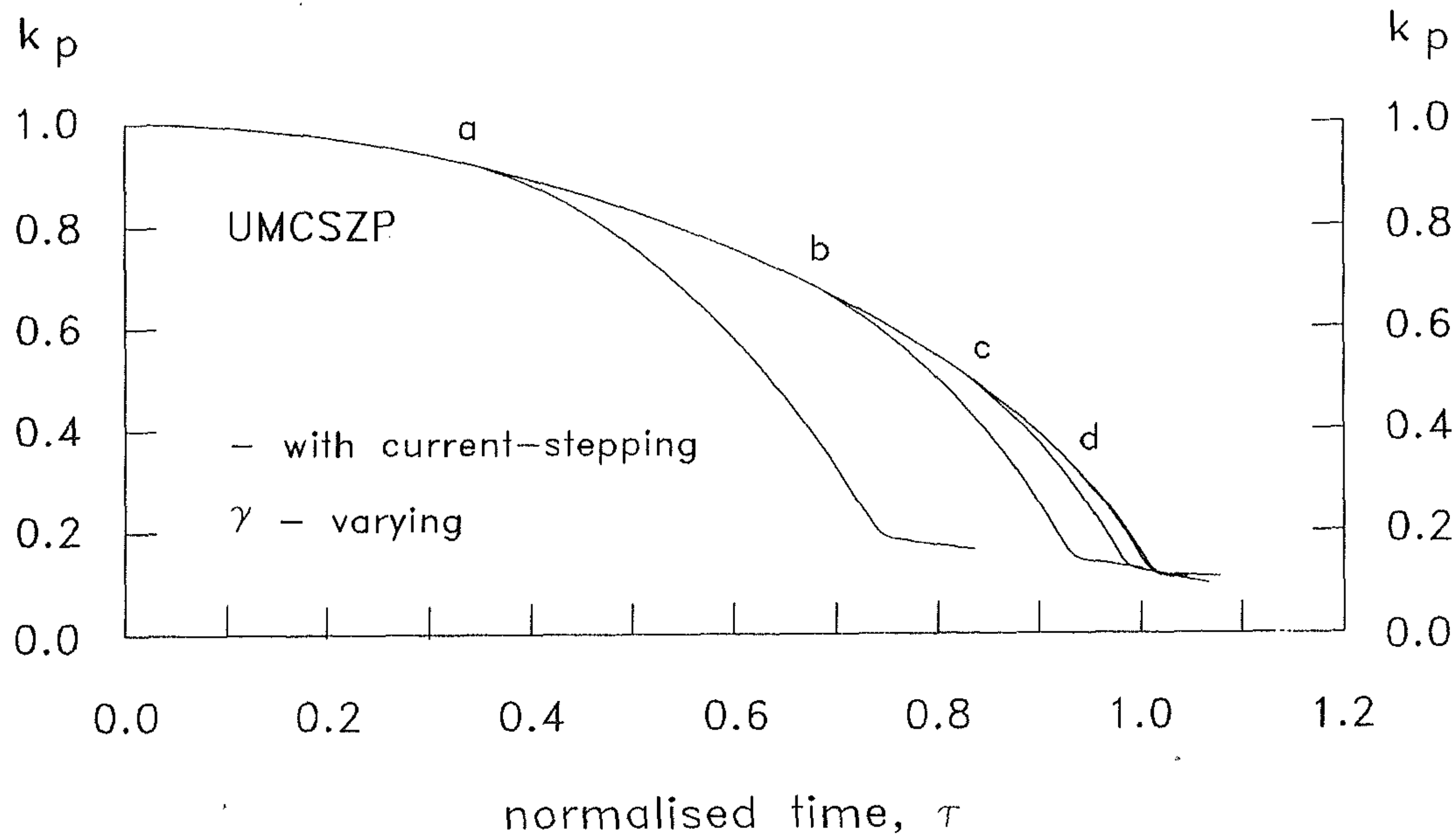


Fig. 5.15 Normalised piston trajectories predicted by the γ - varying model for UMCSZP with various times of current-stepping (0.8 mbar hydrogen).

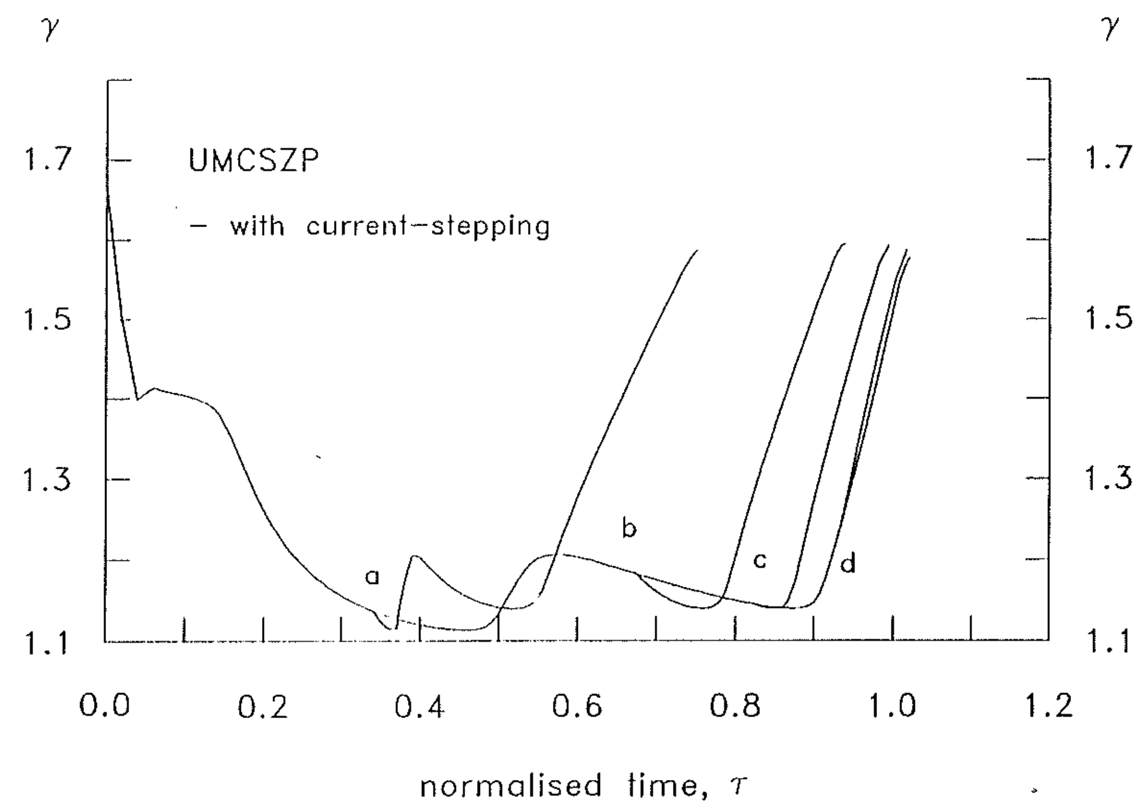


Fig. 5.16 The variation of γ_{eff} for UMCSZP predicted by the γ - varying model for various times of current-stepping (0.8 mbar hydrogen).

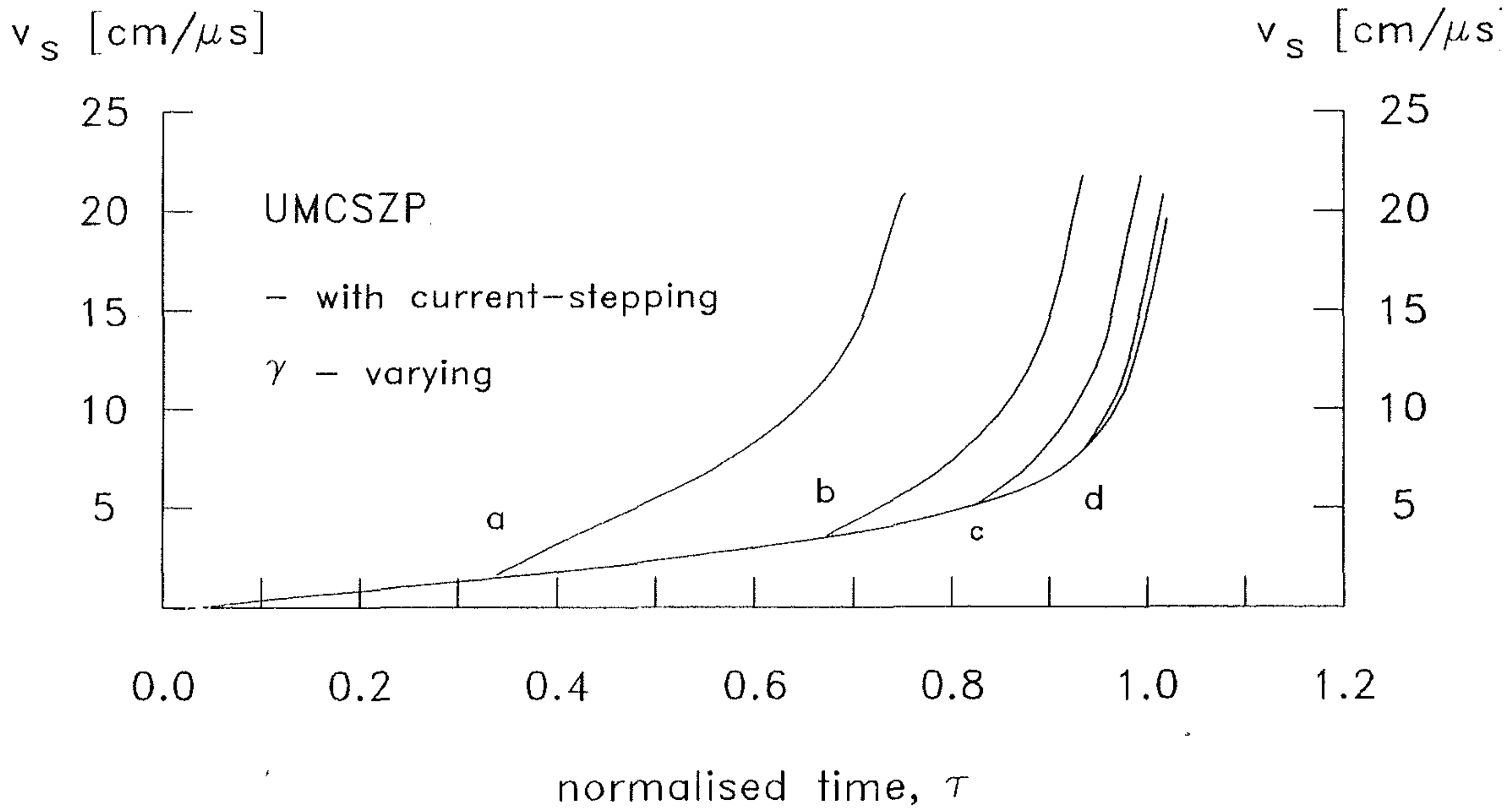


Fig. 5. 17 The variation of shock speed, v_s for UMCSZP predicted by the γ - varying model for various times of current-stepping (0.8 mbar hydrogen).

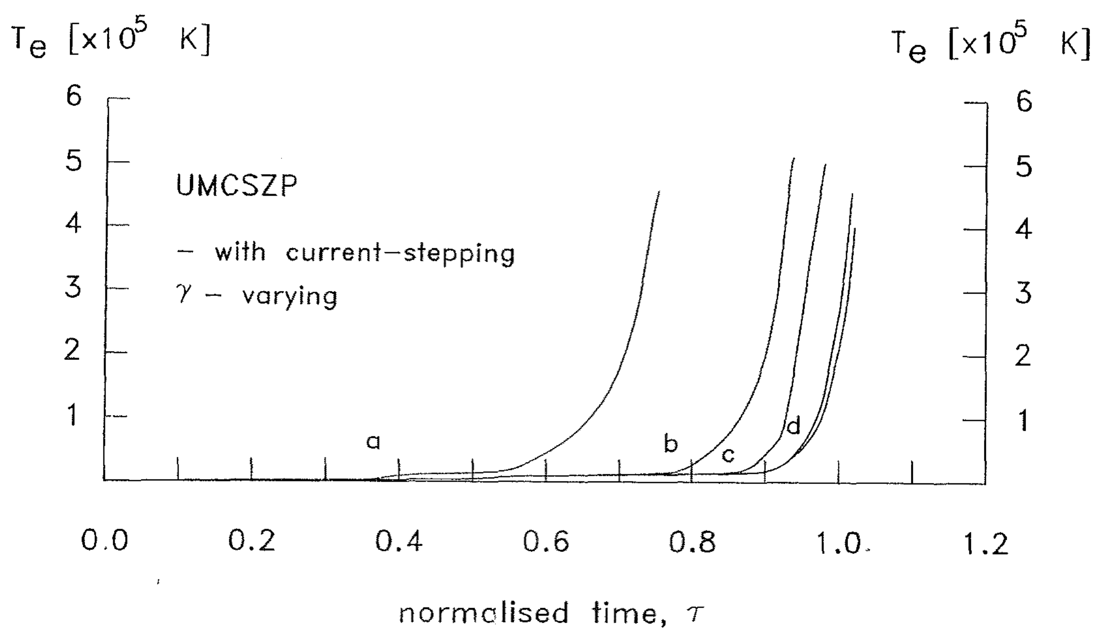


Fig. 5. 18 The variation of plasma temperature, T_e for UMCSZP predicted by the γ - varying model for various times of current-stepping (0.8 mbar hydrogen).

the first compression. Upon current-stepping γ_{eff} rises rapidly towards 5/3 and this has an effect of decreasing the compressibility of the plasma which opposes the enhanced compression of current-stepping. In the UMCSZP system these two factors cancel each other and hence no enhancement in compression is predicted. The minimum radius ratio for the various times of current-stepping predicted by the extended model is shown in Fig. 5.19. The corresponding enhancement factor is shown in Fig. 5.20.

5.7 Comparison with experimental results

The numerical results obtained from the extended current-stepped model are compared to the experimental results obtained. Fig. 5.21 shows the experimental and theoretical radial trajectories with different times of implementation of current stepping for the current-stepped pinch at 0.8 mbar hydrogen. The result for the first bank alone is also superimposed for comparison.

Experimental observations made have shown no enhancement in compression by current-stepping for the UMCSZP system. Instead, a slightly larger final radius is obtained upon current-stepping. This phenomena has been attributed to the observed pulling back effect of the current-sheath at the instant of attachment of the second current to the plasma - upon the switching of the second bank.

The extended current-stepped model also indicates no enhancement upon current-stepping for the UMCSZP system. This is because of the low compressing shock speed, below 5 cm/μs, for the major duration of the first compression. At this shock speed

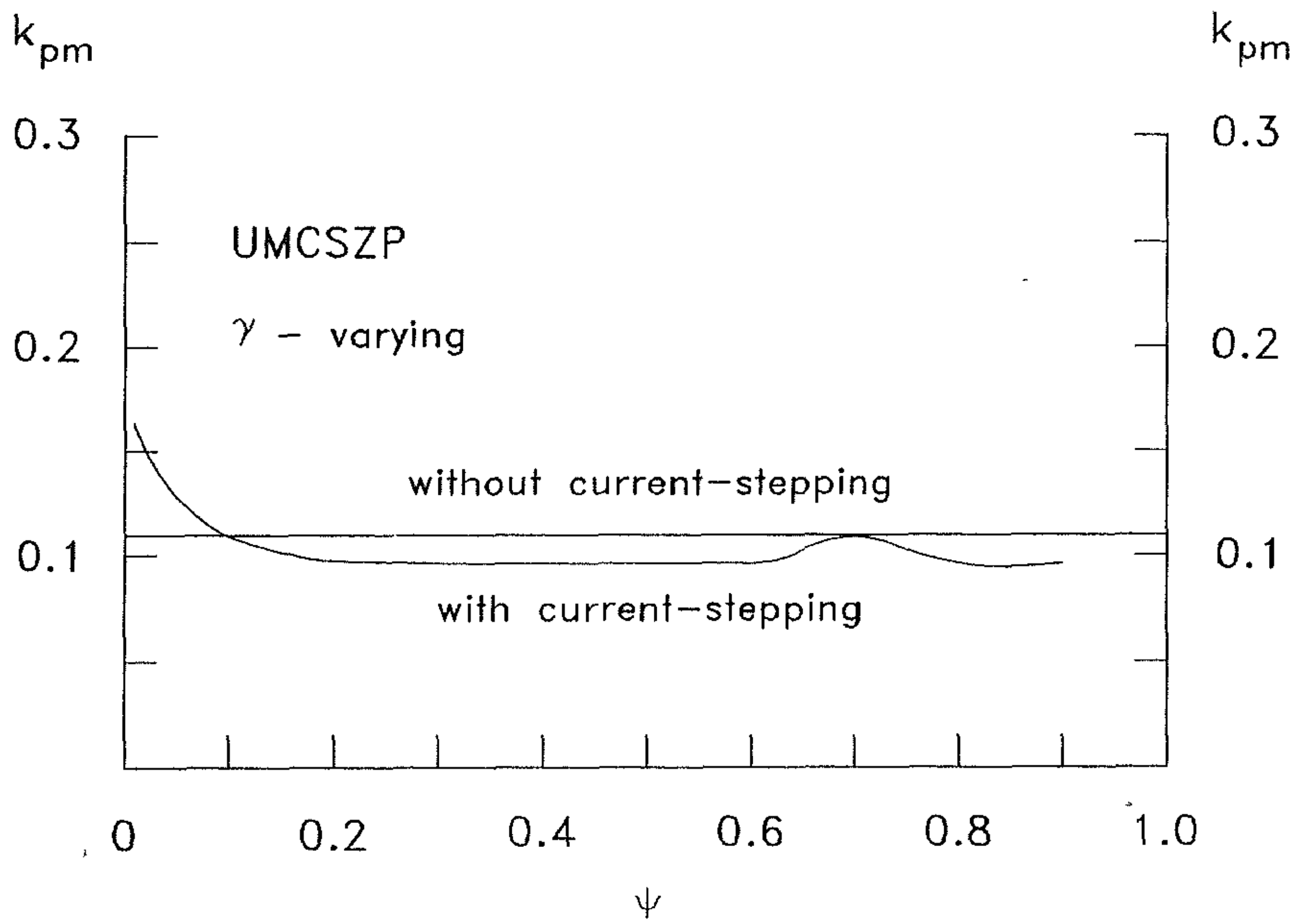


Fig. 5.19 Minimum radius ratio, k_{pm} versus ψ for UMCSZP predicted by γ - varying model.

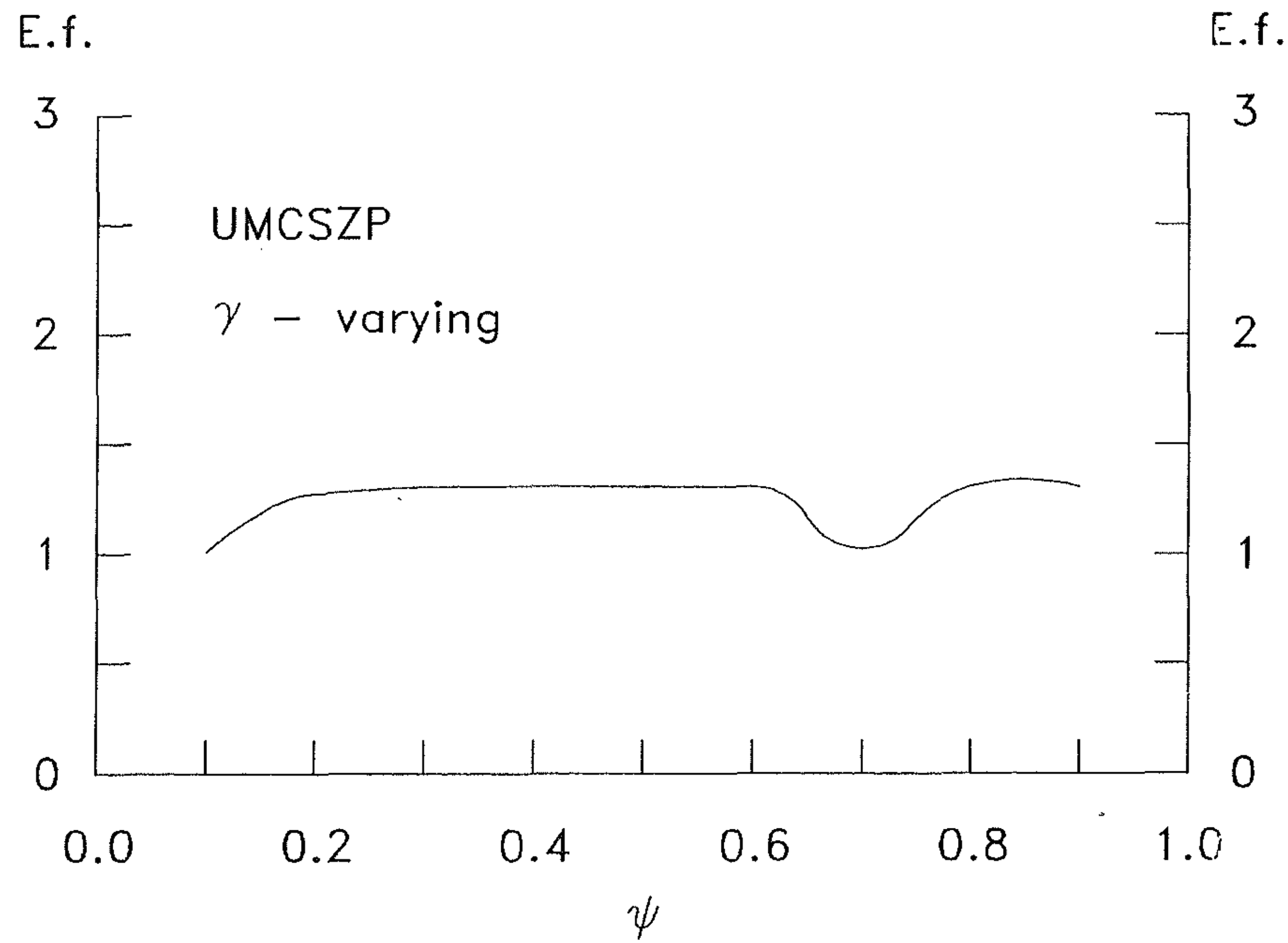


Fig. 5.20 The enhancement factor, E.f. versus ψ for UMCSZP predicted by the γ - varying model.

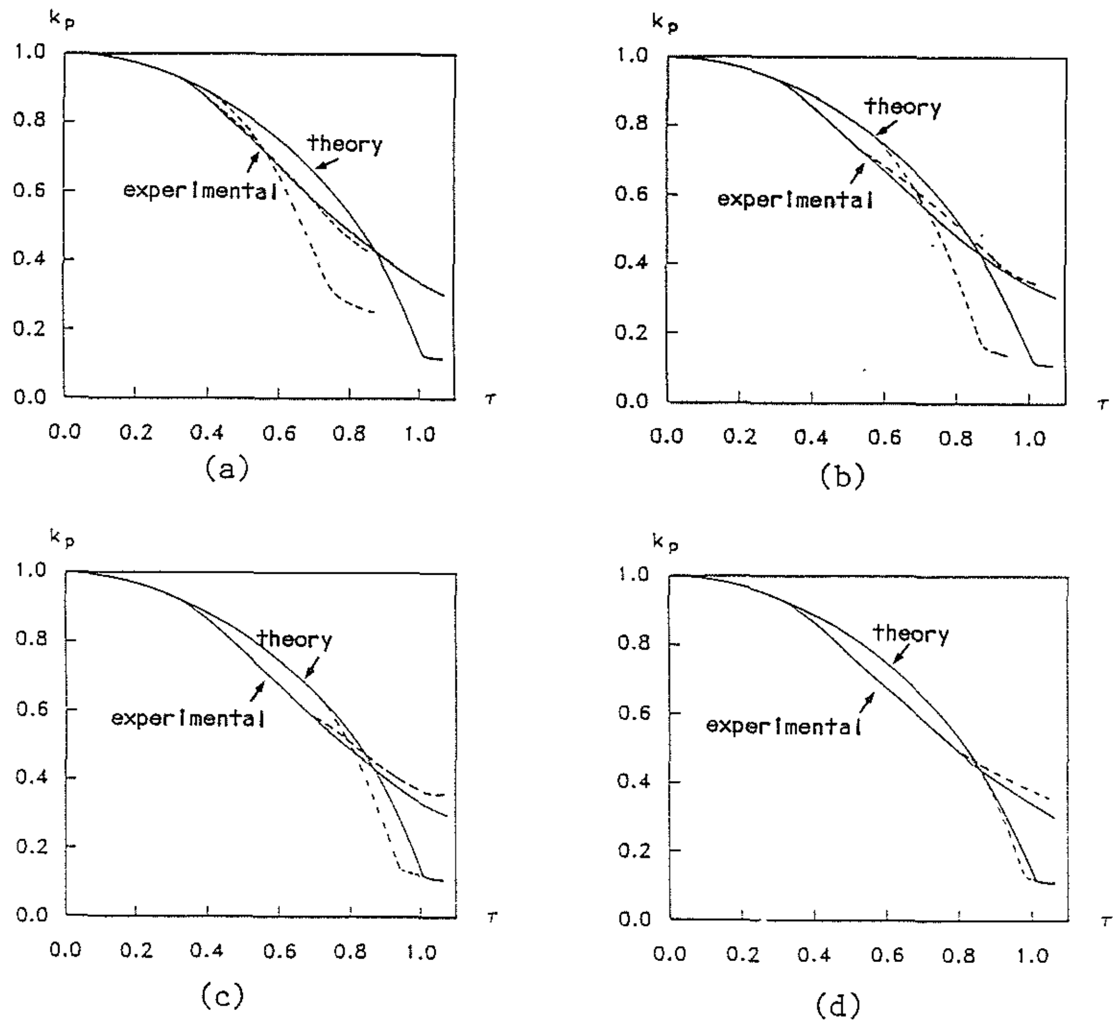


Fig. 5.21 Experimental and theoretical piston trajectories predicted by the γ - varying model of the UMCSZP.

(a) τ_{cs}= 0.42 (b) τ_{cs}= 0.56 (c) τ_{cs} = 0.70 (d) τ_{cs}= 0.79

the degree of ionisation achieved is about 50 % and γ_{eff} is about 1.2. At the instant of current-stepping γ_{eff} rises rapidly from 1.2 to 1.6, greatly decreasing the compressibility of the plasma. This opposes the compression enhancement effect of current-stepping. Hence for the UMCSZP system the enhancement effect of current-stepping is being masked by the γ_{eff}-effect.

The variations of γ_{eff} for single and current-stepped compressions at different times of current-stepping for UMCSZP operated at 0.8 mbar H_2 have been shown in Fig. 5.16. A sharp increase in γ_{eff} can be observed upon current-stepping for the various experimental times of current-stepping.

However, the theory predicts a smaller final radius than that observed experimentally for both without and with current-stepping. This may be due to the assumption adopted in the theory that current flows in a very thin current sheath of infinite conductivity. From the consideration of the plasma flow field-magnetic field coupling represented by the magnetic Reynolds number, $\mathcal{R}_m$ (refer Fig. 5.11) a rather diffusive and resistive plasma except during the final stage of compression is indicated in agreement with experimental deductions. Fig. 5.22 and 5.23 show the theoretical minimum radius ratio and the current-stepping enhancement factor predicted by both the γ-constant and the γ-varying model for UMCSZP at various times of switching of the second bank for a 0.8 mbar H_2 discharge. It appears that with the present set-up the effect of pinch enhancement cannot be observed due to the γ-masking effect.

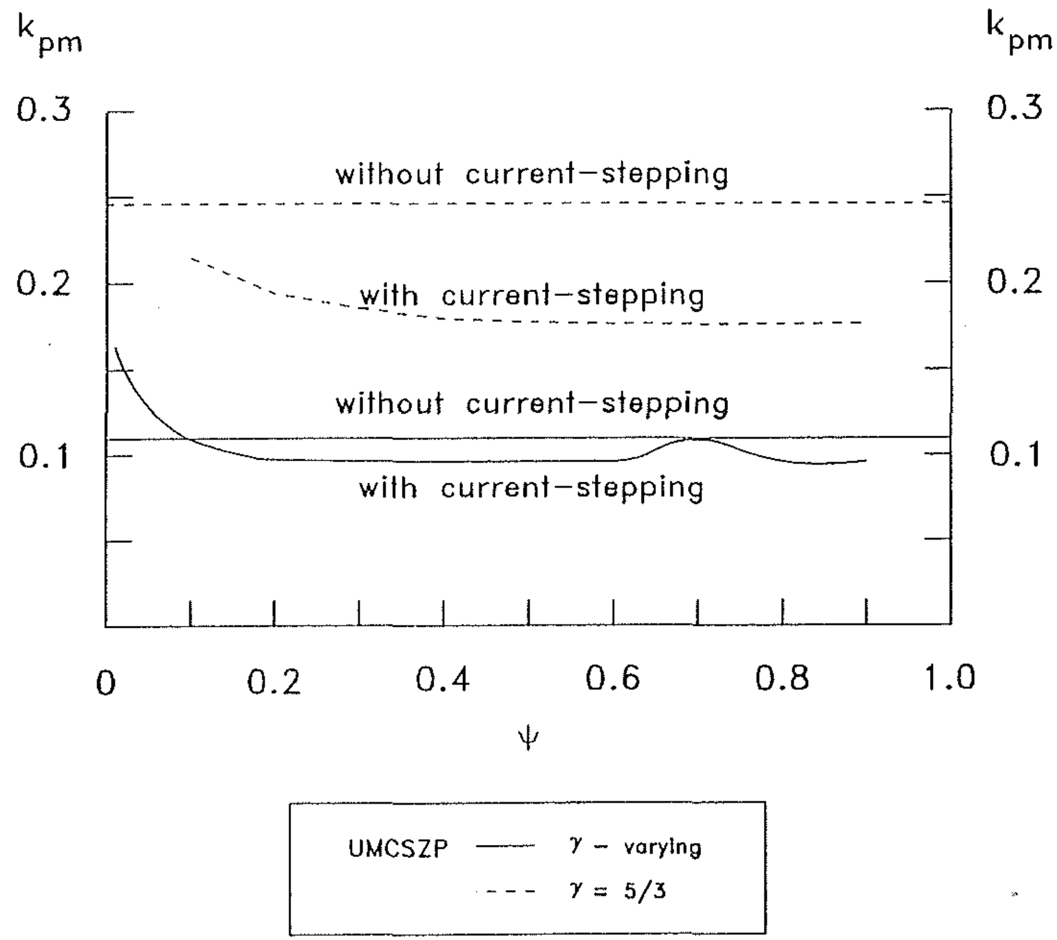


Fig. 5.22 Minimum radius ratio, k_{pm} versus ψ predicted by both the γ - varying and the $\gamma = 5/3$ model.

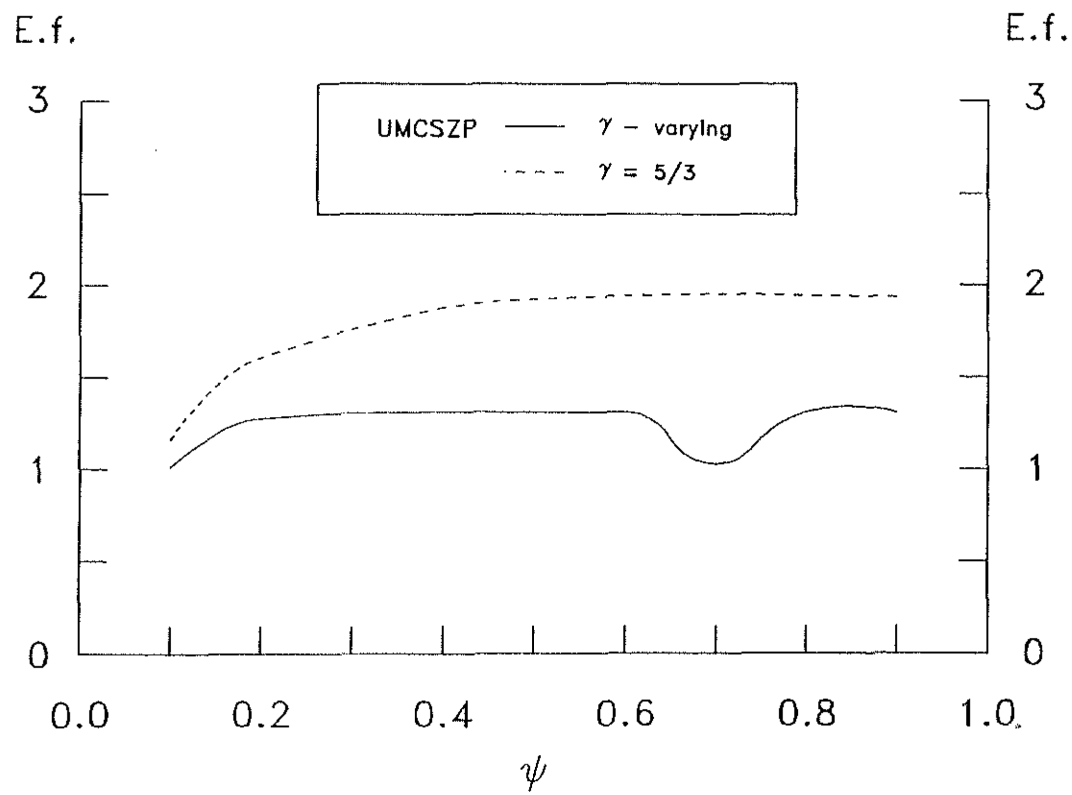


Fig. 5.23 Enhancement factor, E.f. versus ψ predicted by both the γ - varying and the $\gamma = 5/3$ model.

5.8 Discussions

The above studies indicate that to observe the enhancement effect of current-stepping, it is absolutely necessary to minimise the γ-masking effect. This may be done by reducing the sharp increase in γ_{eff} upon current-stepping. This is possible by having γ_{eff} just before current-stepping close to the ideal constant value of 5/3. However the approach of γ_{eff} close to 5/3 is a function of the shock speed. For higher shock speed, γ_{eff} will reach close to 5/3 in a shorter time thus allowing a wider range of time for successful operation of current-stepping. Enhancement of compression by current-stepping is possible if the average compressing shock speed is 10 cm/μs and above. With this average shock speed, the γ_{eff} at the instant of current-stepping is close to 1.6 since the shock speed then is at least 30 cm/μs. The introduction of current-stepping brings γ_{eff} to 1.6 and approaches 5/3. This reduced compression due to the increase in γ_{eff} produces only a small opposing effect of γ-masking and thus allows the enhancement effect of current-stepping to emerge. If the prediction of the theory is correct, successful current-stepping can then be observed.

A survey is made on UMCSZP to study possible regime for successful operation of current-stepping. Since the average shock speed is equal to the initial radius, r_o divided by the characteristic pinch time, t_p, the average compressing shock speed can only be increased by either increasing the initial pinch radius, r_o or reducing the characteristic pinch time, t_p.

From eqn [3.1], the characteristic pinch time, t_p is observed to be dependent on the particle density, ρ_o and the current density, (I_o / r_o^2). The characteristic pinch speed is

$$\frac{r_o}{t_p} \propto \frac{\left[\dfrac{I_o}{r_o}\right]}{\rho_o^{1/2}} \qquad [3.1]$$

For a matched condition, t_p is approximately equal to t_o. However since t_o is a fixed quantity in UMCSZP, to increase the average shock speed we have to increase the initial radius of the pinch, r_o. Therefore to increase the average compressing shock speed from $\simeq$ (3 cm/μs) of the present system to about 10 cm/μs or higher would require an increase in the initial radius by at least 3 times. The initial density will have to be reduced by more than 81 times, see equation [3.1]. The minimum requirement of an initial radius of 20 cm and an initial operating pressure of 10^{-2} mbar is not practical with our present system.

Further it is also predicted by the extended model that there is an increase in the plasma inductance and more importantly the plasma dynamic resistance associated with the increased speed.

From the consideration of energy balance, the electrical power input into the plasma of inductance, $L_p(t)$ as given by

$$P_{input} = I_p V_p = I_p \dot{I}_p L_p + I_p^2 \dot{L}_p \qquad [5.17]$$

exceeds that stored in the plasma associated with the B-field generated

$$P_{stored} = \frac{dW}{dt} = \dot{I}_p L_p + \frac{1}{2} I_p^2 \dot{L}_p \qquad [5.18]$$

by $\frac{1}{2}I_p^2\dot{L}_p$, which is treated as an ohmic loss associated with a dynamic plasma resistance $R_D(t)$ equal to $\frac{1}{2}\dot{L}_p$.

The dynamic plasma resistance $R_D(t)$ can be expressed in terms of the physical dimensions of the pinched column and the speed of the compressing piston. This is given below as

$$R_D(t) = 1 \times 10^{-7} \frac{\ell}{r_p} \dot{r}_p \qquad (\Omega) \qquad [5.19]$$

In the case of our present pinch of length *15 cm*,

$$R_p(t) = 15 \times 10^{-3} \frac{\dot{r}_p}{r_p} \qquad (\Omega) \qquad [5.20]$$

where $\dot{r}_p$ is in cm/μs and r_p is in cm. For a piston speed of $\simeq$ *(20 cm/μs)* and r_p of $\simeq$ *2 cm*, the plasma dynamic resistance is *0.15* Ω.

The extended model predicts a faster compression and a smaller final column of plasma. These lead to a higher dynamic resistance in the plasma. The increase in the plasma dynamic resistance is shown in Fig. 5.24. If the current-step is executed at τ between 0.8 and 1.0, the extended model predicts a dynamic plasma resistance in the range of 15 to 140 mΩ. This indicates the requirement of high impedance current sources to ensure that upon the implementation of current-stepping, most of the current from the second current-stepped source flows through the plasma column and not through the first source while that of the first source continues to supply a major portion of its current to the

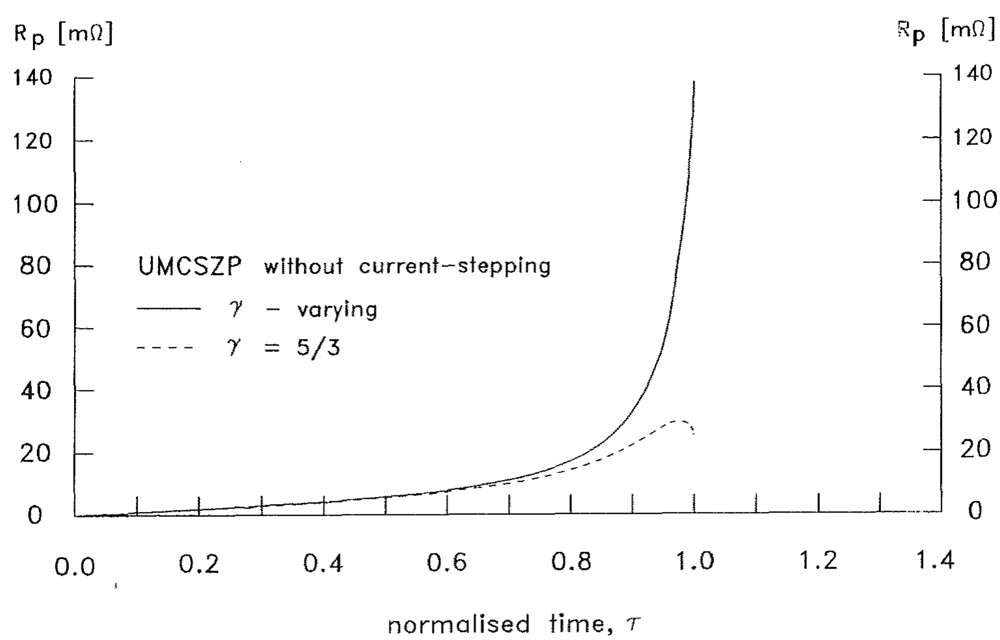


Fig. 5.24 The plasma dynamic resistance, R_p versus normalised time for a 0.8 mbar hydrogen pinch without current-stepping predicted by the γ - varying and the γ = 5/3 model.

pinching column of plasma and not to the second bank. The present current-stepped current sources of LC discharge systems are of low impedance in the region of 20 to 40 mΩ and hence they are not suitable.

5.9 Conclusions

The extended current-stepped model shows that it is not possible to observe the enhancement effect of current-stepping with the present UMCSZP system because of the γ-masking effect. However, if the γ-masking effect is reduced sufficiently, the enhancement effect of current-stepping may dominate over the γ-masking effect. This is possible if the average compressing shock speed of the first compression is increased to beyond 10 cm/μs corresponding to a peak speed of $\geq$ 30 cm/μs just before current-stepping. Then, at the instant of current-stepping, γ_{eff} would have reached a value of above 1.5. Hence the current-stepped results in only a small increase in γ_{eff}.

With the present UMCSZP it is not possible to increase the average shock speed towards 10 cm/μs. Also the current sources are underdamped LCR discharge systems with rather low impedances. These low impedance systems become unsuitable since the new model predicts an increase in the plasma dynamic resistance brought about by the higher compressing speed and the smaller radius at the instant of current-stepping. At the instant of current-stepping, the plasma dynamic resistance is comparable to the source impedances and thus a substantial amount of current is lost through the sources. This reduces the increase in the plasma current upon current-stepping.

A summary of the theoretical and experimental results obtained for UMCSZP operating at 0.8 mbar H_2 is shown in Table 5.1. It is observed that firstly, the γ-effect increases the compressibility of the first compression resulting in a final radius ratio of 0.11 compared to that of the γ-constant prediction of 0.24. Secondly, upon current-stepping, the γ-effect causes the compressibility of the plasma to decrease due to the increase in γ_{eff} and hence opposes the effect of current-stepping. Therefore no enhancement is predicted by the γ-varying model. Upon current stepping the final radius ratio predicted by the γ-varying model remains around 0.11 while for the γ-constant model, the final radius ratio decreases from 0.24 to a minimum of 0.18.

Experimentally, the final radius ratios obtained without current-stepping is 0.30 and with current-stepping is about 0.35. It shows no enhancement in compression upon current-stepping. Moreover, during the attachment of the second bank current with the plasma current sheath upon current-stepping, a pulling back effect of the current sheath is observed . This has been attributed to the poor plasma flow field-magnetic field coupling of the plasma current sheath.

A new system is necessary to observe the enhancement effect in current-stepping. The system must operate in a regime where the γ-masking effect is greatly reduced if not eliminated, that is with an average compressing shock speed of the first compression beyond 10 cm/μs corresponding to a peak shock speed of $\geq$ 30 cm/μs just before current-stepping. With this, the γ-masking effect will be greatly reduced since at the instant of current-stepping the shock speed will be above 30 cm/μs and γ_{eff}

		Theoretical (energy balance slug model		Experimental (from IV signals – corrected)
		k_{pm}		k_{pm}
		$\gamma = 5/3$	γ – varying	
First bank alone		0.24	0.11	0.30
	t_{cs} (μs)	k_{pm-cs}		k_{pm-cs}
		$\gamma = 5/3$	γ – varying	
Current-stepped Z-pinch	0.9	0.21	0.14	0.40
	1.2	0.19	0.11	0.34
	1.5	0.19	0.11	0.36
	1.7	0.18	0.10	0.35
Density compression enhancement factor, $\left[\frac{k_{pm}}{k_{pm-cs}}\right]^2$		(1.3 – 1.8)	–	–

Table 5.1 Theoretical ($\gamma = 5/3$ and γ - varying) and experimental minimum radius ratios, k_{pm} for the gas compressional Z-pinch and k_{pm-cs} for the current-stepped Z-pinch of UMCSZP.

close to 5/3. Hence, the rise in γ_{eff} upon current-stepping is small and the γ-masking effect minimal, allowing the enhancement effect of current-stepping to emerge. With this improvement, the plasma will be heated to a higher temperature and hence its resistivity will also be reduced, thus facilitating the attachment to the current sheath upon current stepping.

Furthermore, high impedance sources are required as the plasma dynamic resistance increases with increasing compressing shock speed and reduced pinch column. The requirement of high impedance current sources leads to the necessity of having voltage sources above 100 kV in order to deliver enough current density to drive the pinch. Thus a high impedance and high voltage source such as the Marx and water-line systems are required. Since the second bank must be faster than the first for current-stepping, a Marx system is chosen for the first bank and a pulse charged water-line for the second stepped current source. A new system comprising of a 3-stage Marx and a pulse charged water-line system is designed to study the enhancement in the compression by a current-step based on the extended current-stepped model. This will be presented in Chapter 6.

CHAPTER 6

The Design of the Marx-Water Line Combination Current-Stepped Z-Pinch, [MLCCSZP]

6.1 Introduction

A Marx-line combination system consisting of a Marx for the first compression and a water-line *[Miller, R.B., 1982]* pulse charged by another Marx for the second or the current-stepped compression, named the Marx-water-line combination current-stepped Z-pinch, MLCCSZP is designed based on the improved energy balance model. A minimum average shock speed of 10 cm/μs with a peak shock speed of 30 cm/μs at the instant of current-stepping for the first compression is suitable for the enhanced compression of the current-step to be visible.

6.2 The design of MLCCSZP

6.2.1 Design Procedures

From the prediction of the extended model, an average compressing shock speed of at least 10 cm/μs for the first compression with a peak shock speed of 30 cm/μs at the instant of current-stepping is required for the enhancement of the compression by the current-step to be observed.

From equation [3.1], the average shock speed is given by

$$V_s = \frac{r_o}{t_p} = \left[\frac{\mu_o(\gamma+1)}{\rho_o} \right]^{1/2} \frac{I_o}{4\pi r_o^2} \qquad [6.1]$$

The first design consideration is to fix the average shock speed. The next consideration is to fix the parameters of the first current source. The requirement of high impedance source in the range of 1 to 2Ω has to be considered because of the high plasma dynamic resistance resulting from the higher compressing speed and the smaller final radius produced. However we limit the maximum operating voltage to 150 kV in order to minimise high voltage breakdown problems. A source of impedance 1Ω at 150 kV is deemed suitable to provide a current rising to 150 kA in 900 ns. The initial radius is fixed at 6 cm. Then with ρ_o = 1.1 x 10 $^{-5}$ kg m^{-3} or an operating pressure of 0.14 mbar, we design t_p equal to 0.6 μs. An average compressing shock speed of 10 cm/μs peaking to about 30 cm/μs at the time of current-stepping can thus be obtained for the compression of the first bank.

Next, the second current source to provide the current-step has to be at least two times faster than that of the first current source. Since the duration of the pinch upon current-stepping is ≃ 100 ns, a source that delivers at least 50 kA in 100 ns is required. A pulse charged water-line current source is chosen for this purpose. The 1.7 Ω water-line is 1.5 m in length with an inner diameter of 15 cm and outer diameter of 20 cm. When charged to 150 kV it can deliver about 52 kA in 100 ns for a matched load.

Further the ion-ion equilibrium time *[Lee, S., 1985b]* given by,

$$t_{i-i} = 1.1 \times 10^7 \frac{T^{3/2}}{n_i \ln(\Lambda)} \qquad [6.2]$$

where $n_i = \Gamma n_o$, is the plasma density; Γ and n_o are the compression ratio, $(\gamma+1)/(\gamma-1)$ and the initial density respectively, and

Λ is the ratio of the Debye length to the 90^0 e-i impact parameter; $\ln(\Lambda) = 10$.

is checked to ensure that the ion-ion equilibrium time is sufficiently small for shocked heating to be effective.

With these considerations the following design parameters are arrived at for the MLCCSZP :

[i] First bank - A 3-stage Marx :

$r_o = 6\ cm$	$\ell = 5\ cm$	$P_o = 0.14\ mbar$
$V_o = 150\ kV$	$L_o = 600\ nH$	$C_o = 600\ nF$
$t_o = 0.6\ \mu s$	$Z = 1\ \Omega$	$I_o = 150\ kA$

$dI_o/dt = 2.5 \times 10^{11}\ A/s$

[ii] Second bank - A Marx-pulse-charged water-line :

Water-line:

$V_o = 150\ kV$	$L_0 = 90\ nH$	$C_o = 30\ nF$
$l_W = 150\ cm$	$D_W = 20\ cm$	$d_W = 15\ cm$
$t_o = 52\ ns$	$Z_W = 1.7\ \Omega$	$I_o = 88\ kA$

$dI_o/dt = 2 \times 10^{12}\ A/s$

For the first bank, a 3-stage Marx with 50 kV per stage is designed to provide the first current source. Another 3-stage Marx also with 50 kV per stage but of lower energy is designed to pulse charge the 150 kV water-line.

6.3 Design of the Marx system

6.3.1 Marx1 - First current source of MLCCSZP

The Marx1 constituting the first bank consists of 3 units of Maxwell capacitors, each *1.8 μF, 60 kV* maximum with an equivalent series inductance of *20 nH*. These capacitors are charged in parallel and discharged in series upon the switching of the first capacitor. They are arranged lengthwise with the anode facing the cathode and placed inside a rectangular mild-steel casing which is grounded. Each of these capacitors rests on 6 pieces of insulator blocks. The inside of the casing is lined with layers of mylar sandwiched between polyethylene sheets to provide the necessary insulation between the capacitors and the earth casing. Each of the capacitors is also similarly insulated. A coaxial configuration is chosen here since a rather low inductance *(600 nH)* is required. A schematic of the capacitors arrangement is shown in Fig.6.1. The first switch, S1 uses a swinging cascade spark gap operating at atmospheric pressure. The second and third switches are also atmospheric spark gaps operating in the overvoltage mode. The output from the third gap is connected to the anode of the Z-pinch. High power $CuSO_4$ resistors are used as earthing resistors.

6.3.2 Marx2 - For pulse-charging of the water-line

The design of Marx2 is an improvement over that of Marx1 and it is also more compact. It makes use of 3 units of Maxwell capacitors each of 0.33 *μF, 100 kV* maximum and with an equivalent series inductance of *25 nH*. These capacitors are stacked vertically one above the other and supported by perspex holders

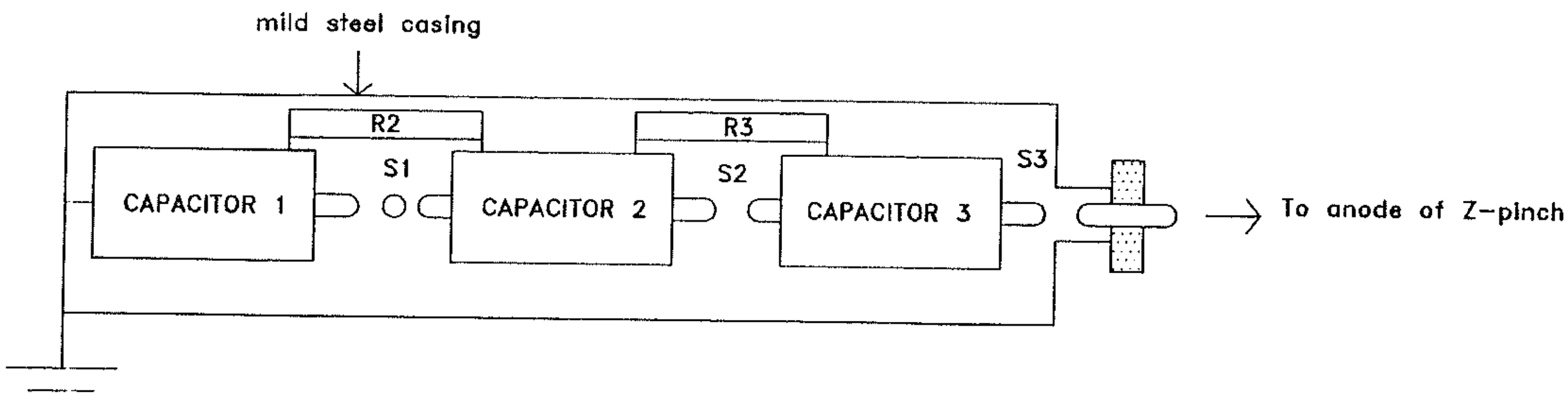


Fig. 6.1 A schematic of the layout of Marx1.

(refer Fig. 6.2). Pressurised spark gaps are used here and they are arranged in rows as shown in Fig. 6.3. This arrangement facilitates the erection of the Marx2 through the irradiation of the second and third gaps upon the switching of the first gap. Again $CuSO_4$ resistors are used as earthing resistors. The capacitors, pressurised spark gap system and the earthing resistors are placed inside a grounded mild steel casing filled with oil to provide the necessary insulation. The output of the Marx is taken through a damping resistor ($CuSO_4$) to charge the water-line. A schematic of the the overall setup of Marx2 is shown in Fig. 6.4.

6.3.3 Design of the water-line

The coaxial water-line is made from brass sheet of thickness approximately 2 mm rolled into a cylindrical tube of diameter *15 cm (d)* for the inner electrode and *20 cm (D)* for the outer electrode. The length is *150 cm* (ℓ). At each end of the rolled brass tube for each of the inner and outer electrodes, a curved brass plate is attached. (see Fig. 6.5). This is to increase the electric field across the insulation and to reduce the electric field parallel to the insulating surface. This prevents flashover across the inner and outer electrodes. A diagram of the water-line is shown in Fig. 6.5. The -electrical parameters of the water-line are as follows :

$$L_W = \frac{\mu_o \ell}{2\pi} ln(D/d) \qquad [6.3]$$

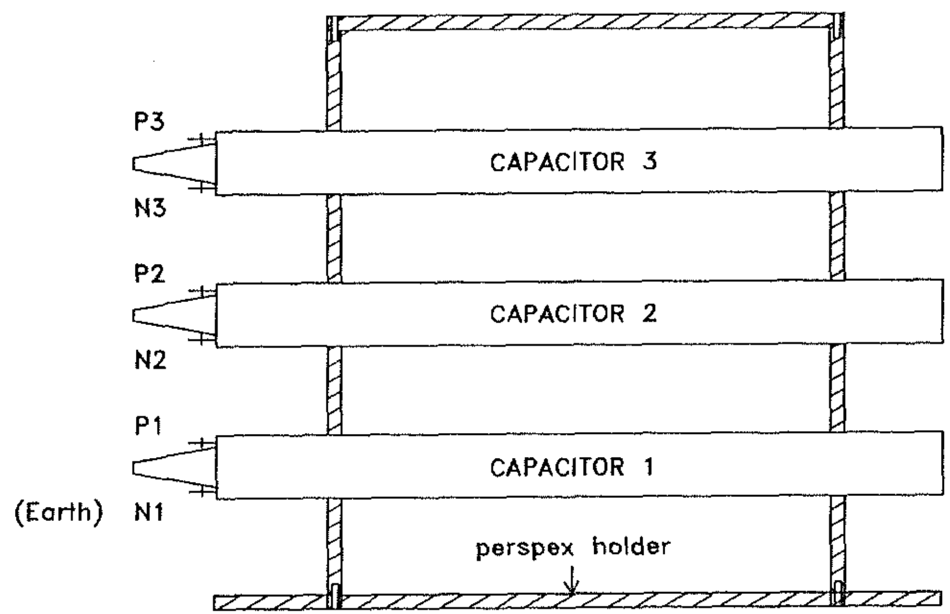


Fig. 6.2 The arrangement of the capacitors of Marx2.

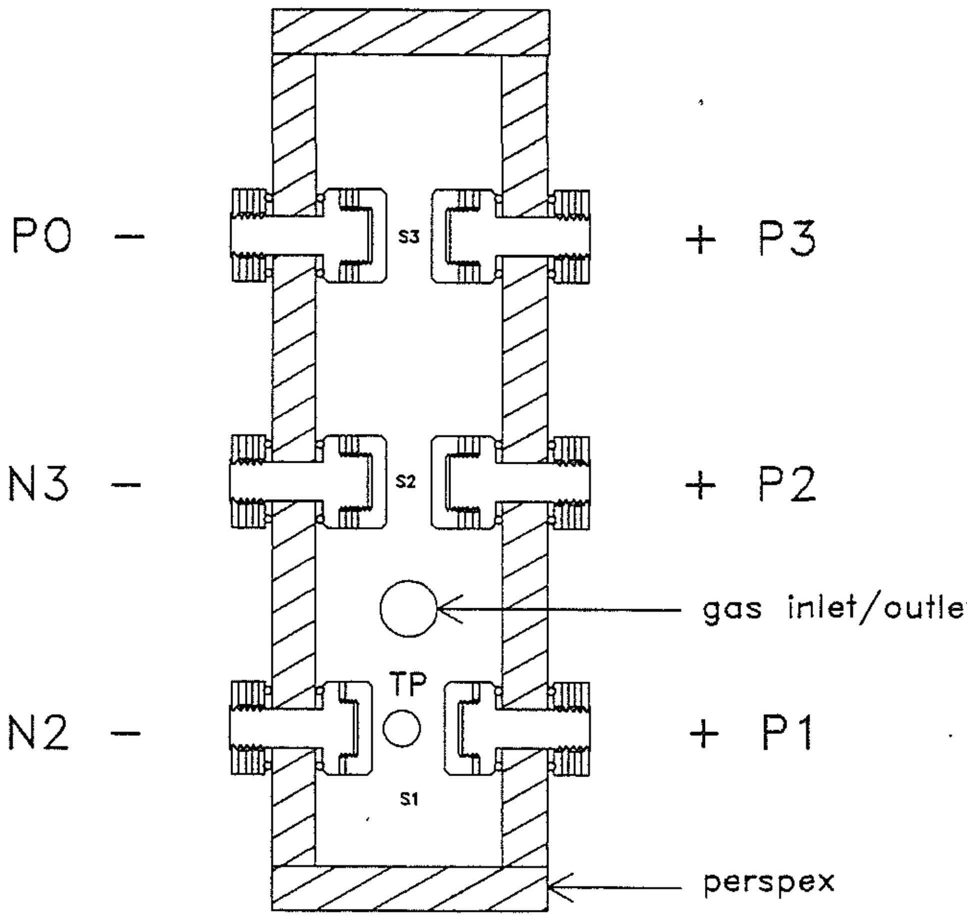


Fig. 6.3 The pressurised spark gaps of Marx2. N_2, N_3 are connected through 500 Ω to ground. PO, from the output gap is connected to the anode flange of the Z-pinch.

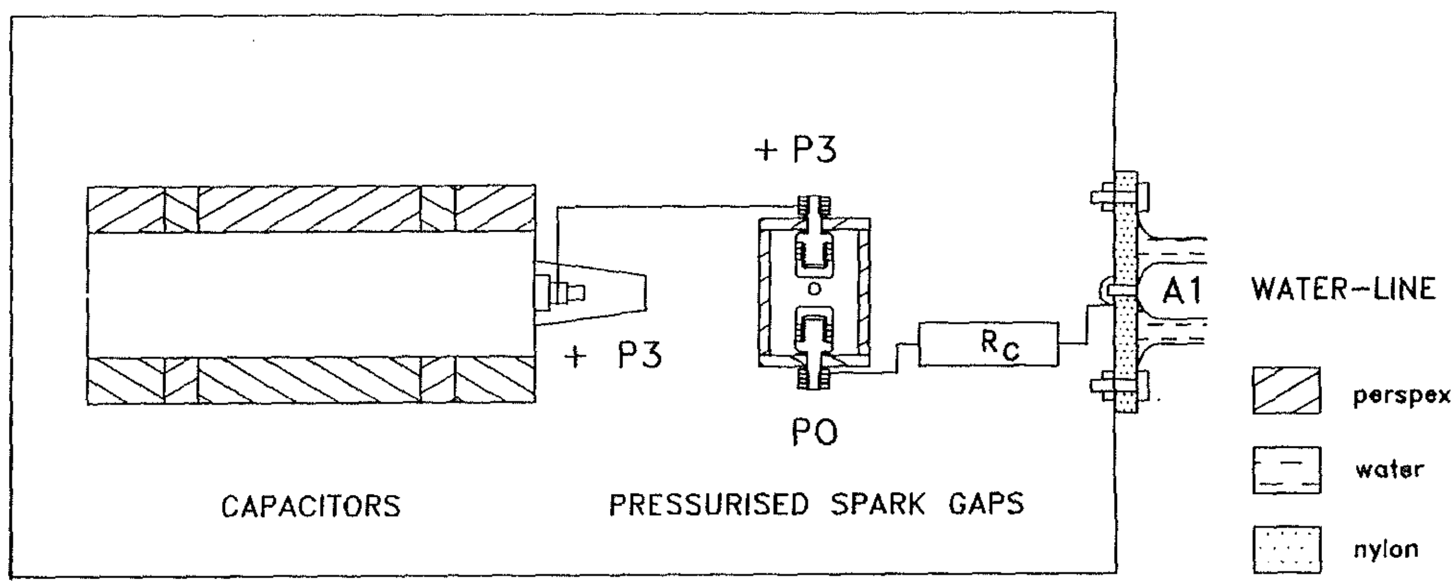


Fig. 6.4 A schematic of the layout of Marx2.

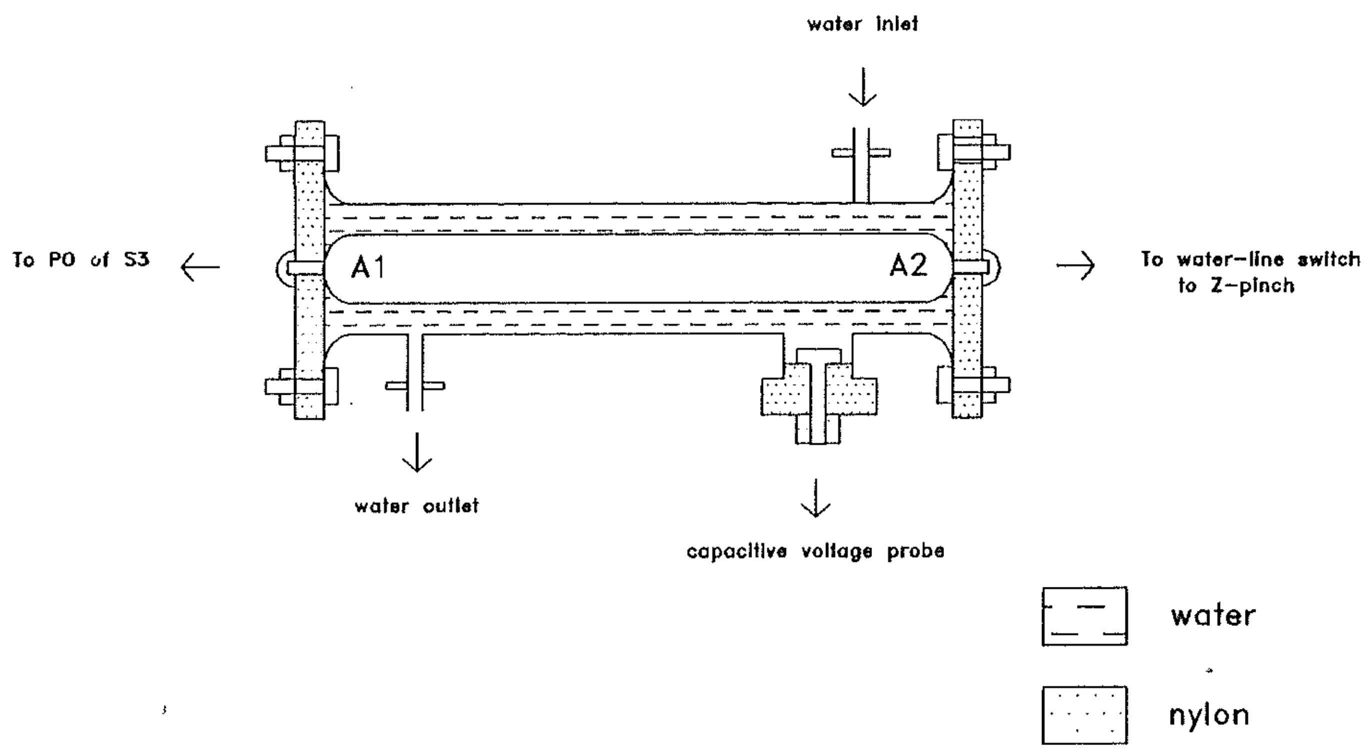


Fig. 6.5 The design of the water-line.

$$C_W = \frac{2\pi\varepsilon_w\varepsilon_o\ell}{ln(D/d)} \qquad [6.4]$$

$$Z_W = \sqrt{(L_W/C_W)} = \frac{1}{2\pi}\left[\frac{\mu_o}{\varepsilon_w\varepsilon_o}\right]^{1/2} ln(D/d) \qquad [6.5]$$

$$\tau_W = \sqrt{L_W C_W} = \frac{\ell\sqrt{\varepsilon_w}}{c} \qquad [6.6]$$

where C_W, L_W, Z_W, τ_W are the capacitance, inductance, impedance and the characteristic transit time of the coaxial water-line respectively. ε_W is the dielectric constant of the line insulator, which in the case of water is approximately 80. μ_o and ε_o are the permeability and permittivity of free space respectively and c the speed of light.

The duration of the current pulse, τ_d is given by

$$\tau_d = 2\tau_W = \frac{2\ell\sqrt{\varepsilon_W}}{c} \qquad [6.7]$$

Water is chosen as the line insulator because of its high dielectric properties which remain stable up to about *1 GHz*, resulting in an increased line capacitance per unit length. The resistivi of the water must be kept at 10^4 Ω m, therefore deionised water must be used and continuously recycled through a resin deioniser.

The breakdown voltage of the line assuming a uniform electric field as given by *Martin [1970]* is reproduced below ;

$$E_{bd}\, t_{eff}^{1/3}\, A^{1/10} = k \qquad [6.8]$$

where E_{bd} is the breakdown field in *MV/cm*, t_{eff} the duration in μs when the field is *63%* of E_{bd} and A the electrode area in cm^2 over which the electric field does not vary by more than *10%* from the maximum value. k is a polarity dependent constant which is *0.3* for a water-line charged positively. If the water-line is charged negatively then k is 0.6.

The maximum voltage that the line has to tolerate is given by

$$V_{max} = \frac{2V_m C_m}{C_m + C_w} \qquad [6.9]$$

where V_m, C_m are the series output voltage and capacitance of the erected Marx used to pulse charge the water-line and C_w the capacitance of the water-line.

6.3.4 Design of the water-line switch

A pressurised spark gap *[Favre, M.B., 1985]* is to be used to switch the pulse charged water-line onto the pinch load. It consists of two brass electrodes placed approximately 1 cm apart and enclosed in a cylindrical nylon casing machined from a solid nylon block. The design is as shown in Fig. 6.6. The estimated inductance of the switch is *100 nH*.

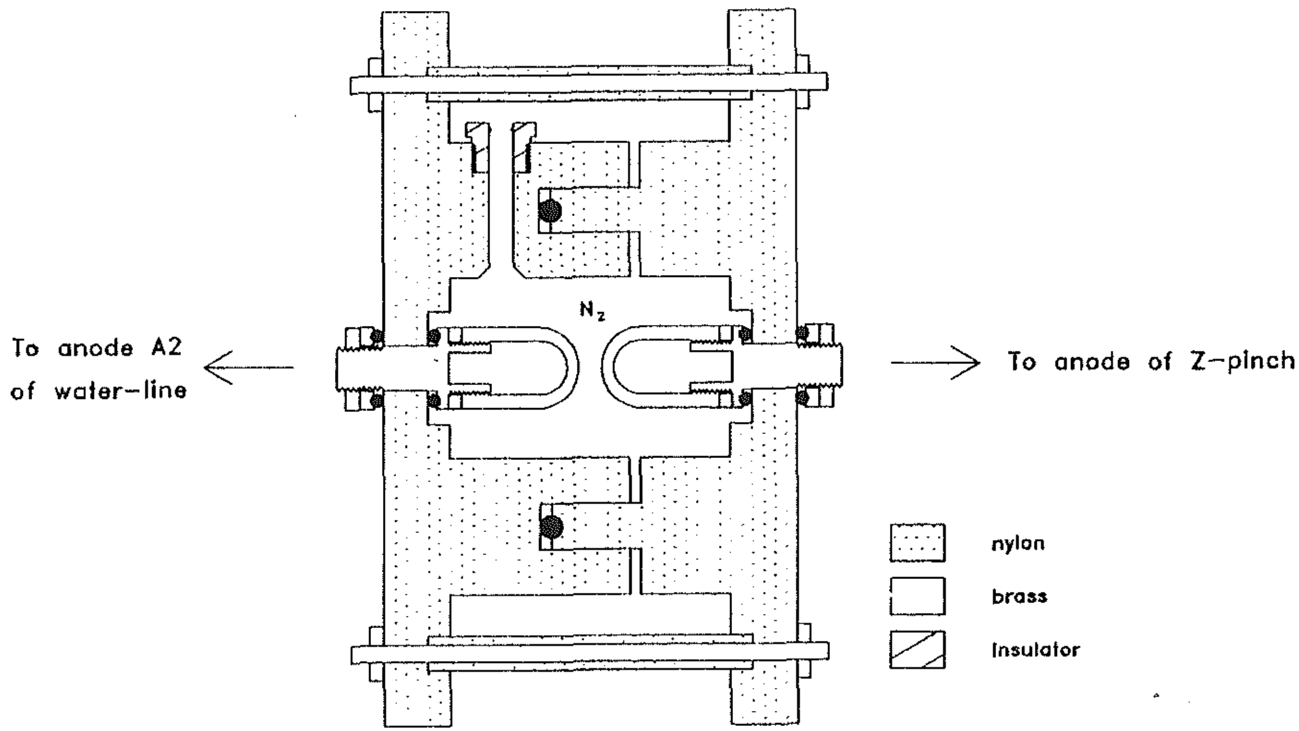


Fig. 6.6 The design of the water-line switch.

6.4 The Circuit equations of MLCCSZP current generator

6.4.1 Charging of the water-line

The water-line is pulse charged by a 3-stage Marx generator. During charging, the water-line behaves as a capacitor of capacitance c_W. The charging process is presented in Fig. 6.7. The Marx upon erection can be represented as a capacitor, C_m with a series inductance, L_m. R_m is included to obtain a critically damped charging source. Applying Kirchoff's law the following circuit equations are obtained.

$$V_{Cm} = V_{Lm} + V_R + V_{CW} \qquad [6.10]$$

$$V_O - \frac{q_O}{C_m} = \dot{I}_m L + IR + \frac{q}{C_m} \qquad [6.11]$$

6.4.2 Discharging of the water-line

The pulse charged water-line discharges through the load upon switching. For the duration of two transit times, $0 < T < 2\tau_W$, the discharge of the water-line through an inductive load can be represented as being driven by a voltage source, $V_O = V_W$ with a series impedance, Z_O equal to the line impedance (see Fig. 6.8). The current delivered by the water-line when switched is given by

$$\dot{I} = \frac{(V_O - IZ_O)}{(L_S + L_L)} \qquad [6.12]$$

where L_S is the inductance of the switch and L_L is the inductance

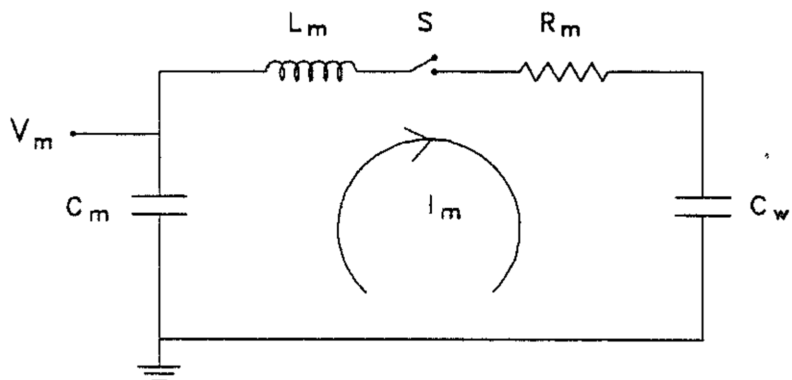


Fig. 6.7 The circuit diagram representing the pulse charged water-line.

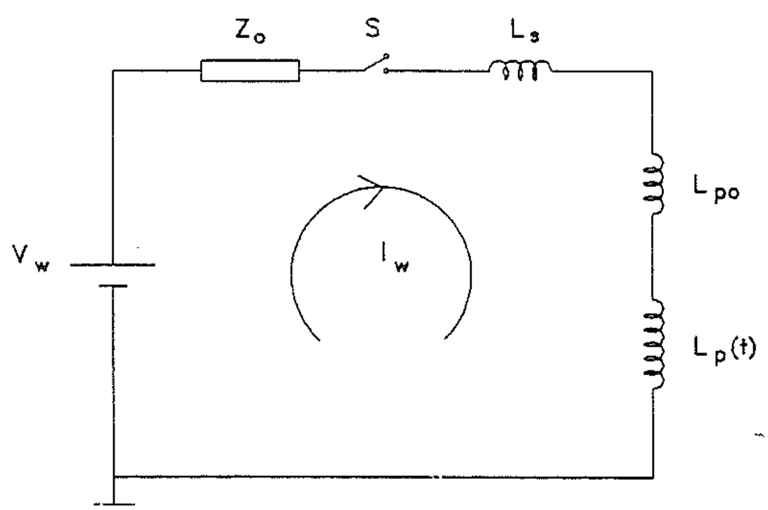


Fig. 6.8 The discharge circuitry of the first bank (Marx1) of MLCCSZP.

of the load.

6.4.3 The MLCCSZP circuit

The circuit representing the current-stepped source of the MLCCSZP is shown in Fig. 6.9.

The first current source delivered by the 3-stage Marx system is represented as a capacitor discharge through a series inductor to the pinch load. The second current source provided by the discharge of the water-line is represented as in section 6.4.2.

The circuit equations describing the MLCCSZP current-stepped sources are as follows :

[a] S_1 on & S_2 off

$$\frac{d}{dt}\left[[L_1 I_1] + [L_{po} + L_p(t)]\, I \right] = V_{10} - \frac{\int I_1 dt}{C_1} \qquad [6.13]$$

$$I = I_1 \qquad [6.13a]$$

[b] S_1 on & S_2 on; (S_0 on)

$$\frac{d}{dt}\left[[L_1 I_1] + [L_{po} + L_p(t)]\, I \right] = V_{10} - \frac{\int I_1 dt}{C_1} \qquad [6.14a]$$

$$\frac{d}{dt}\left[[L_s I_2] + [L_{po} + L_p(t)]\, I \right] = V_{w0} - I_2 Z_o \qquad [6.14b]$$

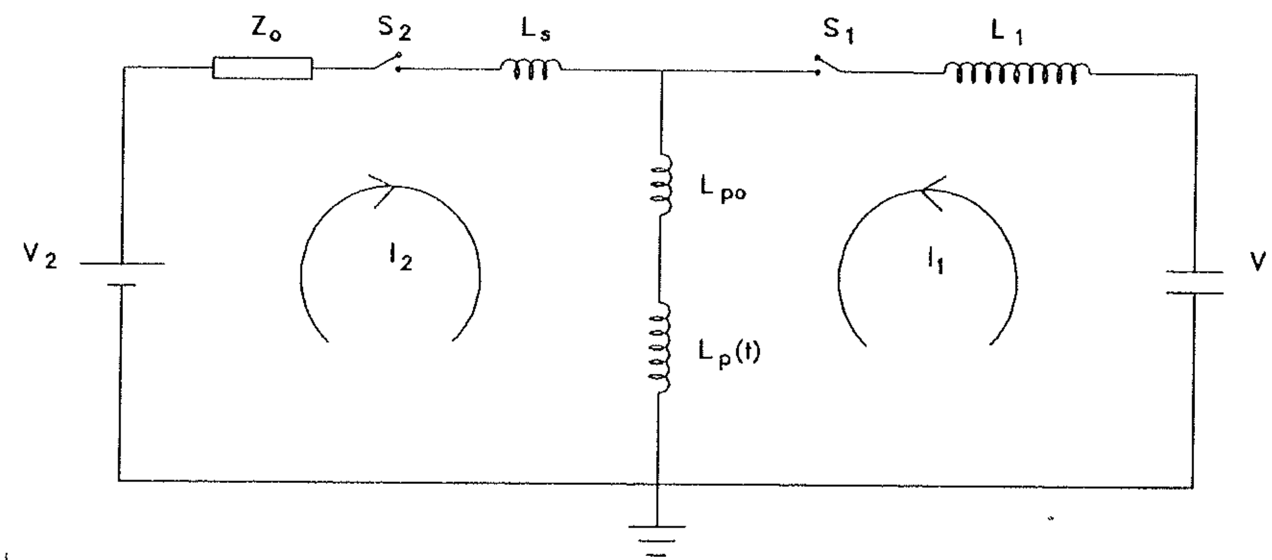


Fig. 6.9 The discharge circuitry of the MLCCSZP.

$$I = I_1 + I_2 \qquad [6.14c]$$

L_s is the inductance of the water-line switch.

When S_1 is switched alone a conventional gas compressional Z-pinch is obtained. Switching of S_2 after a suitable delay will produce the current-stepped Z-pinch.

6.5 Computation procedures

Equations [6.13] to [6.14] are solved numerically together with the extended model representing the plasma dynamics as given in equations [5.14] to [5.16]. For the single compression by the first bank alone, the circuit conditions are represented by equations [6.13] and [6.13a]. In the case of the current-stepped compression by the first bank followed by the second bank, the circuit conditions are represented by equations [6.14a], [6.14b] and [6.14c]. The condition of the water-line during charging is represented by equation [6.11]. The block diagram of the computational procedures is shown in Fig. 6.10.

6.6 Numerical results

A set of numerical results obtained for the MLCCSZP designed using the γ-varying energy balance slug model is presented.

6.6.1 Comparison of the first compression between UMCSZP and MLCCSZP

The variations of γ_{eff} during the first compression

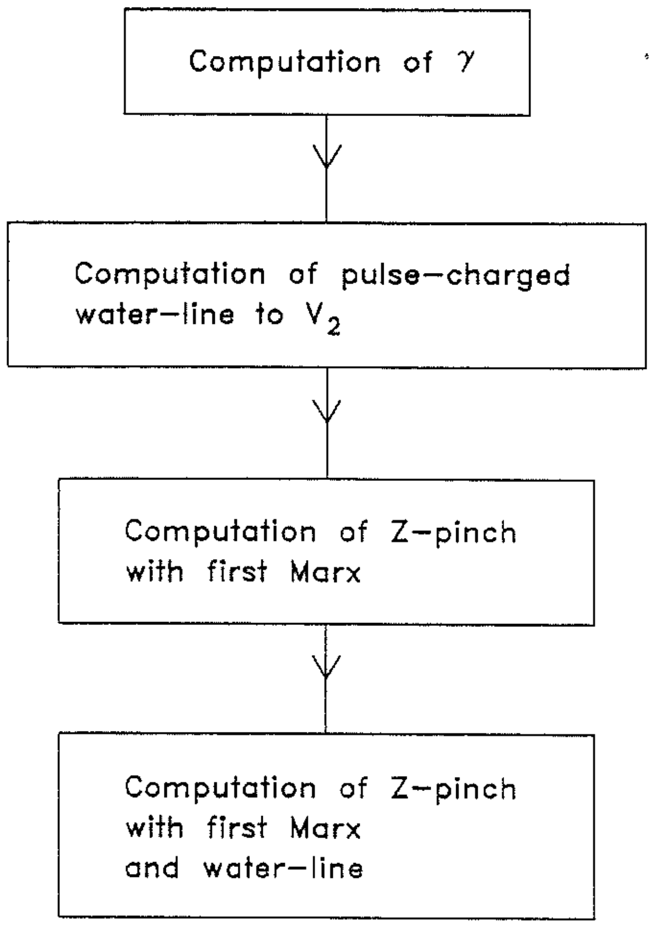


Fig. 6.10 Block diagram of the computation sequences of MLCCSZP.

MLCCSZP and UMCSZP are shown in Fig. 6.11. Comparing to UMCSZP which is shown in broken line, the new MLCCSZP system γ_{eff} approaches 5/3 in a shorter time due to the higher compressing shock speed (refer Fig. 6.12). Its deviation from the compression of an ideal gas with $\gamma = 5/3$ is reduced. The minimum radius ratio predicted by the extended model for the new system is 0.17 and the difference from that predicted by the γ-constant model is reduced from 50% for UMCSZP to 19% for the MLCCSZP. This has an effect in the piston and shock trajectories shown in Fig. 6.13 for UMCSZP and Fig. 6.14 for the MLCCSZP.

There is also a significant improvement in the plasma flow field-magnetic field coupling represented by the magnetic Reynolds number, $\mathcal{R}_m$.and shown in Fig. 6.15.

6.6.2 Current-stepped Z-pinch

With the MLCCSZP system, an enhancement in the compression upon current-stepping is indicated. The increase in the compressing speed of the first compression is sufficient to produce an annular column of plasma with γ_{eff} approaching 1.6, which is relatively closer to 5/3, at the instant of current-stepping. Therefore, upon current-stepping, the increase in γ is small and thus enables the enhanced effect of the current-stepped compression to dominate. The variation of γ_{eff} for different times of current-stepped are shown on Fig. 6.16. The associated shock speed and plasma temperature are shown in Fig. 6.17 and Fig. 6.18. The plasma current, voltage and inductance are shown in Fig. 6.19, Fig 6.20 and Fig. 6.21 respectively. The corresponding shock and piston trajectories of

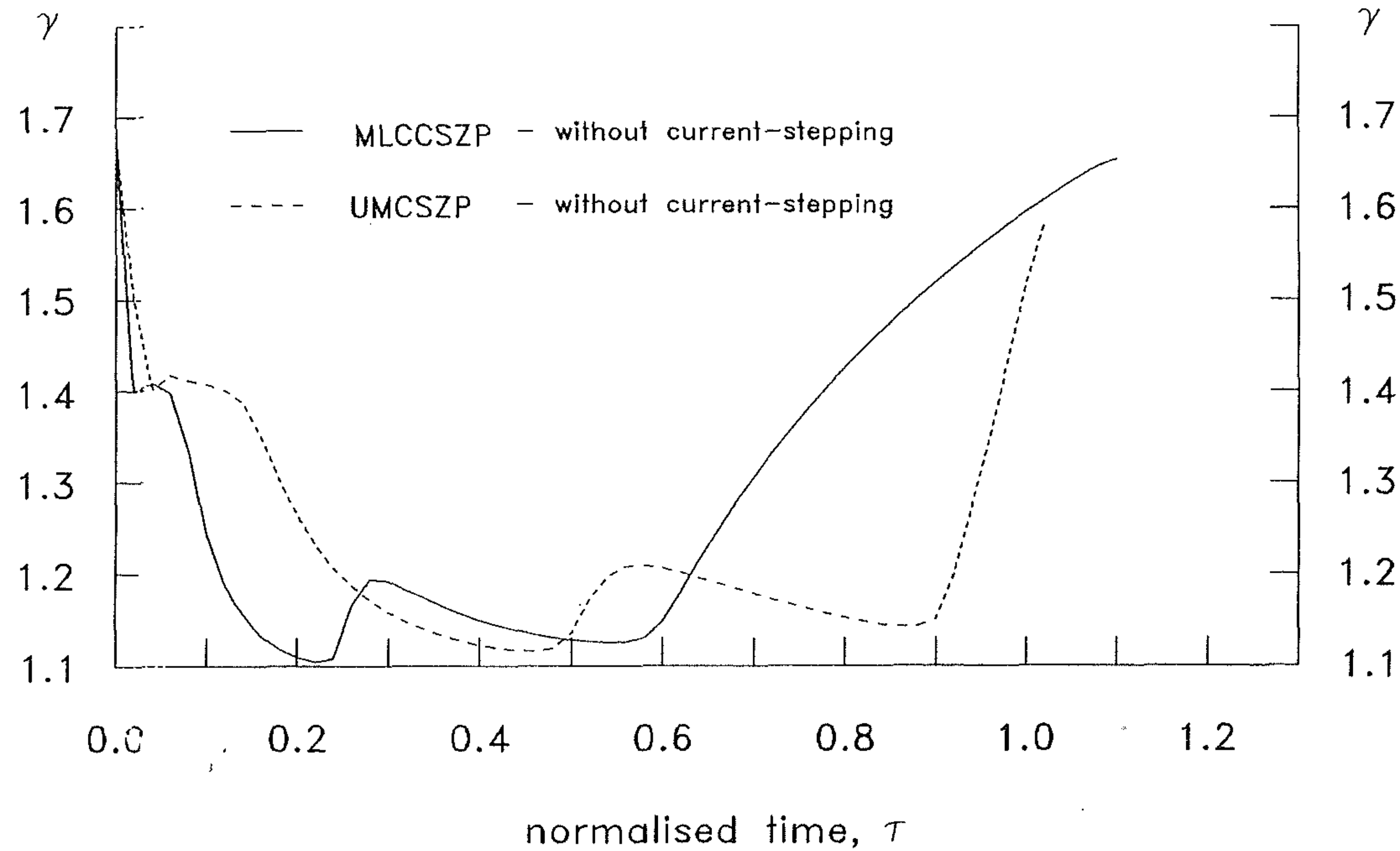


Fig. 6.11 Variation of γ_{eff} of MLCCSZP and UMCSZP without current-stepping.

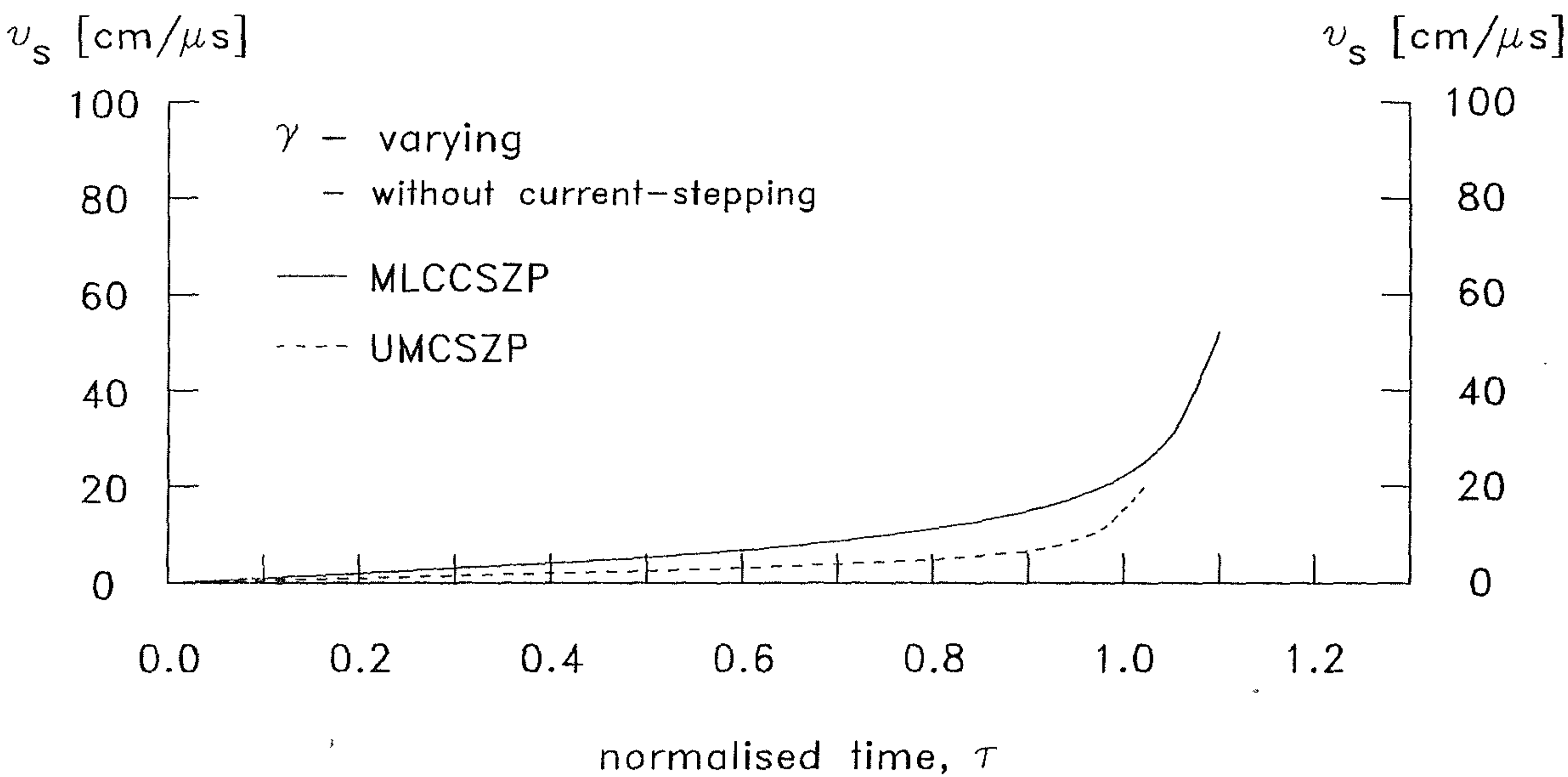


Fig. 6.12 Variation of shock speed, v_s of MLCCSZP and UMCSZP without current-stepping.

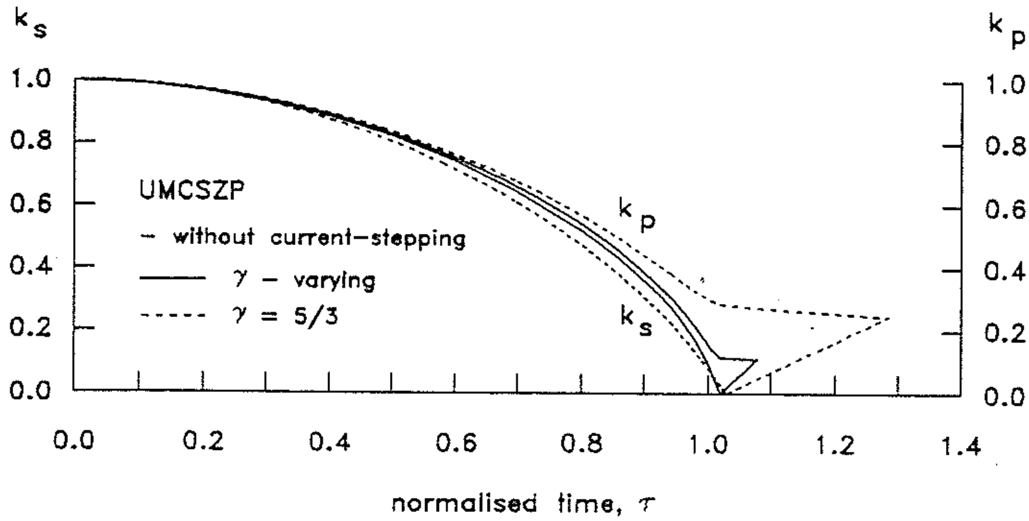


Fig. 6.13 Piston and shock trajectories with γ - varying and $\gamma = 5/3$ of UMCSZP.

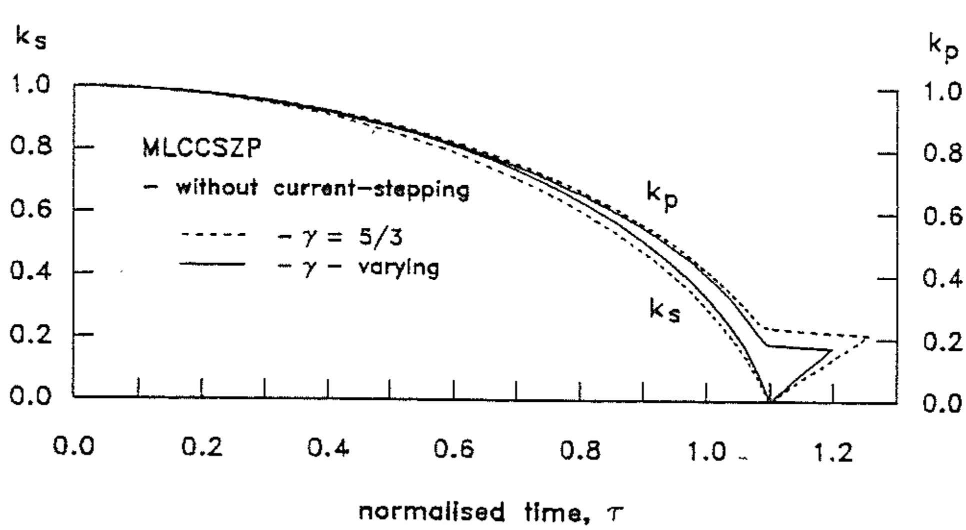


Fig. 6.14 Piston and shock trajectories with γ - varying and $\gamma = 5/3$ of MLCCSZP.

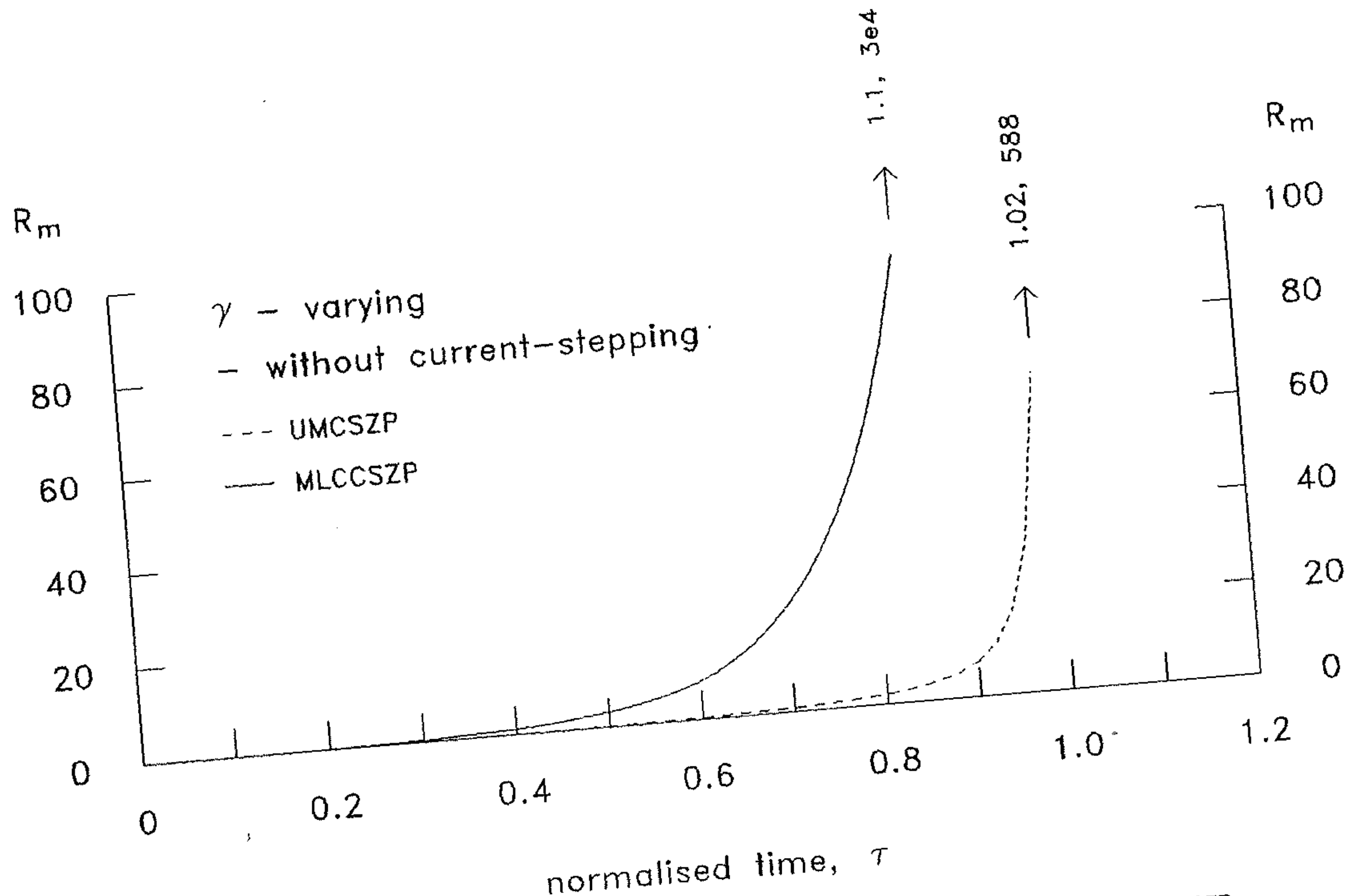


Fig. 6.15 Development of the magnetic Reynolds number, R_m of MLCCSZP and UMCSZP

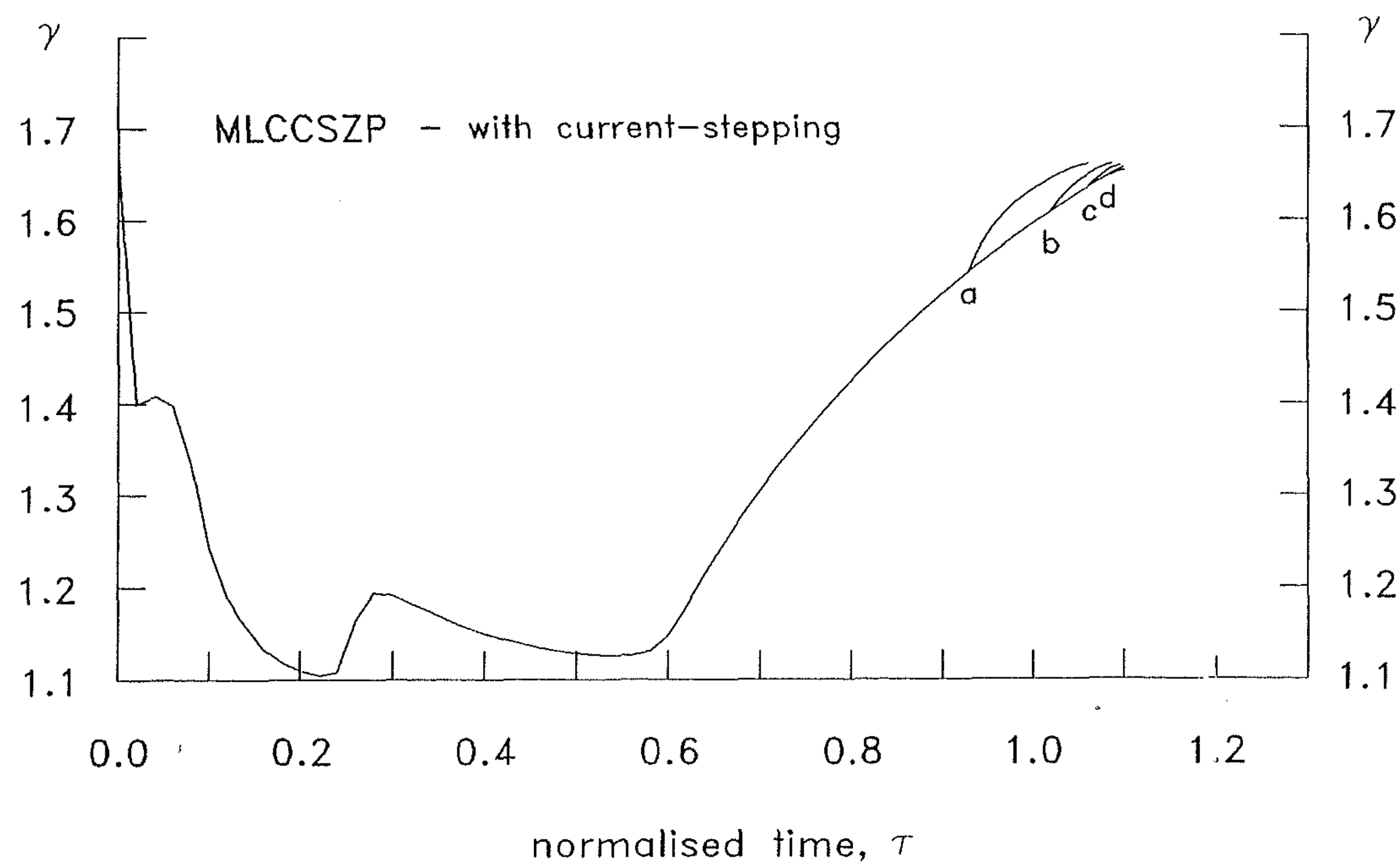


Fig. 6.16 Variation of γ_{eff} of MLCCSZP with various times of current-stepping.

(a : ψ = 0.2, b : ψ = 0.4, c:: ψ = 0.6, d : ψ = 0.8).

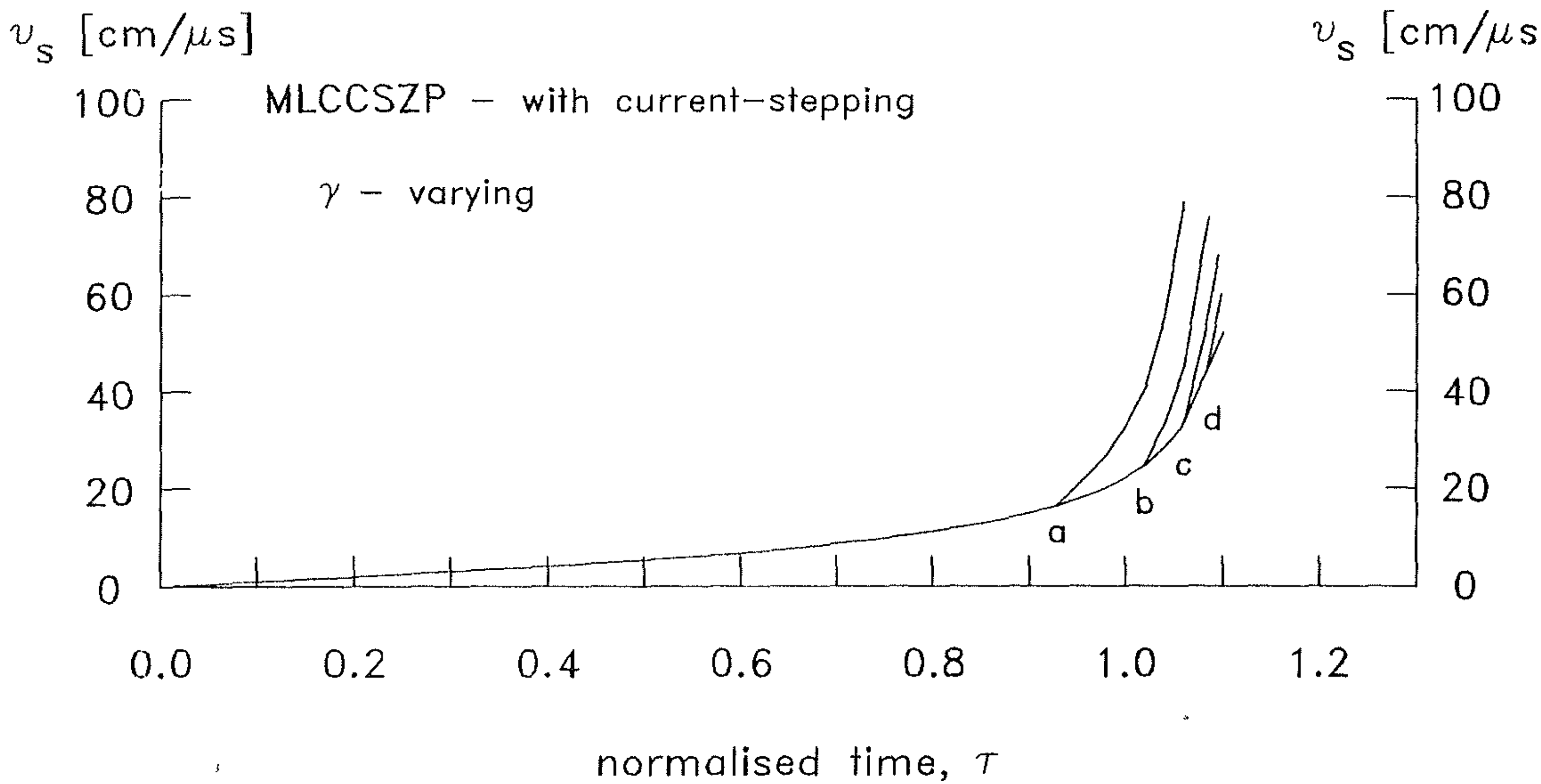


Fig. 6.17 Variation of shock speed, v_s of MLCCSZP with various times of current-stepping.

(a : $\psi = 0.2$, b : $\psi = 0.4$, c : $\psi = 0.6$, d : $\psi = 0.8$).

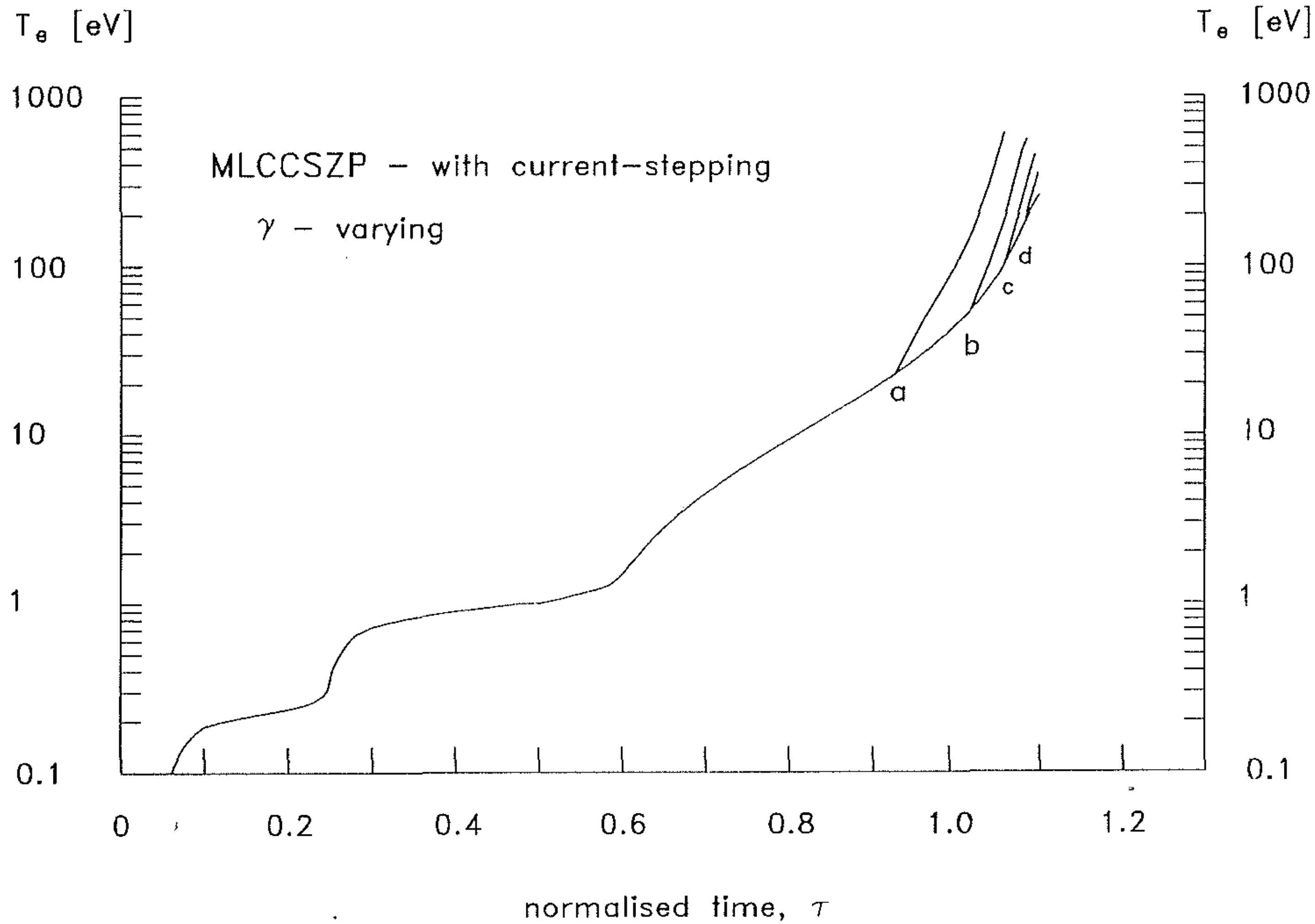


Fig. 6.18 Variation of plasma temperature, T_e of MLCCSZP with various times of current-stepping. (a : $\psi = 0.2$, b : $\psi = 0.4$, c : $\psi = 0.6$, d : $\psi = 0.8$).

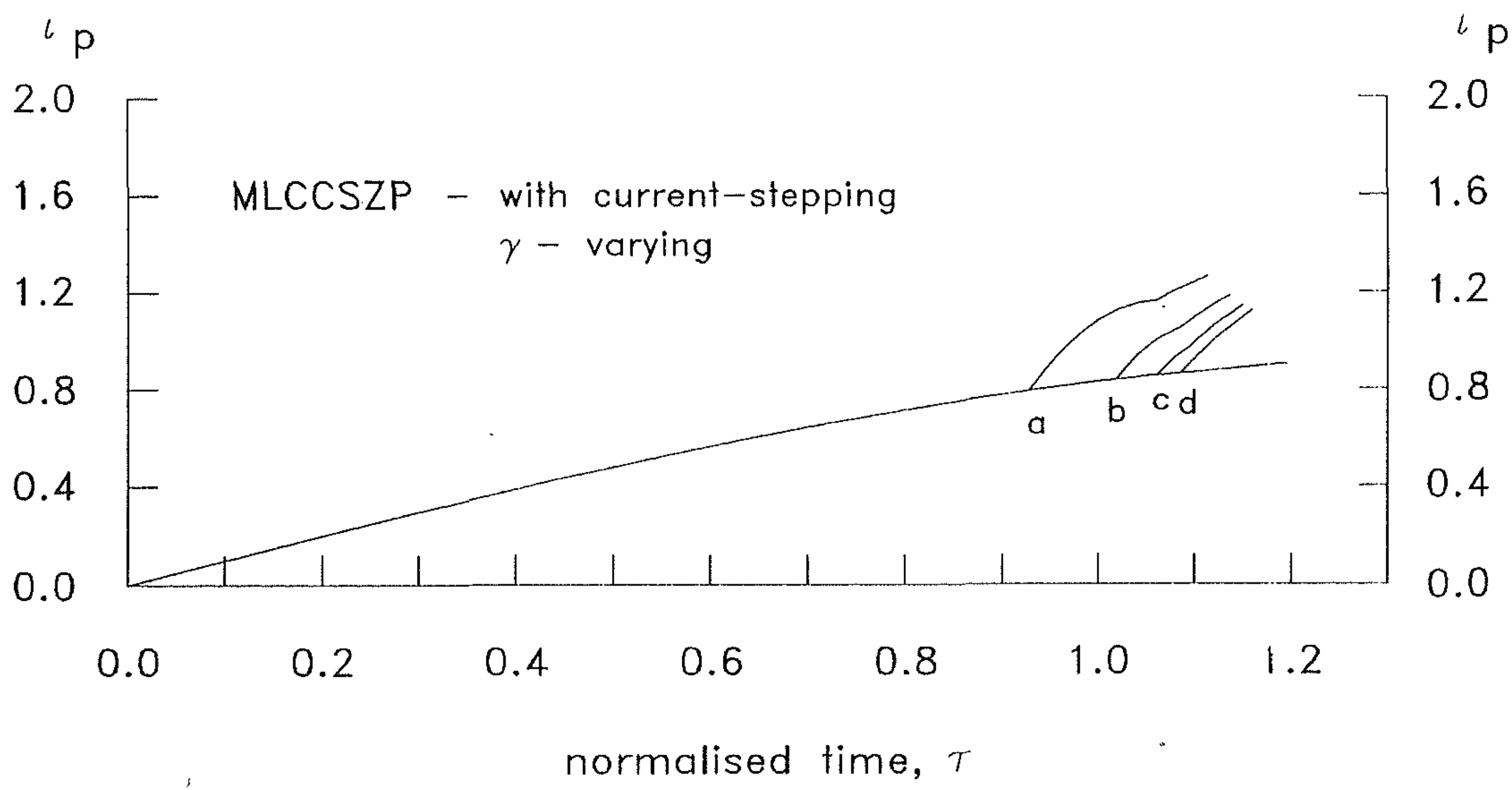


Fig. 6.19 Normalised plasma current, ι_p of MLCCSZP with various times of current-stepping. (a : ψ = 0.2, b : ψ = 0.4, c : ψ = 0.6, d : ψ = 0.8).

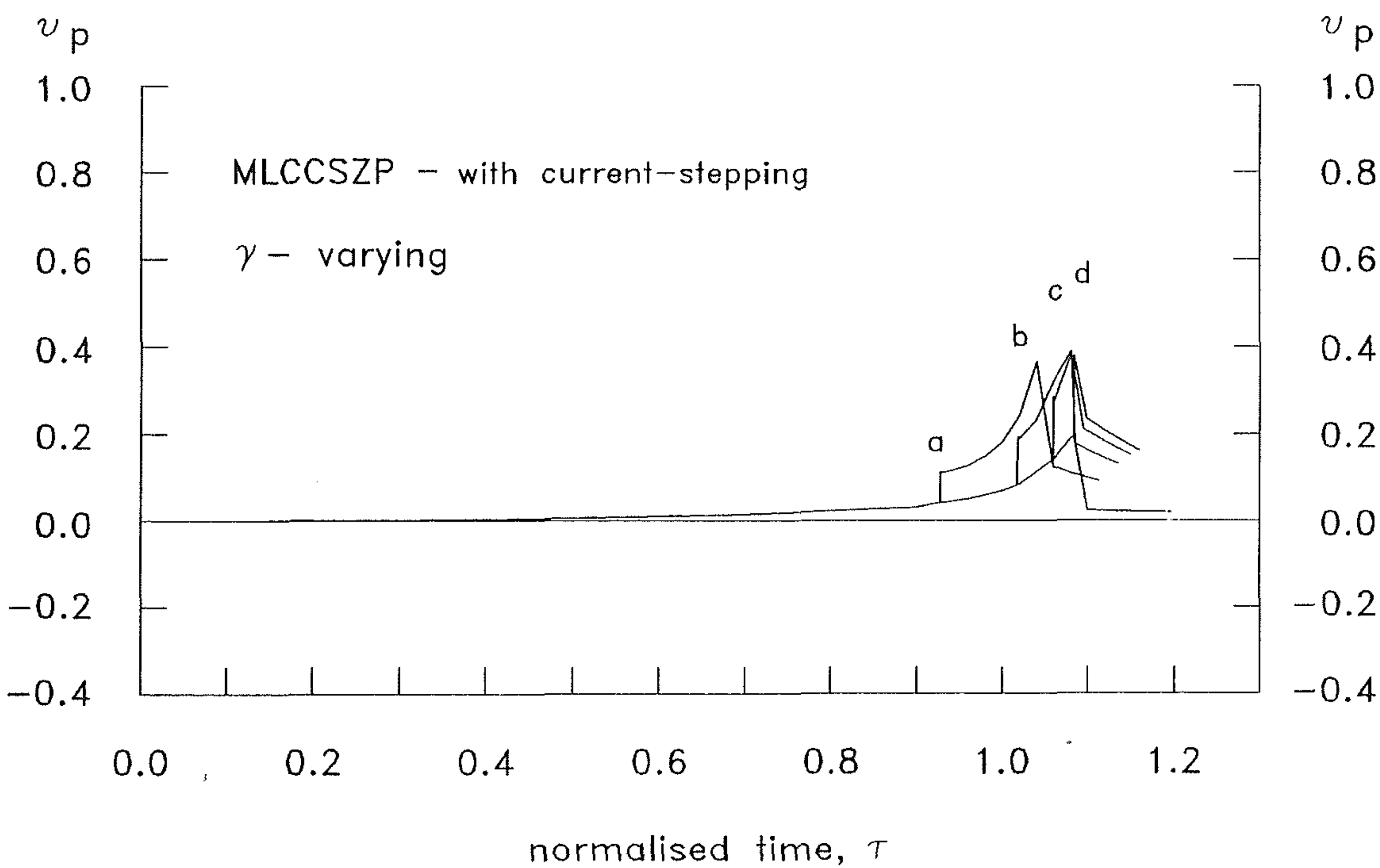


Fig. 6.20 Normalised plasma voltage, ν_p of MLCCSZP with various times of current-stepping.
(a : $\psi = 0.2$, b : $\psi = 0.4$, c : $\psi = 0.6$, d : $\psi = 0.8$).

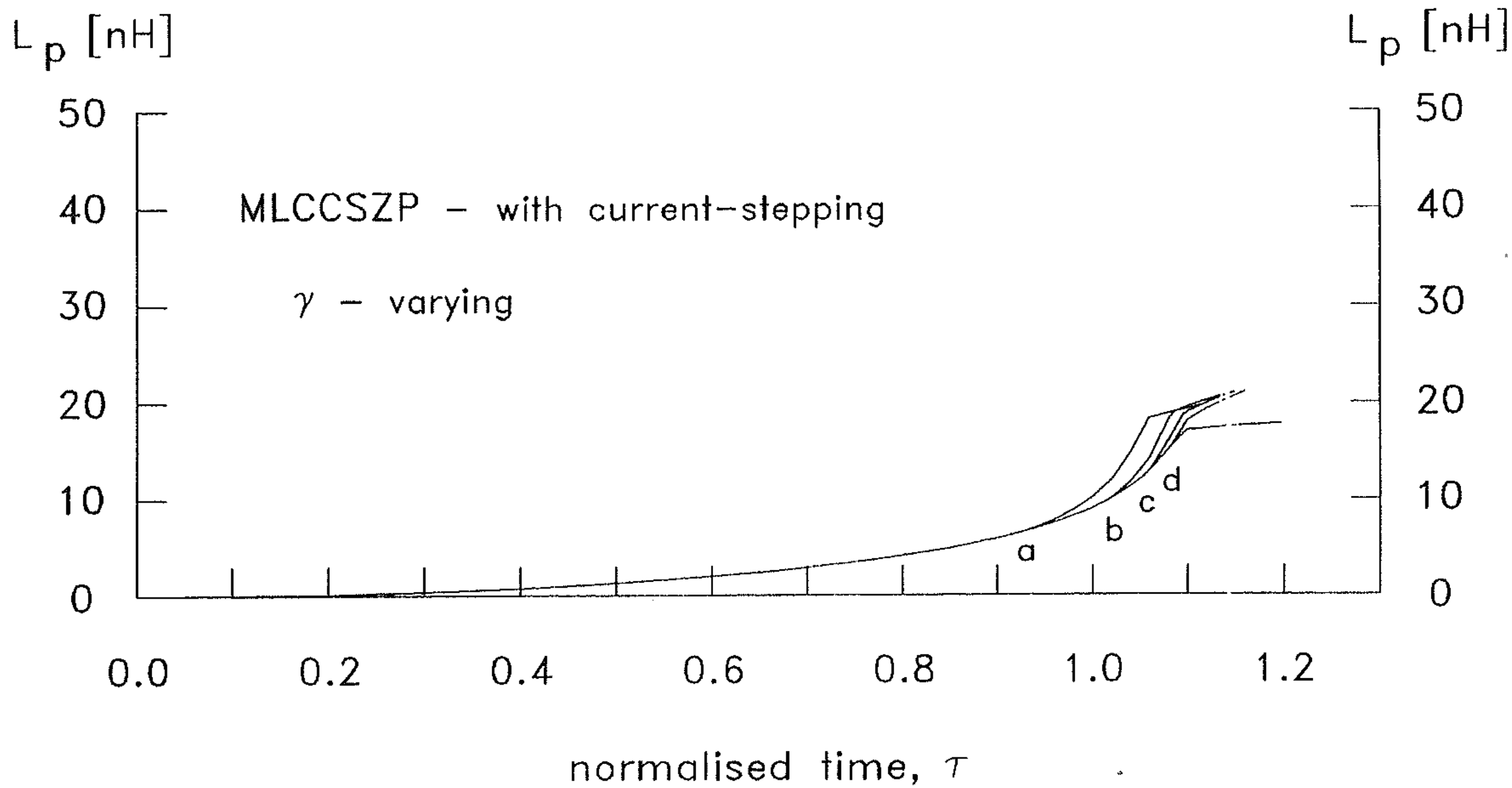


Fig. 6.21 Plasma inductance, L_p versus normalised time, τ of MLCCSZP with various times of current-stepping. (a : $\psi = 0.2$, b : $\psi = 0.4$, c : $\psi = 0.6$, d : $\psi = 0.8$).

the enhanced compression are shown in Fig. 6.22 and Fig. 6.23.

The enhanced compression factor predicted by the MLCCSZP is shown in Fig. 6.24. There is a marked improvement compared to that of UMCSZP shown in broken line.

6.7 Experimental set-up of MLCCSZP

The circuit equation for Marx1 is as shown in Fig. 6.25. The load resistors are high power $CuSO_4$ resistors. The capacitors are charged in parallel by a 60 kV, 30 mA power supply unit. The power supply unit is designed to charge both Marx units together. The circuit diagram of the power supply unit is as shown in Fig. 6.26. The output gap of the third capacitor is connected to the anode of the pinch machine. The anode of the pinch machine is also connected via a parallel plate to the self breakdown gap of the water-line which switches the water-line onto the pinch load. The other end of the water-line is connected to the output gap of Marx2, its charging source. A schematic of the layout is as shown in Fig. 6.27

6.8 The trigger pulse generator

The triggering pulse of 40 kV, 40 ns rise-time is provided by a 50 stage cascaded SCR unit. The circuit of the cascaded SCR switch is as shown in Fig. 6.28. When a trigger pulse is introduced to the gate of the lowest SCR which has its cathode connected to ground, this SCR switches on and swings the anode to ground. This causes the lowest triggering capacitor to discharge through R_T through the gate to switch the next higher SCR and so on. Once the chain of cascaded SCRs becomes fully

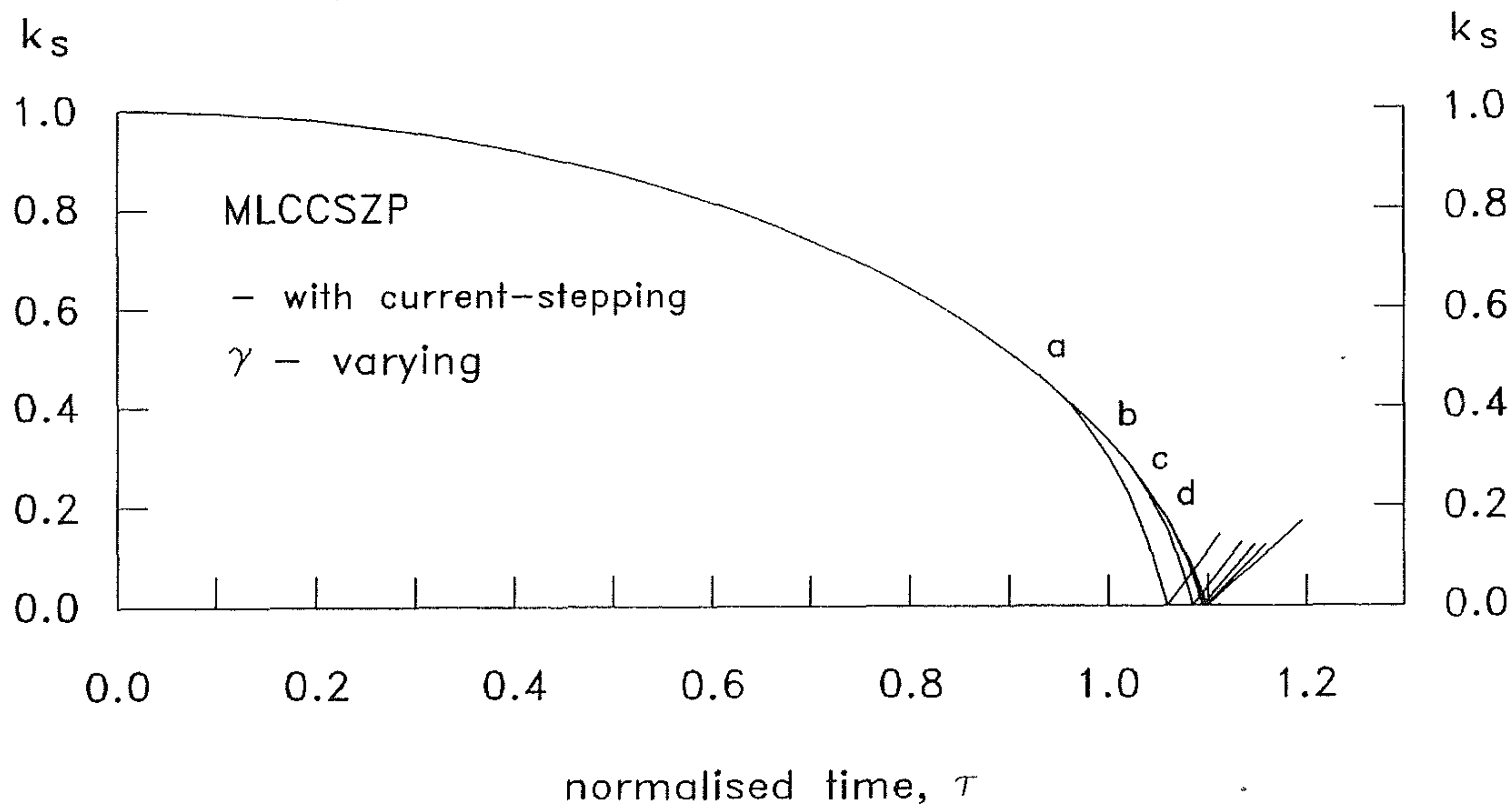


Fig. 6.22 Normalised shock trajectories, k_s of MLCCSZP with various times of current-stepping.

(a : $\psi = 0.2$, b : $\psi = 0.4$, c : $\psi = 0.6$, d : $\psi = 0.8$).

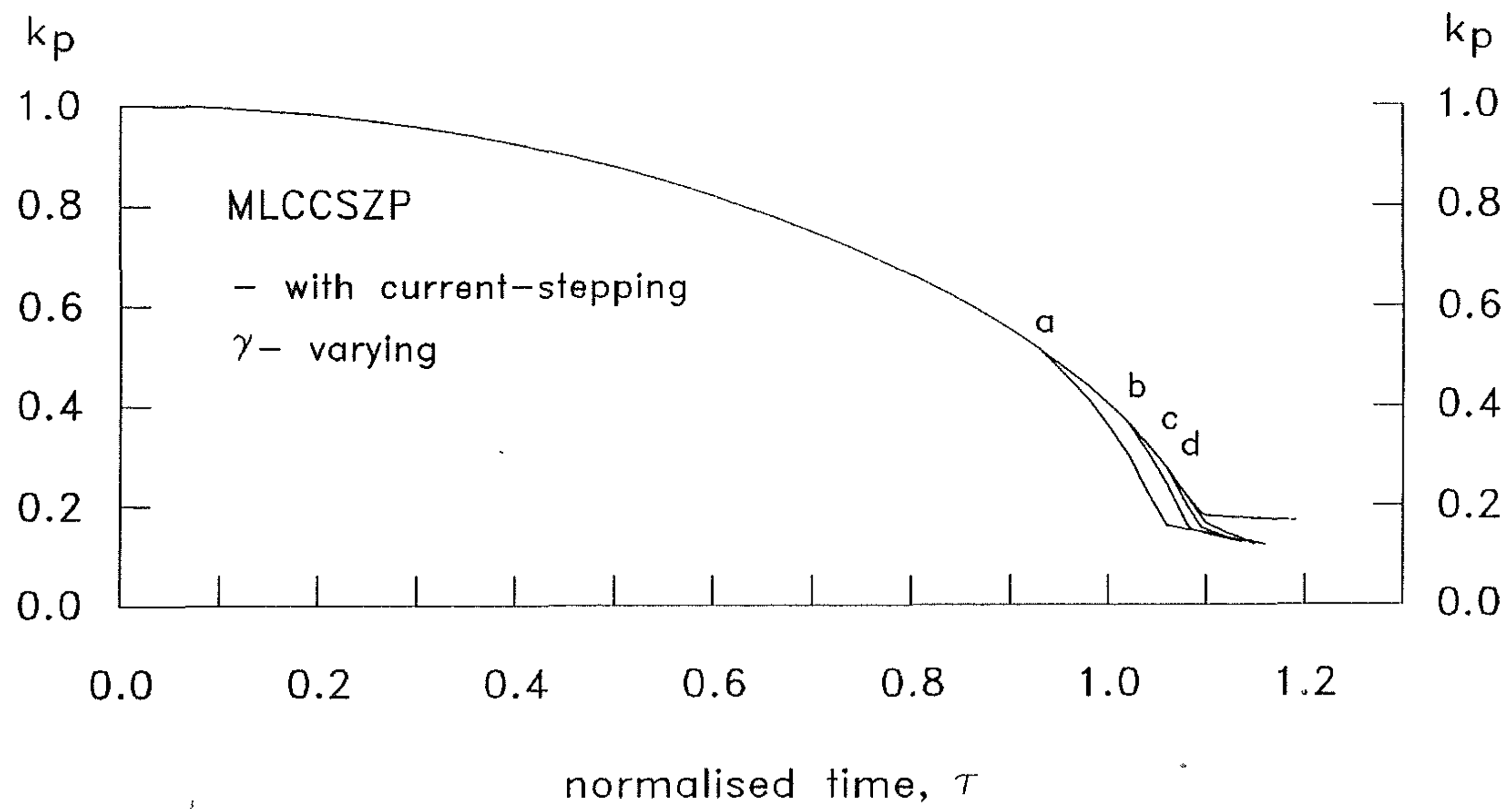


Fig. 6.23 Normalised piston trajectories, k_p of MLCCSZP with various times of current-stepping.
(a : $\psi = 0.2$, b : $\psi = 0.4$, c : $\psi = 0.6$, d : $\psi = 0.8$).

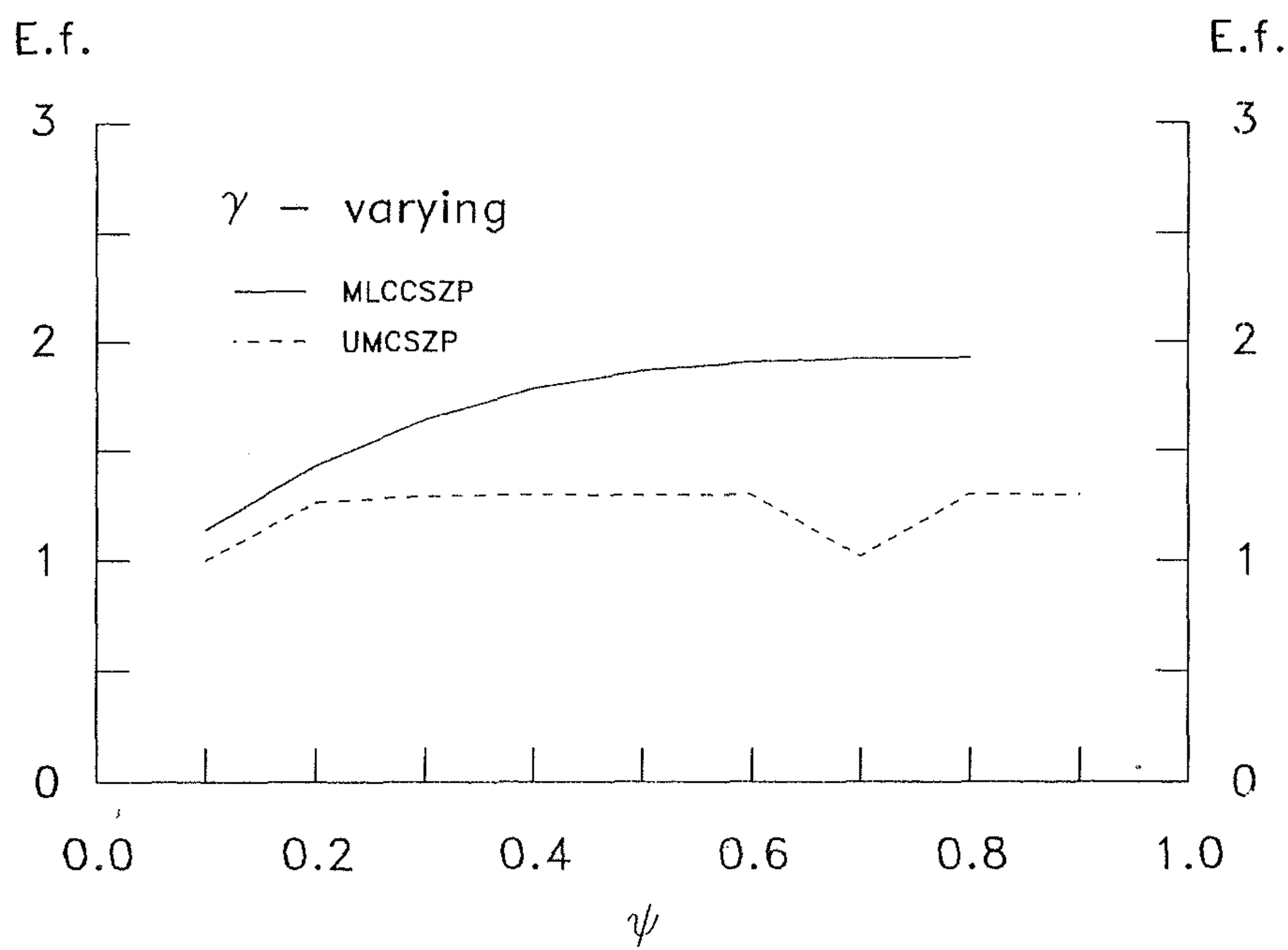


Fig. 6.24 Compression enhancement factor, E.f. versus ψ of MLCCSZP and UMCSZP.

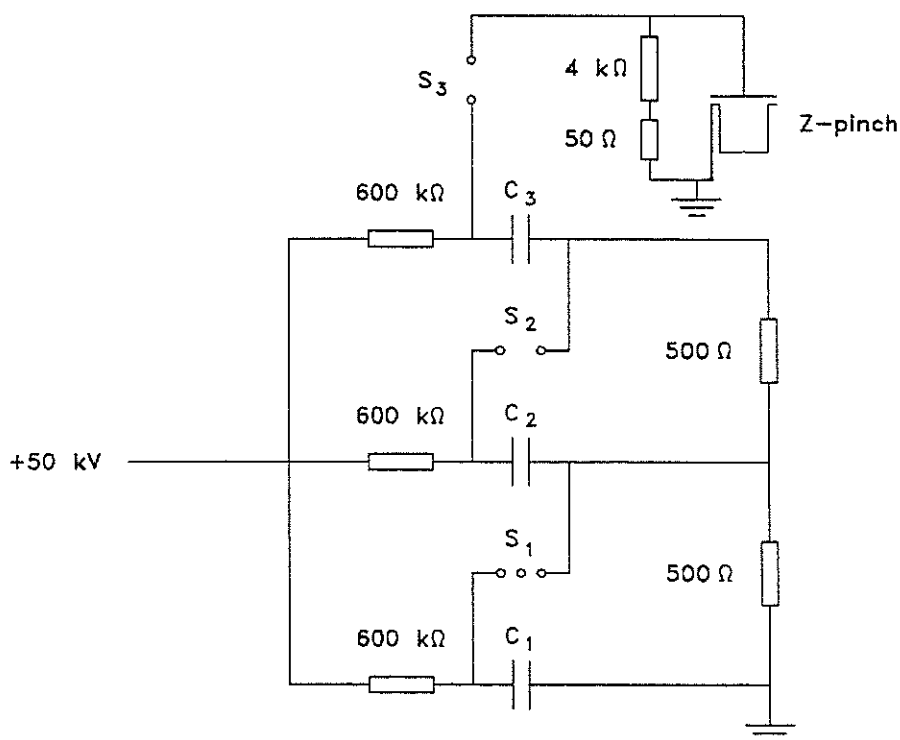


Fig. 6.25 The circuit diagram of Marx1.

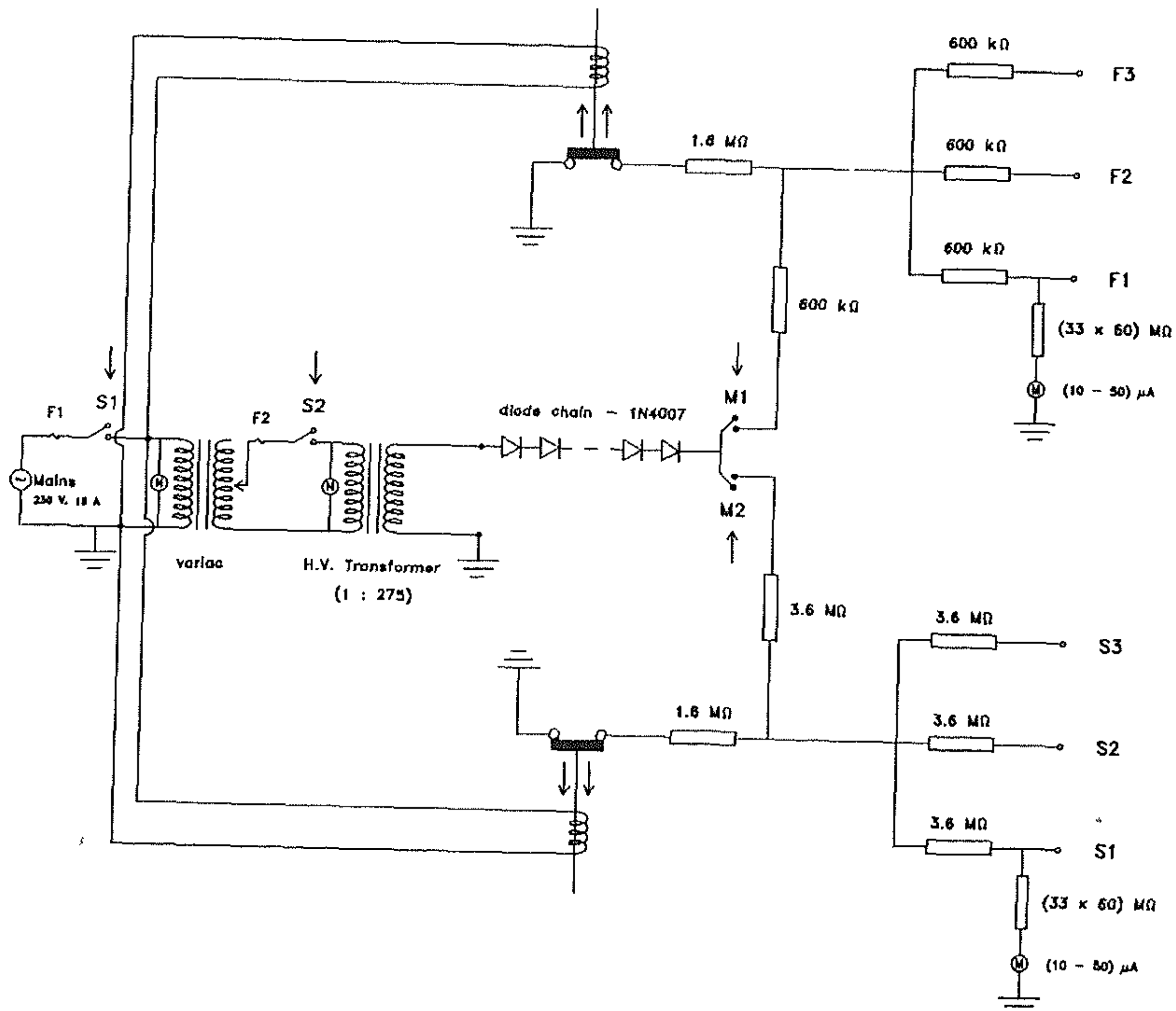


Fig. 6.26 The circuit diagram of the 50 kV charger for Marx1 and Marx2.

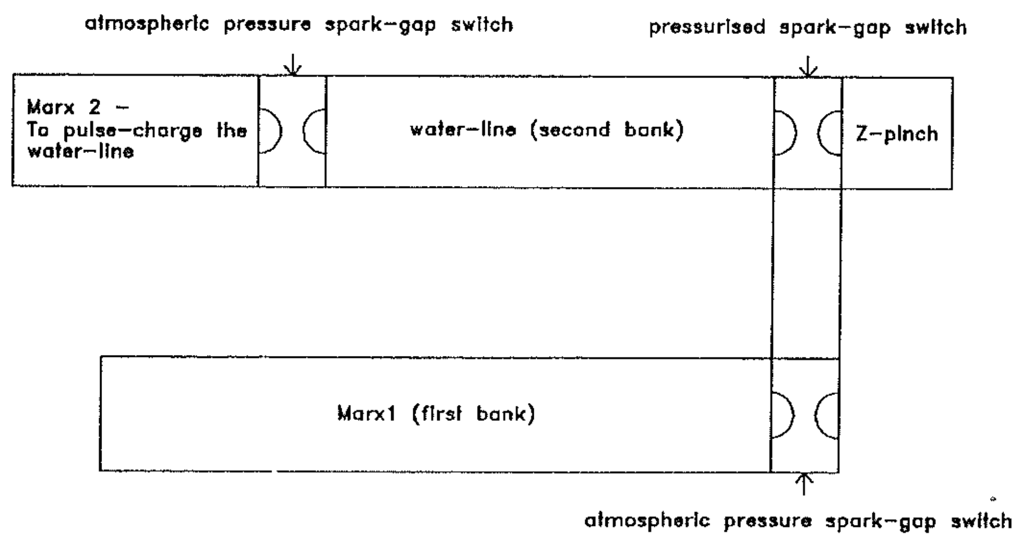


Fig. 6.27 A schematic layout of the MLCCSZP system.

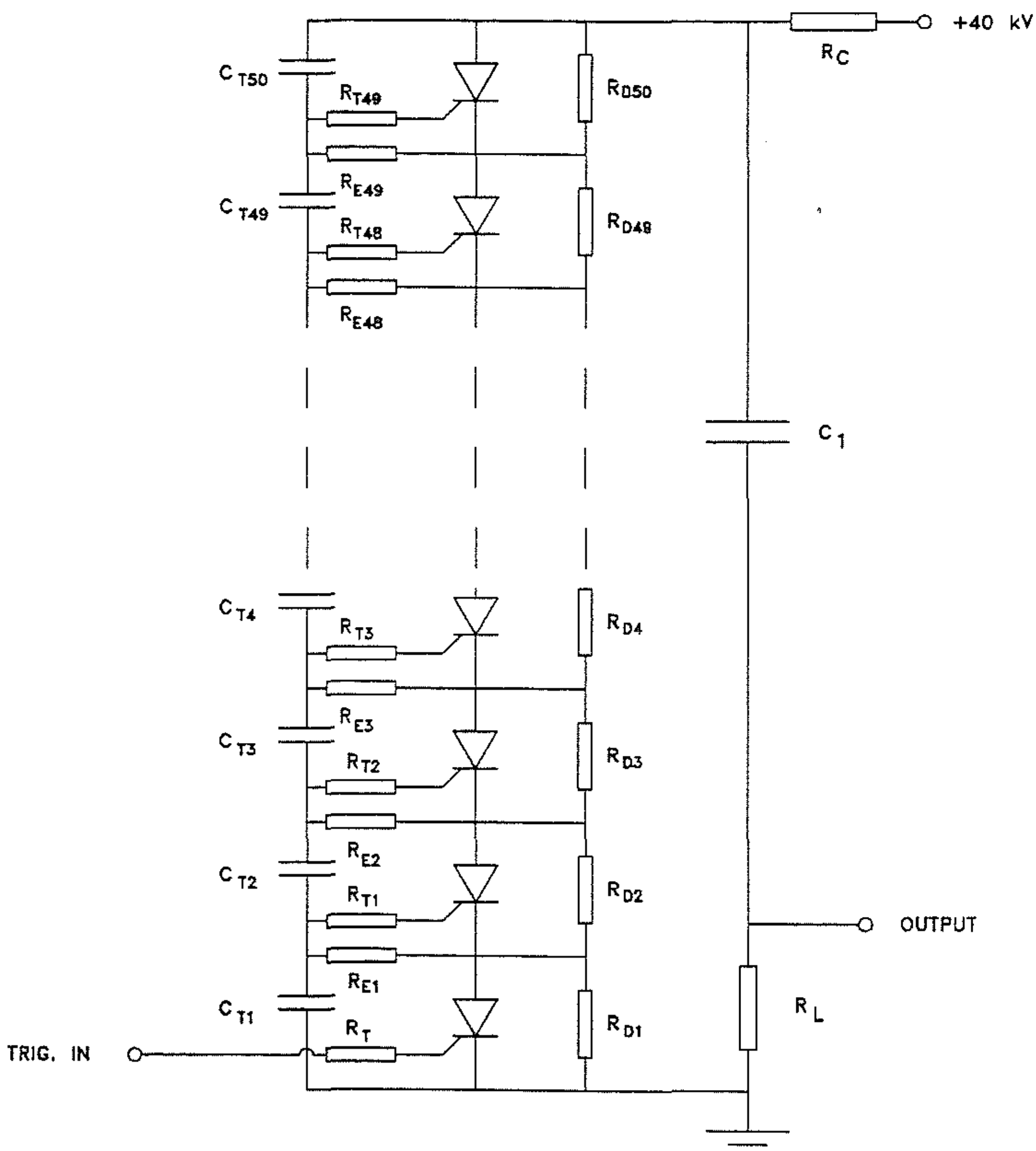


Fig. 6.28 The circuit diagram of the 40 kV, 40 ns pulse generator. (C_T = 0.01 μF (1 kV), R_D = 33 MΩ, R_L = 2 kΩ, R_T = 100 kΩ, R_E = 1Ω, C_1 = 0.01 μF (40 kV)).

conducting, the discharge time of the storage capacitor, C_s is determined by its capacitance, the load resistance and the current dependent resistance of the SCR. The magnitude of the load resistor is limited by the current and power handling ability of the SCR. The rise time of the voltage pulse is determined by the magnitude of the voltage being switched and the circuit configuration.

The output pulse of the cascaded SCR measured across a load of 2 kΩ is as shown in Fig. 6.29.

6.9 Electrical measurements

The output voltage of the Marx1 is measured using a high power resistive voltage probe *[Saw, S.H., Wong, C.S. & Lee, S., 1991]*. It consists of a series of resistive dividers cascaded together. The first section consists of a resistive divider whereby R_1 and R_2 attenuate the voltage to a few thousand volts. The high power resistor R_1 is made from $CuSO_4.5H_2O$ solution. It has been found to be reliable if the resistivity of the solution is kept between 30 and 10^4 Ωcm *[Bishop, A.E. & Edmonds, G.D., 1965]*.

The choice of R_1 and R_2 are determined by the resistivity and the line impedance respectively. Since the probe is used in the pulse discharge environment it is preferable to carry a large signal on the line to the measuring station to increase the signal to noise ratio. Further attenuation to a suitable value for the input channel of the scope is via in-line attenuators. A schematic of the high power voltage probe system is as shown in Fig. 6.30.

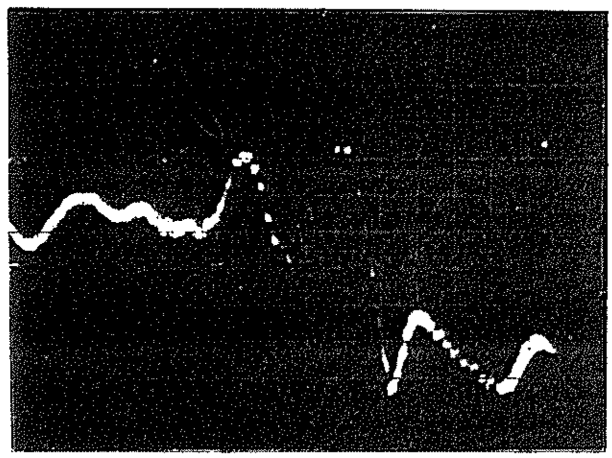
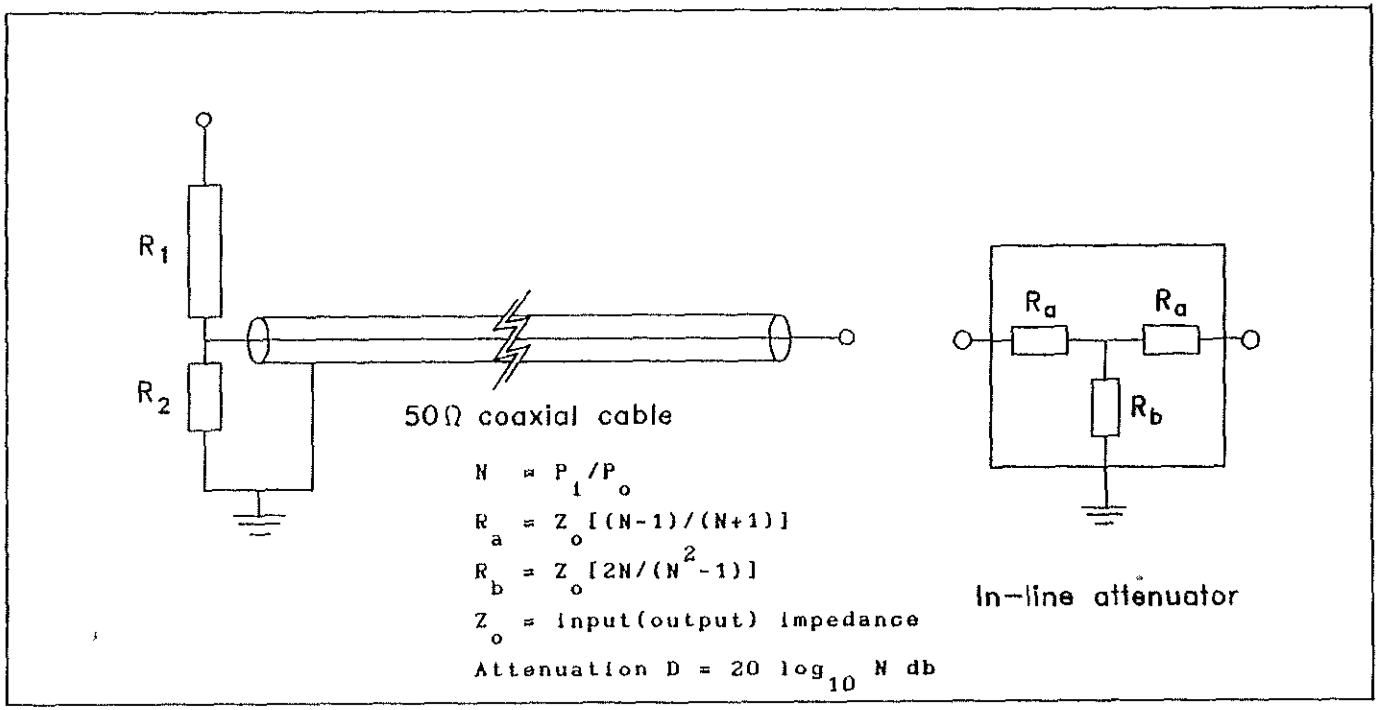


Fig. 6.29 A typical pulse of the cascaded SCR pulse generator. (scale : 13 kV/div, 50 ns/div).



$N = P_i / P_o$

$R_a = Z_o[(N-1)/(N+1)]$

$R_b = Z_o[2N/(N^2-1)]$

Z_o = input(output) impedance

Attenuation $D = 20 \log_{10} N$ db

Fig. 6.30 The high voltage probe system.

6.9.1 Construction of the high power probe

The $CuSO_4$ resistor R_1 shown in Fig. 6.31 is made from a standard solution of 1N diluted to the required resistivity. It is contained in a reinforced rubber water-hose with copper electrodes and nylon rings to secure the copper electrodes to the water-hose. Copper electrodes are chosen to maintain the concentration of the $CuSO_4$ solution as well as to prevent corrosion of the electrodes which will contaminate the solution. The $CuSO_4$ solution is filled into the water-hose through a small opening at one end of the electrode using a syringe. In the preparation of the copper sulphate solution deionised water or distilled water is used. The area of the electrode, length of the water-hose and resistivity of the solution are determined by the desired value of R_1 and the power it has to handle under fault conditions. Generally 1 cm^3 can safely absorb 100 J of energy. In our design, it is required to absorb a maximum energy of 7 kJ under fault conditions; when for example the pinch tube does not breakdown. This sets the minimum volume of $CuSO_4$ solution required to 70 cm^3. Normally the probe is designed to take at least twice the energy. As such the following parameters are chosen; area, $A = 5\ cm^2$, length, $\ell = 50\ cm$ and resistivity, $\rho = 400\ \Omega cm$. This gives a *25 kJ, 4 kΩ* resistor as R_1. With the value of R_2 chosen as 50 Ω, an attenuation factor of *80* times is obtained from this resistive divider stage. Hence for a *150 kV* pulse, about *1900 V* is carried by the line to the measuring station. The second stage of the attenuation makes use of home-made in-line attenuators since commercially available attenuators are of insufficient wattage. These in-line

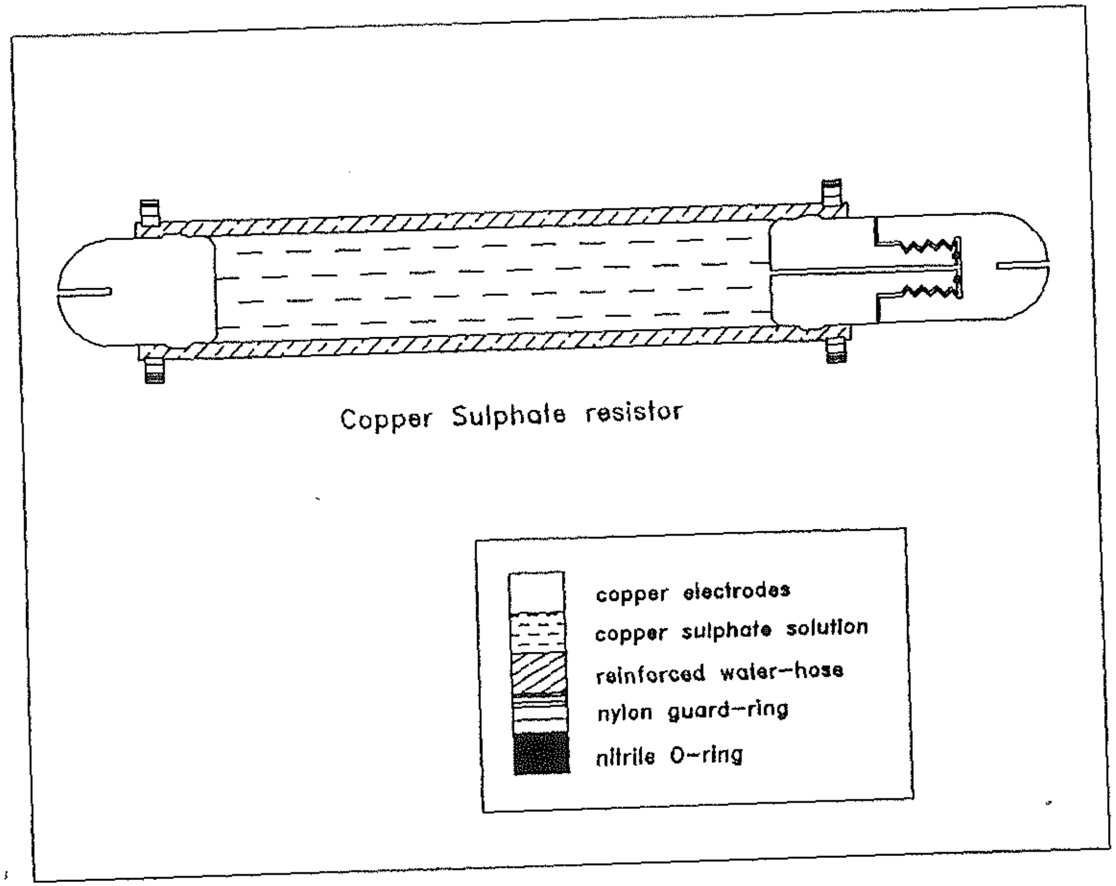


Fig. 6.31 The design of the high power resistor.

attenuators are designed to see an impedance of $50\ \Omega$, at both ends and thus can be placed in series to produce further attenuation. The resulting attenuation factor is given by the sum of the attenuation ratios measured in dB.

The current signal is measured by means of a Rogowski coil placed between the output gap of Marx1 and the anode of the Z-pinch machine.

6.10 Preliminary testing of Marx1

Initially each of the capacitors are charged in stages to 50 kV to test the insulation. Next, the first gap, S_1 (refer Fig. 6.1) which is to be triggered is set. The first capacitor is then charged to 50 kV with the second capacitor connected through a 500 Ω $CuSO_4$ resistor to ground. Upon being triggered S_1 switches, connecting the anode of the first capacitor to the cathode of the second capacitor through the 500 Ω $CuSO_4$ resistor to ground. Similarly the second and third gaps are set and tested individually to hold 50 kV and to break down at 55 kV by charging each of the capacitors one at a time. (shorting the other two capacitors to prevent inductive charging).

The Marx1 is initially tested at 90 kV. It is gradually increased to 120 kV. Fig. 6.32 shows a typical output voltage signal of 120 kV when the Marx1 is discharged through a load of 75 Ω. The operating voltage of the Marx1 will eventually be increased to 150 kV.

Further experimental studies planned for future work will be presented in Chapter 7.

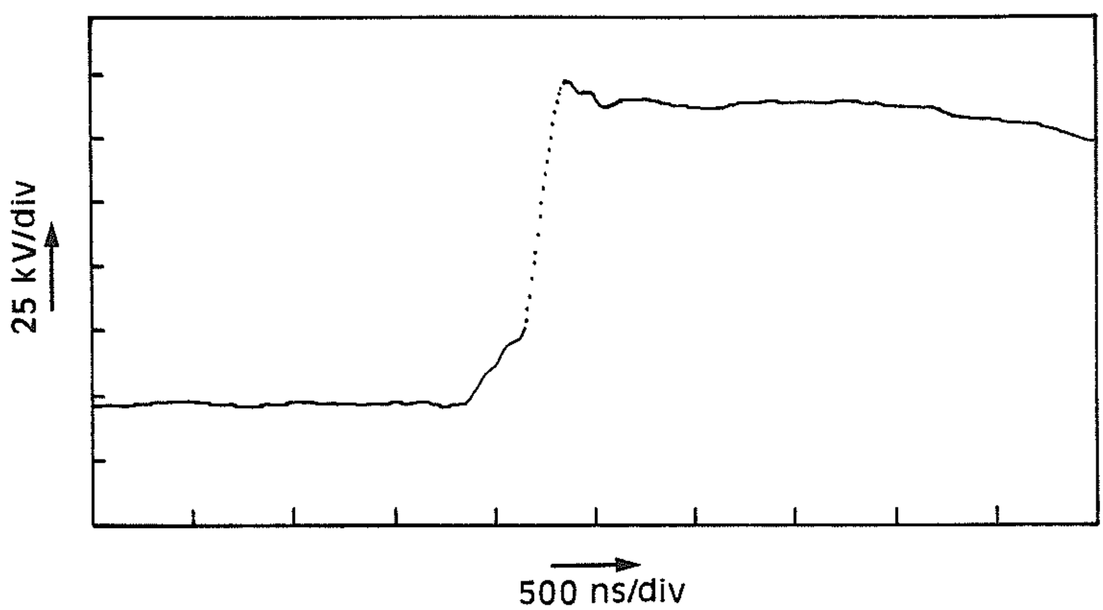


Fig. 6.32 The output voltage signal of Marx1 at 120 kV discharging through a load of 75 Ω.

CHAPTER 7
CONCLUSION

7.1 Introduction

The current-stepping technique was first introduced by *Lee [1984]*. It has been proposed as one of the various means of producing a dense plasma as a prelude to radiation cooling with eventual applications towards pinch fusion reactor. A numerical study has been conducted by *Ali (1985)* using the energy balance slug model assuming γ-constant. Experimental studies take off from thence with the design and construction of the UMCSZP. Analysis of the results obtained leads to the improvement of the energy balance slug model and the design of a new system, the MLCCSZP.

7.2 Review of the experimental results

The first attempt to observe the enhancement effect in the current-stepped compression in a hydrogen Z-pinch using UMCSZP indicates that there is no enhancement upon current-stepping. The deviation of the experimental results from that predicted by the model is attributed to the breakdown of the assumptions adopted in the model. These assumptions adopted in the model do not hold in UMCSZP. It is found necessary to improve the model to include the varying effective specific heat ratio, γ_{eff} associated with the plasma.

From the extended theory, it is found that no enhancement effect will be observed as it is being masked by the thermodynamic and electromagnetic effects of the plasma. The first compression occurs with an average shock speed of 4 cm/μs. Upon current-stepping the computation shows a sharp rise in γ_{eff} leading to a decrease in the compressibility of the plasma. Hence the enhancement effect of the current-stepped compression is being masked. To allow the enhancement effect of current-stepping to emerge, it is necessary for the first compression to have an average compressing shock speed exceeding 10 cm/μs. With these average shock speeds, the shock speed at the instant of current-stepping is close to 30 cm/μs and γ_{eff} is close to 5/3. Therefore the rise in γ_{eff} brought about by current-stepping is also small. Hence the decrease in the compressibility of the plasma is small. Thus the γ-masking effect is reduced enabling the enhancement effect of current-stepping to emerge and become obvious.

To operate at the necessary high speed regime a new system comprising of a Marx-Water-line combination current source, MLCCSZP is designed and constructed on the extended γ-varying model to investigate the enhancement effect of current-stepping.

7.3 Future plans

At the moment of writing this thesis, the MLCCSZP is still under construction. The first Marx bank, Marx1 has been set-up and tested up to *120 kV*. The new pinch unit has been constructed. However, it is yet to be connected to Marx1. The water-line and the second Marx, Marx2 used to pulse charge the water-line is under construction. The Marx has probably never been used to drive a gas compressional Z-pinch before although the water-line has. The combination of both the Marx and the water-line has certainly never been attempted before and presents a considerable technological challenge. However, it would be an interesting attempt to demonstrate the pinch compression enhancement of the current-step.

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